# QE1O1 – GENERAL STANDALONE CRITERIA AND THEIR QED3 APPLICATIONS

Abstract. This plan separates general criteria from their QED3 applications in the field-strength existence argument. In particular, the reflection-positivity criterion gives covariance away from the full diagonal without closedness. The stronger model-specific conclusion requires the additional closed-form argument recorded in the application.

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## 1. Introduction and target statement

The source is QE1M1. B1 gives only the stated off-diagonal covariance conclusion. A2 retains the additional M17–M18 argument used in the full target Z. The general criteria, their applicability checks and the final model statement have separate linked pages.

## Z. QED3 field-strength existence: final assembly

Model-specific. Overview Target.

Statement to establish. Fix $0 < \rho < 1$ and $0 < \bar{m} \le 1$. For the chamber cutoff D0, there are an odd blocking scale $L$, a positive charge ceiling, and a particular real bare mass sequence $m_{0,N}(e)$ independent of the outer volume, with $|m_{0,N}(e)| \le (e\sqrt{\eta})^{3/4-8\varepsilon}/8$, $\eta = L^{-N}$ and $0 < \varepsilon \le 1/200$, such that volume-first field-strength limit laws exist for every fixed admitted $e > 0$. Every such parent law is closed, Gaussian dominated and satisfies tensor E0′–E3. Almost every translation-ergodic component satisfies tensor E0′–E4. For a fixed positive probe supplied by B7 and a possibly smaller positive charge ceiling, the components of that same law have the quantitative positive fourth cumulant stated in A3.

Assembly. Q1 and B3 give the limit law in A1. Q3 chooses the bare sequence used by both branches. A2 applies the general component and rotation criteria, then uses its QED3 strengthening M17–M18 for the full-Schwartz conclusion. Q4 and B5 give A3, with B7 fixing the sign. Intersect the two full-measure sets for the same parent. The result is subsequential existence; it asserts neither uniqueness nor convergence of every simultaneous cutoff sequence.

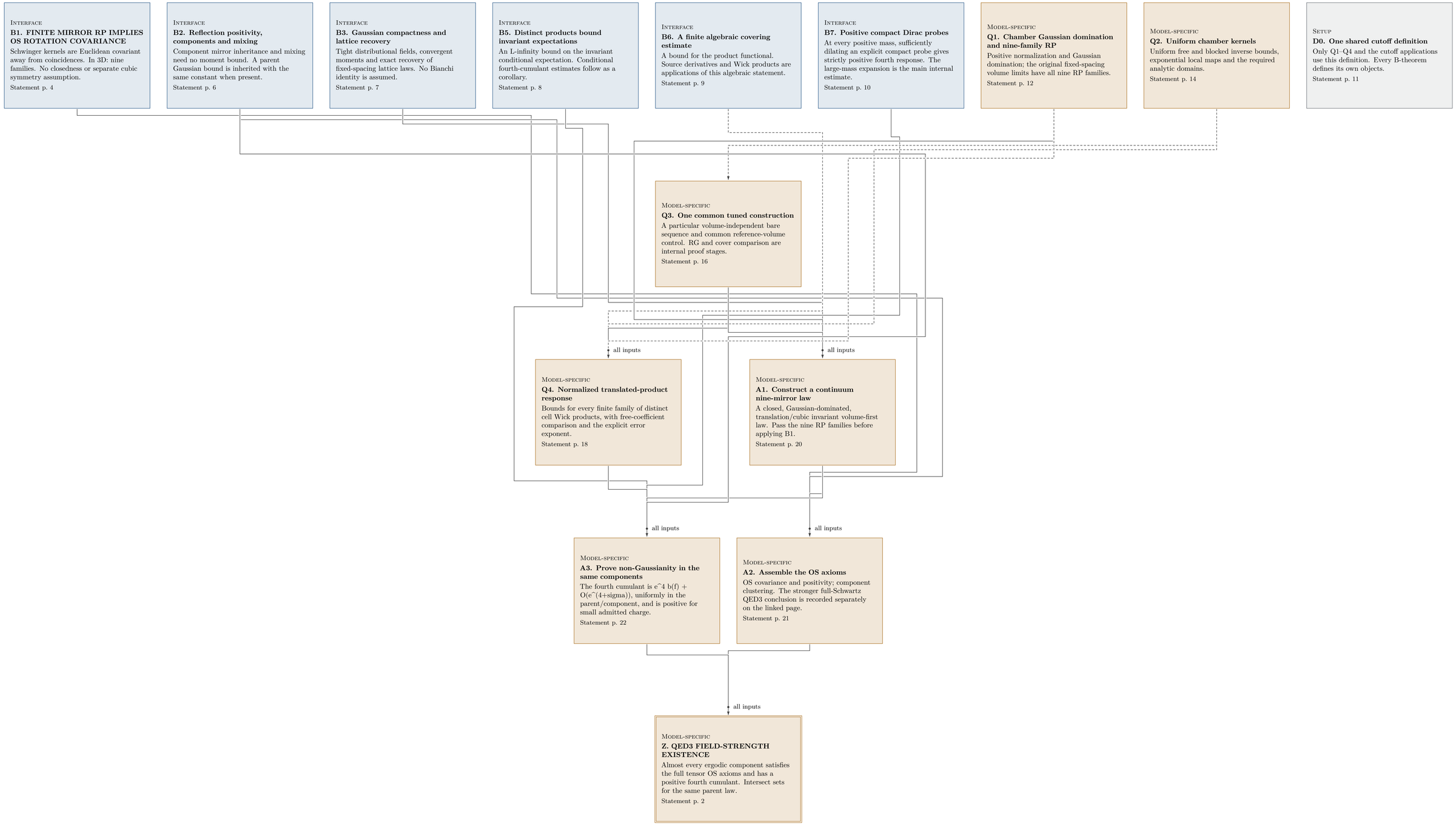
Internal proof details. Z01 records the complete parameter scheme and source assembly.

Source. Manuscript Theorem 1.1 and §27, pp. 1–2, 258–261.

# QE1O1 – General standalone criteria and their QED3 applications

Definitions/setup · Model-specific · Interface (general)

Full hypotheses, source locations and proof estimates appear on the linked statement pages.




Solid arrows supply inputs to an application; dashed arrows identify use inside a proof. Every incoming branch at a junction is required together with the standing hypotheses on the statement page. Click a box for its statement. Return links are navigation, not proof dependencies.

## 2. Coordinate and diagonal positivity imply OS covariance

### B1. Coordinate and diagonal positivity imply OS covariance

Interface. Overview Target.

Statement. Let $d \geq 2$ and let $E$ be a finite-dimensional real inner-product space carrying a continuous orthogonal representation $\rho : O(d) \to O(E)$. In each coordinate plane assume that its complexified rotation action is a sum of the characters $e^{im\theta}$ with $m \in \{-2, -1, 0, 1, 2\}$. Put $\mathcal{S} = \mathcal{S}(\mathbb{R}^d; E)$. For $T(x) = Qx + a$ use the pullback

$$T^* f(x) = \rho(Q^{-1}) f(Qx + a).$$

Let $\mu$ be a Borel probability law on the weak dual $\mathcal{S}'$, and write $\Phi(f) = \mathbb{E}_\mu e^{iF(f)}$. Assume Gaussian domination and translation invariance:

$$\mathbb{E}_\mu e^{F(f)} \leq e^{C\|f\|_2^2} \quad (f \in \mathcal{S} \text{ real}), \qquad C > 0, \qquad \Phi(f(\,\cdot + a)) = \Phi(f) \quad (a \in \mathbb{R}^d).$$

For each hyperplane

$$H \in \mathcal{H} = \{\{x_j = 0\} : 1 \leq j \leq d\} \cup \{\{x_j + x_k = 0\}, \{x_j - x_k = 0\} : j < k\},$$

let $\Theta_H$ be its reflection. Assume that for every finite family of real Schwartz tests $f_\alpha$ supported in one open side of $H$ and every $z_\alpha \in \mathbb{C}$,

$$\sum_{\alpha,\beta} \overline{z_\alpha} z_\beta \Phi(f_\alpha - \Theta_H^* f_\beta) \geq 0.$$

Then the moments $S_N(f_1, \ldots, f_N) = \mathbb{E}_\mu \prod_j F(f_j)$ exist, and for every Euclidean isometry $T$,

$$S_N(T^* f_1, \ldots, T^* f_N) = S_N(f_1, \ldots, f_N) \quad \text{whenever the } f_j \text{ have pairwise disjoint supports.}$$

Equivalently, the tempered tensor kernel of $S_N \circ (T^*)^{\otimes N} - S_N$ is supported on

$$\Delta_N = \{(x_1, \ldots, x_N) : x_j = x_k \text{ for some } j \neq k\}.$$

This is off-diagonal OS covariance. Closedness of $F$ and a separate cubic-symmetry assumption are absent.

Use. For $d = 3$ the hypothesis involves nine mirror directions. Scalar fields, vector fields, exterior forms of any degree and two-tensors satisfy the representation condition. The conclusion allows contact terms on $\Delta_N$; full Schwartz covariance and covariance of the law are stronger conclusions requiring an additional argument.

Proof outline. Translation invariance and RP imply invariance under the stated reflections and hence under signed coordinate permutations. The coordinate and diagonal reconstructions give the sharp spectral cone. Bounded field insertions then give ordered holomorphic moment kernels. First work with configurations whose successive planar displacement vectors lie in a common cone of opening less than $\pi/2$. For tests near these configurations the rotation defect $D_N(\theta) = S_N(R_\theta^* f_1, \dots, R_\theta^* f_N) - S_N(f_1, \dots, f_N)$ extends through the whole complex angle cylinder with growth at most $e^{3N|\operatorname{Im}\theta|}$. Let $M_N$ be the largest absolute nonzero Fourier frequency occurring for some such test tuple near a narrow configuration, and set it to zero if none occurs; $M_0 = 0$. The bound gives $M_N \le 3N$, and quarter-turn symmetry gives $M_N \in 4\mathbb{Z}$. The continued reflection pairing is positive; Cauchy–Schwarz on a split into $a$ and $b$ fields gives $2M_{a+b} \le M_{2a} + M_{2b}$. A $45^\circ$ change of frame multiplies a leading squared-norm coefficient by $(-1)^{M_{2a}/4}$. Positivity in both frames forces $M_{2a} \in 8\mathbb{Z}$. Thus $M_2 = 0$. The even-order increments are nondecreasing multiples of eight; a positive increment would contradict $M_{2a} \le 6a$. All frequencies therefore vanish. Separating-cut analyticity propagates the identity to every configuration outside $\Delta_N$. This argument never removes contact terms.

Internal proof details. Use the completed Yang–Mills rotation paper's reflection reconstruction, positive-kernel deformation, ordered-kernel and narrow-configuration proofs, followed by its geometric reductions. The earlier QED3 two-point Bianchi and polygon proof is a different route and is not the proof of this criterion.

Source. Yang–Mills rotation manuscript, Theorem A, TeX 467–600; reductions 613–827; narrow growth and two-frame positivity 1569–2028; reflection invariance 2088–2136. The expanded supplied growth proof is sources/ym-rotation/S4-section7-growth-fragment.tex.

## 3. Reflection positivity, components and clustering

### B2. Reflection positivity, components and clustering

Interface. Overview Target.

Statement. Let $V$ be a finite-dimensional real inner-product space with an orthogonal representation $\rho : O(3) \to O(V)$. Let $\lambda$ be a probability law on $\mathcal{S}'(\mathbb{R}^3; V)$, with its evaluation sigma-algebra. For $g(x) = Qx + a$ use

$$(g \cdot f)(x) = \rho(Q) f(Q^{-1}(x - a)), \qquad (g \cdot F)(f) = F(g^{-1} \cdot f).$$

Assume translation invariance, and invariance and reflection positivity at the planes

$$x_j = 0 \ (1 \le j \le 3), \qquad x_j - x_k = 0, \quad x_j + x_k = 0 \ (1 \le j < k \le 3).$$

Reflection positivity means $\mathbb{E}_\lambda[\overline{A(r \cdot F)} A(F)] \ge 0$ for bounded character sums with compact smooth tests in an open side of the plane. Invariance at these reflections already implies invariance under the cubic group.

Let $\mathcal{I}$ be the all-translation invariant sigma-algebra, $P = \mathbb{E}_\lambda[\,\cdot \mid \mathcal{I}]$, and $U(a)A(F) = A(\tau_{-a} \cdot F)$. Then

$$\langle A, U(a)B \rangle_{L^2(\lambda)} \longrightarrow \langle A, PB \rangle_{L^2(\lambda)} \quad (|a| \to \infty), \qquad A, B \in L^2(\lambda).$$

In the translation-ergodic decomposition $\lambda = \int \lambda_\xi \, d\nu(\xi)$, almost every component, on one common full-measure set, is translation ergodic, invariant and reflection positive at every real translate of these nine mirror directions, and mixing in every direction:

$$\mathbb{E}_{\lambda_\xi}[\overline{A}\, U(a)B] \longrightarrow \overline{\mathbb{E}_{\lambda_\xi} A}\, \mathbb{E}_{\lambda_\xi} B.$$

No moment bound or field equation is required for these conclusions.

If, in addition, for some $C \ge 0$,

$$\mathbb{E}_\lambda e^{F(f)} \le \exp(C\|f\|_2^2/2) \qquad (f \in \mathcal{S}(\mathbb{R}^3; V) \text{ real}),$$

then almost every same component satisfies the bound with the same $C$, simultaneously for all real tests. Any specified measurable event of full parent measure also has full measure in almost every component.

Use. The first assertion supplies clustering and mirror inheritance independently of ultraviolet moment estimates. The additional Gaussian clause supplies those estimates when the application has them. A linear Bianchi identity is not a hypothesis.

Proof outline. Construct the coordinate reflection Hilbert spaces. The coordinate/diagonal cone gives $H_j \ge |P_i|$, hence every coordinate invariant projection equals the all-translation projection. Semigroup decay is uniform in transverse translations and gives the all-direction limit. Invariant weights have representatives in either half-space and are fixed by its reflection. Applying positivity to $\sqrt{h}A$ proves conditional reflection positivity. Countable determining character algebras give one component set. The Gaussian clause is a separate cube-average argument: translated $L^2$ overlaps of a fixed test tend to zero.

Internal proof details. The finite coefficient space enters only the character algebra and inner products. Orthogonal $\rho$ replaces the exterior-square matrices without changing the spectral, conditional or overlap arguments. The dimension of physical space remains three. Closedness, when wanted, is merely one choice of the last full-measure event.

Source. Manuscript TeX 4893–5194, especially `prop:all-direction-mixing`, and 5200–5329. The Gaussian argument starts only at `eq:component-exponential-average`. The finite-$V$ replacement is a transcription of those component-independent arguments, not a theorem about the existence of a Yang–Mills field law.

## 4. Gaussian lattice fields: compactness and recovery

### B3. Gaussian lattice fields: compactness and recovery

Interface. Overview Target.

Statement. Let $V = \mathbb{R}^r$ with $r < \infty$, and choose fixed offsets $t_\alpha \in [0, 1/2]^3$, one for each component. For $0 < \eta \le 1$, $n \in \mathbb{N}$ and $R = \eta n \ge 1$, let $u_\alpha(x)$, $x \in (\mathbb{Z}/n\mathbb{Z})^3$, have a law satisfying, with one common $C \ge 0$,

$$\mathbb{E}e^{\sum_{\alpha,x} b_\alpha(x)u_\alpha(x)} \le \exp\left(\frac{C}{2}\sum_{\alpha,x}|b_\alpha(x)|^2\right) \quad (b \text{ real}).$$

Extend $u$ periodically and set $F^\sharp_\alpha = \eta^{-3/2}u_\alpha(x)$ on $\eta(x + t_\alpha + [-1/2, 1/2)^3)$. Let $h_k$ be the scalar Hermite functions of $1 + |x|^2 - \Delta$, with $\lambda_k = 2|k|_1 + 4$, and define

$$\|F\|^2_{-4} = \sum_{\alpha,k} \lambda_k^{-8}|F(h_k e_\alpha)|^2.$$

The step-field laws are uniformly tight on the Hilbert space $H_{-4}$ defined by this norm. Fixed-mesh volume limits exist, and the family of all such limits is uniformly tight.

At fixed $\eta$, along a sequence with $n \to \infty$, if step-field laws converge to $\mu^\eta$ on $H_{-4}$, the periodic coefficient laws converge along the same sequence, in the product topology, to a law $\nu^\eta$ on the infinite lattice, and

$$\mu^\eta = (\sharp_\eta)_*\nu^\eta.$$

The componentwise step map is defined and Borel on a set of full $\nu^\eta$ measure. The law $\nu^\eta$ obeys the same Gaussian inequality for finitely supported coefficients.

If $\eta \to 0$, $R \to \infty$, and the step-field laws converge weakly, their limit satisfies

$$\mathbb{E}e^{F(f)} \le \exp(C\|f\|_2^2/2) \quad (f \in \mathcal{S}(\mathbb{R}^3; V) \text{ real}).$$

Every finite-family generating function converges locally uniformly; consequently all joint moments and cumulants converge.

Use. This criterion contains no determinant, lattice field equation, coefficient representation of rotations or reflection positivity. It supplies the limiting random distributions and the exact fixed-mesh lattice law behind each volume limit.

Proof outline. Cell averaging contracts the coefficient $\ell^2$ norm. The periodized test is bounded by one continuous weighted $L^2$ seminorm, uniformly for $R \ge 1$. Sum Hermite variances to bound a stronger negative oscillator norm, then use its compact embedding into $H_{-4}$. For exact recovery take a smooth bump inside each averaging cell with the required integral. Bump evaluations recover every lattice coordinate continuously; countably many closed linear relations recover the original step distribution. Exponential uniform integrability passes generating functions and derivatives.

Internal proof details. The finite sum over $r$ components changes only the compactness constant. The cell offsets do not change the contraction, periodization or bump inverse. The intermediate $H_{-3}$ estimate has a constant depending on $C$ and $r$; the numerical bound 61 is its normalized three-component QED3 case. For plaquettes, take $r = 3$, $t_{ij} = (e_i + e_j)/2$. The discrete Bianchi passage and the equivariance of these particular offsets are additional QED3 specialization statements; they are not assumptions of this compactness criterion.

Source. Manuscript TeX 4430–4550 and 4551–4615, followed by the fixed-spacing inverse and recovery argument at 4691–4794. The coefficient argument there is componentwise. The separate cochain/Euclidean-action passage is 4616–4690.

## B5. Distinct translated products bound conditional expectations

Interface. Overview Target.

Statement. Let $T_y$ be a measure-preserving $\mathbb{Z}^d$ action, $d \geq 1$, on a probability space, and set $U(y)G = G \circ T_{-y}$. Let $\mathcal{I}_{\mathrm{lat}}$ be its invariant sigma-algebra and $\mathcal{I} \subset \mathcal{I}_{\mathrm{lat}}$. Suppose $G$ is real, belongs to every finite $L^p$, and, for every even $J \geq 2$ and pairwise distinct $y_1, \dots, y_J \in \mathbb{Z}^d$,

$$\left| \mathbb{E} \prod_{j=1}^{J} U(y_j)G \right| \leq u^J, \qquad u \geq 0.$$

Then

$$\left\| \mathbb{E}[G \mid \mathcal{I}_{\mathrm{lat}}] \right\|_\infty \leq u, \qquad \left\| \mathbb{E}[G \mid \mathcal{I}] \right\|_\infty \leq u.$$

In particular let $X$ be real with all finite moments and $\mathbb{E}[X \mid \mathcal{I}] = 0$. For $g > 0$, $v, c_2, c_4 \in \mathbb{R}$, put

$$G_m = g^{-m} H_m(X; v) - c_m, \quad H_2(x; v) = x^2 - v, \quad H_4(x; v) = x^4 - 6vx^2 + 3v^2, \qquad m = 2, 4.$$

If the preceding distinct-product hypotheses hold for $G_2, G_4$, with $u_2, u_4$, their conditional fourth cumulant obeys the exact bound

$$\left| g^{-4} \kappa_4(X \mid \mathcal{I}) - (c_4 - 3c_2^2) \right| \leq u_4 + 6|c_2|u_2 + 3u_2^2 \quad \text{almost surely,}$$

where $\kappa_4(X \mid \mathcal{I}) = \mathbb{E}[X^4 \mid \mathcal{I}] - 3(\mathbb{E}[X^2 \mid \mathcal{I}])^2$.

Use. For $C > 0$, $\sigma > 0$, $b \in \mathbb{R}$, small-coupling rates are a specialization: if $0 < g \leq 1$, $u_4, (1 + |c_2|)u_2, |c_4 - 3c_2^2 - b| \leq Cg^\sigma$, then

$$|\kappa_4(X \mid \mathcal{I}) - g^4 b| \leq (8C + 3C^2) g^{4+\sigma}.$$

An ergodic decomposition is needed only to interpret these conditional expectations as component moments.

Proof outline. Average $G$ over increasing lattice cubes and expand a fixed even moment. Repeated-index tuples have vanishing relative count; Hölder controls them, while distinct tuples obey the hypothesis. The mean ergodic limit and Fatou bound every even moment of the invariant conditional expectation. Let the order tend to infinity, then use the tower property for $\mathcal{I}$. For the corollary write $d_m = \mathbb{E}[G_m \mid \mathcal{I}]$ and use

$$g^{-4} \kappa_4 = c_4 + d_4 - 3(c_2 + d_2)^2.$$

Internal proof details. No Gaussian hypothesis, field space, reflection, lattice ergodicity or coupling asymptotic is required for the primary criterion. All finite moments are used at arbitrarily high even orders. Zero conditional mean is required when naming the displayed fourth-moment combination a cumulant.

Source. Manuscript TeX 24983–25058 and 25145–25187, `lem:UV-invariant-product-bound` and `eq:UV-cumulant-identity`. Only even orders occur in the actual averaging proof.

6. A finite algebraic covering estimate

## B6. A finite algebraic covering estimate

Interface. Overview Target.

Statement. Fix integers $N, p \geq 1$ and $r > 0$. In

$$\mathcal{J} = \mathbb{C}[t_1, \dots, t_N]/(t_1^{p+1}, \dots, t_N^{p+1}), \qquad \|P\|_r = \sum_{\alpha_c \leq p} |P_\alpha| r^{|\alpha|},$$

let $\phi_c$ be one-variable polynomials with $\|\phi_c\|_r \leq A$, $A \geq 1$. Let $\ell_c$ be one-variable linear functionals satisfying

$$\ell_c(\phi_c) = 0, \qquad |\ell_c(t_c^j)| \leq D r^j \quad (0 \leq j \leq p), \qquad D \geq 0.$$

Let $\nu$ be a finite complex measure of total variation at most one. Suppose that the following identity holds in $\mathcal{J}$:

$$\Phi = \prod_{c=1}^{N} \phi_c(t_c) \int \prod_{\varnothing \neq S \subset \{1,\dots,N\}} (1 + k_S(\omega; t))\, d\nu(\omega) + L.$$

The coefficients of $k_S$ are measurable and it depends only on variables in $S$. For $0 < s < 1$, $\kappa > 0$, $\lambda \geq 0$, assume

$$\|k_{\{c\}}\|_r \leq \kappa, \quad \|k_S\|_r \leq \kappa^N \ (|S| \geq 2), \quad \|L\|_r \leq \lambda^N, \quad N\kappa^{1-s} \leq \tfrac{1}{2},$$

uniformly in $\omega$. Then, with $\ell = \bigotimes_{c=1}^{N} \ell_c$,

$$|\ell(\Phi)|^{1/N} \leq D\big(A \exp(1) \kappa^s + \lambda\big).$$

Use. The fourth Taylor jet is the case $p = 4$. If $\phi_c(0) = 1$, $m \in \{2, 4\}$ and $c_{m,c} = \phi_c^{(m)}(0)$, the functionals $(\partial_{t_c}^m - c_{m,c})|_{t_c=0}$ annihilate the reference factors. When $\Phi$ is a normalized source jet, their tensor product is the expected product of centered Wick polynomials: for $g > 0$, $v \geq 0$ and real $X_c$ with all exponential moments, take

$$\Phi = \left[ \mathbb{E} \exp\left( g^{-1} \sum_c t_c X_c - \frac{v}{2g^2} \sum_c t_c^2 \right) \right]_{\leq 4 \text{ in each } t_c}.$$

Then $\ell(\Phi) = \mathbb{E} \prod_c [g^{-m} H_m(X_c; v) - c_{m,c}]$, where $H_m(x; v) = \partial_z^m e^{zx - vz^2/2}|_{z=0}$. Probability enters only in this application.

Proof outline. The coefficient norm is submultiplicative and $|\ell(P)| \leq D^N \|P\|_r$. Expand the error factors. If their assigned sets do not cover some cell $c$, its untouched factor $\phi_c$ is annihilated. Give singletons cost one and larger sets cost $N$. Every surviving family has cost at least $N$. Retain $\kappa^{sN}$ and sum the remaining costs:

$$(1 + \kappa^{1-s})^N (1 + \kappa^{(1-s)N})^{2^N - 1 - N} \leq \exp(1).$$

The reference-factor norm is at most $A^N$, the measure has total variation at most one, and the remainder contributes at most $(D\lambda)^N$. Taking the $N$th root gives the assertion.

Source. QED3 manuscript TeX 23390–23545. The supplied covering proof uses only coefficient submultiplicativity, the functional norm, annihilation of the reference factor and the integral norm bound. Its fourth-order, identical-factor, derivative-functional and positive-measure specializations are not used in those steps. T07 and T15 record the Wick application; T12–T14 establish its model-specific factorization hypotheses.

## B7. Positive compact probes for the three-dimensional Dirac response

Interface. Overview Target.

Statement. Let $\gamma_j = \operatorname{diag}(\sigma_j, -\sigma_j)$ with the Pauli matrices $\sigma_j$, and set

$$S_m = (m + \gamma \cdot \partial)^{-1} \quad \text{on } L^2(\mathbb{R}^3; \mathbb{C}^4).$$

For every $m > 0$, there is a real $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ such that

$$-6 \operatorname{Tr}(S_m i\gamma \cdot d^*\beta)^4 > 0.$$

The fourth power is trace class and its trace is real. More specifically, one can start with any real compact smooth $\beta_0$ satisfying $dd^*\beta_0 \not\equiv 0$. Then $\beta_\ell(x) = \ell^2 \beta_0(x/\ell)$ has the displayed positivity for all sufficiently large $\ell$.

Use. Choose a single positive compact probe before the lattice and interacting approximations and before fixing the probe-dependent charge interval.

Proof outline. For $a \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, the principal internal estimate is

$$Q_{4,M}(a) := -\tfrac{1}{4} \operatorname{Tr}(S_M i\gamma \cdot a)^4 = \frac{1}{240\pi M^5} \int |\operatorname{curl} a|^4 + O_a(M^{-7}), \quad M \geq 1.$$

The Yukawa kernel justifies the trace, reality and mass derivative. The local gauge reduction and quartic homogeneity identify the leading antisymmetric first derivative; the supplied constant-field heat kernel computes its positive coefficient. Unitary dilation gives

$$Q_{4,m}(a_\ell) = \ell^8 Q_{4,m\ell}(a), \qquad a_\ell(x) = \ell a(x/\ell).$$

When $a = d^*\beta_0$ and $dd^*\beta_0 \neq 0$, the leading integral is strictly positive.

Internal proof details. The numerical coefficient and remainder are central proof estimates, not additional hypotheses of the existence statement. The source's explicit bump is a convenient example of $\beta_0$, not a restriction on the theorem. The doubled three-dimensional Dirac representation and abelian real connection remain part of the statement; no nonabelian or higher-dimensional sign theorem is claimed.

Source. Manuscript TeX 16905–17301, Proposition `prop:positive-probe`, particularly `eq:large-mass-quartic` and `eq:probe-positive-dilation`. The large-mass estimate is proved for every compact real $a$, and the dilation identity is exact.

## 8. The chamber cutoff: the one shared model definition

### D0. The chamber cutoff: the one shared model definition

Setup. Overview Target.

Definition. Fix $0<\rho<1$, weights $(w_0,w_1,w_2,w_3)=(6,6\sqrt{2},6\sqrt{2},6)$, and a point $v_0$ in $C=\{0<x_3<x_2<x_1<1/2\}$. The group $W=\mathbb{Z}^3\rtimes O_{\mathrm{cub}}$ acts on chambers. Write $s_i$ for reflection in the four walls of $C$, with inward normals $e_3,(e_2-e_3)/\sqrt{2},(e_1-e_2)/\sqrt{2},-e_1$. Fermion sites are $gv_0$ and their neighbours are $gs_iv_0$. The photon projection is $\pi(k+rv_0)=k$. The pair notation $(c,g)$ means the site $c+gv_0$, with $c\in\mathbb{Z}^3$ and $g\in O_{\mathrm{cub}}$. Use $\gamma_j=\mathrm{diag}(\sigma_j,-\sigma_j)$, where $\sigma_j$ are the Pauli matrices, and $\gamma_v=\sum v_j\gamma_j$. For a real photon link cochain $B$, put $B(x,x)=0$ and define

$$(D^{\mathrm{ch}}[B]\psi)(x)=\frac{\rho}{2}\Big(\sum_i w_i\Big)\psi(x)-\sum_{i=0}^{3}\frac{w_i}{2}(\rho-\gamma_{(x'-x)/|x'-x|})e^{iB(\pi x,\pi x')}\psi(x'),\qquad x'=gs_iv_0\ (x=gv_0).$$

Equivalently write $D_q[A]=D^{\mathrm{ch}}[qA]$. On a torus $\mathbb{Z}^3/\Gamma$, where $\Gamma\subset\mathbb{Z}^3$ has rank three and $\min_{0\neq\gamma\in\Gamma}|\gamma|_\infty\geq 4$, spinors have a prescribed character $\chi:\Gamma\to\{\pm1\}$. The photon variables are real links $A$ and closed integer plaquette cochains $z$, modulo $(A,z)\sim(A+2\pi k/q,z-dk)$, $k$ an integer link cochain, $q>0$. Their curvature is $u=dA+2\pi z/q$. Count integer cochains and integrate links over a fundamental phase cell. The unnormalized weight is

$$e^{-\|u\|_{\ell^2}^2/2}\det(D^{\mathrm{ch}}[qA]+\mu),\qquad \mu\in\mathbb{R}.$$

The norm counts one positive orientation of each plaquette. The first model theorem proves that this has a strictly positive finite mass; its normalized curvature law will be denoted $\nu_{\Gamma,q,\mu}$. Write $Z(\beta)$ for the same unnormalized partition integral with $\exp(-\|u-\beta\|_{\ell^2}^2/2)$ in place of the Maxwell weight. The cubic unit torus is $C_n=C(n)=\mathbb{Z}^3/(n\mathbb{Z})^3$.

For the continuum construction let $L\geq 5$ be odd, $N,T$ nonnegative integers, and take cubic sides $n=8hL^{T+N}$, antiperiodic spinors in all three directions, and

$$\eta=L^{-N},\quad R=\eta n=8hL^T,\quad h=L^{\lfloor\log_L\bar m^{-1}\rfloor},\quad q=e\sqrt{\eta},\quad \mu=\bar m\eta+m_{0,N}(e),\quad 0<\bar m\leq 1.$$

The physical field is $\eta^{-3/2}u$ on cubes of side $\eta$ centred at the physical plaquette centres, separately for the three two-form components. A volume-first law means first taking $T\to\infty$ at fixed $N$, then a subsequential limit as $N\to\infty$. The cell average used to sample a continuum test is $(J_\eta f)(x)=\eta^{-3}\int_{[-\eta/2,\eta/2]^3}f(x+y)\,dy$, componentwise; on a torus first periodize $f^R(x)=\sum_{k\in\mathbb{Z}^3}f(x+Rk)$.

Scope of this definition. The general theorems about random fields, invariant conditional moments and finite jets define their own objects and do not refer to this regulator.

Source. Manuscript §2 and Theorem 1.1. The finite geometric and normalization details are in M01.

## 9. All-flux Gaussian domination and nine reflection families

### Q1. All-flux Gaussian domination and nine reflection families

Model-specific. Overview Target.

Objects and hypotheses. Use the chamber regulator above, with the standard weights, any $q > 0$, and any real lattice mass $\mu$. Coordinate torus sides are at least four. On the covering lattice let $\Omega = \mathbb{R}^{P(\mathbb{Z}^3)}$, where $P(\mathbb{Z}^3)$ is the set of positively oriented unit plaquettes, with its product topology. For a reflection $r$, the positive-side sigma-algebra is generated by plaquette curvatures whose entire plaquette lies in the chosen open half-space. The map $r^*F$ includes the sign of the induced action on oriented plaquettes.

Claim. All partition functions are finite and strictly positive. On every coordinate torus and for every real two-cochain $\beta$,

$$0 < Z(\beta) \le Z(0), \qquad \mathbb{E}\exp\langle \beta, F\rangle \le \exp(\|\beta\|_{\ell^2}^2/2).$$

For complex $z_j$ and real $\beta_j$, this also gives

$$\left|\mathbb{E}\exp\sum_j z_j\langle\beta_j, F\rangle\right| \le \exp\left(\frac{1}{2}\left\|\sum_j (\operatorname{Re} z_j)\beta_j\right\|^2\right).$$

Finite-torus reflection positivity holds at each coordinate mirror $x_\mu = c$, $c \in \frac{1}{2}\mathbb{Z}$: for bounded functions $G$ of plaquettes wholly contained in its open half-period slab $c < x_\mu < c + n_\mu/2$,

$$\mathbb{E}[\overline{G(r^*F)}G(F)] \ge 0.$$

This assertion concerns a half-period slab, rather than the image of an infinite half-space.

There is also the following finite-volume product estimate. Take coordinate cells $w(c + (0,1)^3)$, where $w \in \frac{1}{2}\mathbb{Z}_{>0}$, and a rectangular carrying torus with side lengths $ws_\mu \in \mathbb{Z}_{\ge 4}$, each $s_\mu$ even. A real or complex scalar curvature observable is admissible here if it is bounded, or a polynomial in finitely many curvatures, and its plaquettes are contained in a box strictly inside one open cell. Reflect an observable $O$ in the base cell into every cell, using conjugation at each reflection, to obtain $O^{[c]}$. With $N = s_1 s_2 s_3$, define

$$\|O\|_{\mathrm{fold}} = \left(\mathbb{E}\prod_{c \bmod s} O^{[c]}\right)^{1/N}.$$

The expression under the root is nonnegative. For observables $O_c$ in distinct cells, reflected back to the base cell in this norm, the law obeys

$$\left|\mathbb{E}\prod_c O_c\right| \le \prod_c \|O_c\|_{\mathrm{fold}}.$$

For a fixed fold, the unnormalized partition function as a function of one even side count $s_\mu$ is $\operatorname{Tr}(Y^*Y)^{s_\mu/2}$, with $Y$ independent of that count. Thus its $1/s_\mu$ power is nonincreasing; when $Y \ne 0$, its logarithm is convex. These are the precise finite-volume facts used in the response transfer.

Fix an odd integer $L \ge 5$. Extend each cubic law periodically to $\Omega$, and let its side tend to infinity along $n = 8L^T$, $T \to \infty$, or a cofinal subsequence. These laws are tight. Every subsequential limit $\nu$ is supported on $dF = 0$, obeys the same Gaussian bound for every finitely supported $\beta$, is invariant under all lattice translations, signed coordinate permutations and $F \mapsto -F$, and satisfies

$$\int \overline{G(r^*F)}G(F)\, d\nu(F) \ge 0$$

for every $G$ in the $L^2(\nu)$ positive-side sigma-algebra and every mirror in the following nine families:

$$x_\mu = c, \quad c \in \tfrac{1}{2}\mathbb{Z}; \qquad x_\mu \pm x_\nu = c, \quad c \in \mathbb{Z}, \quad \mu < \nu.$$

The diagonal assertion concerns the same law $\nu$ obtained from coordinate cubic tori; it does not replace $\nu$ by a limit on a different torus family.

Proof work. Construct the positive crossing form for the chamber Grassmann integral. Cut-link integration enforces a nonnegative Gauss-law indicator. Identify the form with positive slice operators and their flux-summed traces. Schatten's trace inequality, applied in two coordinate directions, yields the Gaussian bound including all flux sectors.

For diagonal positivity compare equal-volume coordinate and diagonal adapted tori at fixed $q, \mu, \rho, w$, with an error in log partition functions that is $o(n^3)$. For each local trigonometric polynomial, amplify its tilt in two overlapping diagonal frames. Regularized field insertions and elimination of the outer tilted layer construct a Gaussian-dominated diagonal-positive comparison state. Convex separation places the original cubic limit in the compact convex set of such states. A limit on diagonal adapted tori alone would not do this.

Proof tools. The regulator; finite-dimensional Grassmann algebra; Schatten trace inequalities and separation of a compact convex set from a point. Small charge is not required. Comparison errors may depend on the fixed charge and mass; this argument does not justify a simultaneous volume and mesh limit.

Source. Current manuscript, Sections 2–4, pp. 6–47; the final statement is `thm:diag-lattice-limits`, TeX 4130–4183. Gaussian domination is `thm:slab-gaussian`, TeX 2404–2465; adapted positivity is `thm:reg-adapted-positivity`, TeX 1203–1222. Rectangular folds are TeX 1922–1943 and 24230–24284. Proof tasks, rather than unverified additional claims, describe the intervening comparison and elimination.

## 10. Uniform chamber and Maxwell block kernels

### Q2. Uniform chamber and Maxwell block kernels

Model-specific. Overview Target.

Objects and hypotheses. For the standard-weight regulator use the mesh $a = L^{-k}$, $k \geq 0$, with pairing $(a^3/48)\sum_{c,g}$. On mesh-scaled neighbours define $D_A = a^{-1}D_q[aA] = a^{-1}D^{\mathrm{ch}}[qaA]$. Let $Jv(c,g) = v(c)$, $J^*u(c) = 48^{-1}\sum_g u(c,g)$, and $Q = 1 - JJ^*$.

Here $L \geq 5$ is odd and $0 < \alpha < 1$. The unit cubes are $\Delta_z = \{c : |c - z|_\infty < 1/2\}$, $z \in \mathbb{Z}^3$; cubes of side $L$ are centred at $L\mathbb{Z}^3$. The covariant mean from a fine site to its unit-cube centre uses the phase obtained by summing the arithmetic mean of the six coordinate-order path sums through the nested $L$-adic block centres. A path sum has the factor $qa$. Writing this phase as $\varphi_A(z,c)$, set

$$(\mathcal{Q}_k(A)u)(z) = \frac{a^3}{48}\sum_{c\in\Delta_z,g}\varphi_A(z,c)u(c,g), \qquad (\mathcal{Q}_k^\sharp(A)v)(c,g) = \varphi_A(z,c)^{-1}v(z).$$

For $z$ in the $L$-cube centred at $z'$, set

$$(P_L(A)v)(z) = U_A(z',z)^{-1}L^{-3}\sum_{|z''-z'|_\infty<L/2}U_A(z',z'')v(z''),$$

where $U_A$ uses the same six-order path convention. Define $\Pi_1(A) = \mathcal{Q}_k^\sharp(A)\mathcal{Q}_k(A)$ and $\Pi_L(A) = \mathcal{Q}_k^\sharp(A)P_L(A)\mathcal{Q}_k(A)$. For $k = 0$ the analogous $\Pi_L$ includes the chamber-label mean. Put $\gamma_5 = \left(\begin{smallmatrix} 0 & I_2 \\ I_2 & 0 \end{smallmatrix}\right)$, $b_L = L^{-1}$, and

$$b_k = \frac{1-L^{-1}}{1-L^{-k}}\ (k \geq 1), \quad \alpha_y = b_k i\gamma_5 y(b_k + i\gamma_5 y)^{-1}, \quad \beta_y = b_k^2 b_L(b_k + b_L + i\gamma_5 y)^{-1}(b_k + i\gamma_5 y)^{-1}.$$

For $0 \leq \mu \leq 1$, let

$$F_y(A) = \begin{cases} D_A + \mu + \alpha_y\Pi_1(A) + \beta_y\Pi_L(A), & k \geq 1, \\ D_A + \mu + b_L\Pi_L(A) + i\gamma_5 y, & k = 0. \end{cases}$$

For $k \geq 1$, $y \in \mathbb{R}\cup\{\infty\}$, with $\alpha_\infty = b_k$, $\beta_\infty = 0$; for $k = 0$, $y \in \mathbb{R}$.

Claim: fermions. There are positive $C, \gamma, c$, depending only on $L, \rho, \alpha$, with the following properties. The free covering-space Green kernel $G = (D_0 + \mu)^{-1}$, $0 < \mu a \leq 1$, and its mass-zero limit obey

$$|G(x,x')| \leq C(a + |c - c'|_\infty)^{-2}.$$

For neighbouring chamber sites $x, \widetilde{x}$,

$$|G(x,x') - G(\widetilde{x},x')| \leq Ca(a + |c - c'|_\infty)^{-3}.$$

With $d_X(x,\widetilde{x}) = \max(a, |c - \widetilde{c}|_\infty)$, the general difference bound is

$$|G(x,x') - G(\widetilde{x},x')| \leq Cd_X(x,\widetilde{x})^\alpha\big[(a + |c - c'|_\infty)^{-2-\alpha} + (a + |\widetilde{c} - c'|_\infty)^{-2-\alpha}\big].$$

For real $A$, the heavy block also obeys

$$\|(QD_AQ + t)^{-1}\|_{2\to2} \leq (t + g_{\mathrm{h}}/a)^{-1}, \quad t > -g_{\mathrm{h}}/a, \qquad g_{\mathrm{h}} = \rho\lambda_{\mathrm{int}}/2 > 0.$$

Here $\lambda_{\mathrm{int}}$ is the smallest positive eigenvalue of $\sum_{i=0}^2 w_i(1 - R_i)$ on the 48 chamber labels, where $(R_iv)(g) = v(gs_i)$, $0 \leq i \leq 2$, and $s_i$ are the first three wall reflections in D0.

For the blocked kernel there is a power $M$ of $L$ and, for every $\varepsilon > 0$, a $q_0 > 0$, such that the following holds uniformly. The torus side $\ell/(LM)$ is an odd integer at least 13, its spinor character is any unitary character, $0 < q \leq q_0$, and $A = A_0 + A_1$, with $A_0$ real and $\|A_1\|_\infty < q^{-1+\varepsilon}$. On every cube of side $11M$, suppose

$$A_0 = B_0 + d\lambda, \qquad B_0, \lambda \text{ real}, \qquad \|B_0\|_\infty < q^{-1+\varepsilon}, \quad d\lambda(c,c') = a^{-1}(\lambda(c') - \lambda(c)).$$

Then $S_y(A) = F_y(A)^{-1}$ exists and is analytic. Define its exponentially weighted block-row norm by

$$\|T\|_\gamma = \sup_z \sum_{z'} e^{\gamma d(z,z')} \|\mathbf{1}_{\Delta_z} T \mathbf{1}_{\Delta_{z'}}\|_{\infty\to\infty}.$$

Distances are periodic sup distances of unit-cube centres. For distinct $x, \widetilde{x}$ with $0 < d_X(x,\widetilde{x}) < 1$, put

$$\delta_{\alpha,A} v(x,\widetilde{x}) = \frac{U_A(x,\widetilde{x}) v(\widetilde{x}) - v(x)}{d_X(x,\widetilde{x})^\alpha}.$$

The phase uses the six-order paths; it equals 1 for an intracluster pair. The output-row norm groups pairs by the unit block containing $x$. Then

$$\|S_y(A)\|_\gamma \le C w_k(y), \qquad \|\delta_{\alpha,A} S_y(A)\|_\gamma \le C, \qquad w_k(y) = \begin{cases} 1, & k \ge 1, \\ (1+|y|)^{-1}, & k = 0. \end{cases}$$

At level zero its pointwise kernel satisfies

$$|S_y(A; x, x')| \le C(1+|y|)^{-1} e^{-c\, d(c,c')}.$$

Constants and $q_0$ are independent of mesh, side, mass $\mu$, twist and $y$.

Claim: the Maxwell mean field. For periodic photon cochains on $a\mathbb{Z}^3/\ell\mathbb{Z}^3$, use the pairing $a^3 \sum$ and mesh-normalized $d$. Let $\mathcal{Q}_k$ be the cochain average: its scalar average is $a^3 \sum_{c\in\Delta_z}$; on a bond component it is

$$(\mathcal{Q}_k \mathscr{A})_\mu(z) = a^3 \sum_{c\in\Delta_z} \sum_{j=0}^{a^{-1}-1} a \mathscr{A}_\mu(c + j a e_\mu).$$

Put $\delta = d^T$ and let $\mathcal{R}_k$ project orthogonally onto $(\delta d)(\ker \mathcal{Q}_k)$ in scalar cochains. The minimizer $H_k^{\mathrm{ph}} A$ of $\|d\mathscr{A}\|_2^2$, under $\mathcal{Q}_k \mathscr{A} = A$, $\mathcal{R}_k \delta \mathscr{A} = 0$, exists uniquely, is real linear and preserves constants. With $\Delta_k = (dH_k^{\mathrm{ph}})^T dH_k^{\mathrm{ph}}$,

$$|H_k^{\mathrm{ph}}(b,b')| + |\partial H_k^{\mathrm{ph}}(b,b')| + |\delta_\alpha \partial H_k^{\mathrm{ph}}((b,\widetilde{b});b')| \le C_\alpha e^{-\gamma_H d(b,b')},$$
$$|\Delta_k(b,b')| \le C_\Delta e^{-\gamma_H d(b,b')}, \qquad \|\Delta_k\|_{2\to 2} \le C_\Delta^{(2)}.$$

Here $\partial$ is a mesh-normalized output difference, and $\delta_\alpha v = (v(\widetilde{b}) - v(b))/|c-\widetilde{c}|_\infty^\alpha$ for parallel output bonds with $0 < |c-\widetilde{c}|_\infty \le 1$. These constants are independent of $k, \ell$.

Proof work. The chamber verification has three stages: the internal graph gap and cluster Poincaré inequality; the light–heavy Schur symbol, with a four-dimensional Dirac singularity and uniformly invertible 188-dimensional heavy block; and folded local inverses plus a convergent parametrix walk. The last stage uses local real gauges, rather than assuming the connection globally small. The photon proof uses explicit averaging symbols, alias sums and a uniform complex strip. The positive fluctuation square root is obtained subsequently from positive-matrix locality, after verifying the axial gap and finite-support coordinate basis.

Proof tools. The regulator; Fourier analysis of finite-dimensional symbols; local gauge covariance; a conditional parametrix-walk lemma. Block weights, six-path averages and the chamber gap are model verifications, not an unnamed collection of assumptions.

Source. Current manuscript, Sections 9–10, pp. 78–102: Propositions 9.1–9.7, the local inverse and walk proofs, and the Maxwell minimizer and Fourier-kernel proof. Exact anchors: TeX 6888–7261, 7270–7808, 7810–8220; the real-gauge domain is TeX 7586–7604.

## 11. A common tuned bare sequence and pressure comparison

### Q3. A common tuned bare sequence and pressure comparison

Model-specific. Overview Target.

Objects and hypotheses. Use the standard-weight chamber regulator. Fix $0 < \rho < 1$ and a positive physical reference mass $0 < \bar{m} \le 1$. There exist an odd integer $L \ge 13$, a power $M \ge L^2$ of $L$, $0 < \varepsilon \le 1/200$, and $0 < e_G \le 1/3$ for which the following construction is possible. These choices depend on $\rho$, and $e_G$ may also depend on $\bar{m}$.

For $0 < e \le e_G$, define the physical scales

$$h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \qquad v_K = -\log(e\sqrt{h}), \qquad r_K = \min\{L^i : i \in \mathbb{Z}_{\ge 0},\ L^i \ge v_K^2\},$$
$$R_{\mathrm{top}} = hMr_K, \qquad R^* = R_{\mathrm{top}}/L, \qquad \eta = L^{-N}, \qquad K = N + \log_L h,$$
$$a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad \mu_j = \bar{m}a_j, \quad r_j = \min\{L^i : i \ge 0,\ L^i \ge (-\log e_j)^2\}.$$

The family on which the common construction is performed is

$$\mathcal{T}(e) = \{hL^t : t \in \mathbb{Z}_{\ge 0},\ hL^t \le R_{\mathrm{top}}\}.$$

For $R$ in this family, let $k_*(R)$ be the first level $j \ge 1$ for which $R/a_j = Mr_j$, once the fine mesh is sufficiently small. Write $a_* = a_{k_*}$, $e_* = e_{k_*}$ and $\ell_* = R/a_*$. The construction proves the existence of this stop and the mesh threshold $N_{\min}(R, e)$; they are conclusions, not assumptions of a renormalization framework.

A directly measurable output. For a unit torus of side $n = R/\eta$, let $\mathcal{F}_n = dC^1(C_n; \mathbb{R})$ be the exact-curvature space. Denote its centred Gaussian law with covariance the identity on $\mathcal{F}_n$ by $\gamma_n$. For $u \in \mathcal{F}_n$, let $A(u)$ be the unique mean-zero transverse bond field satisfying $dA(u) = u$. For $\theta \in [-\pi, \pi]^3$, set $H_{\theta,\mu} = \theta_\mu/(qn)$, where $q = e\sqrt{\eta}$. Define the unnormalized zero-flux matter factor

$$z_{r,n}(q, \boldsymbol{\mu}) = \int_{[-\pi,\pi]^3} \frac{d\theta}{(2\pi)^3} \int_{\mathcal{F}_n} \det(D_q[A(u) + H_\theta] + \boldsymbol{\mu})\, d\gamma_n(u).$$

The subscript $r$ denotes the fixed regulator. This definition does not divide by a free determinant. Its free comparison is

$$z_{r,n}^{\mathrm{free}} = \int_{[-\pi,\pi]^3} \det(D_q[H_\theta] + \bar{m}\eta) \frac{d\theta}{(2\pi)^3}.$$

Claim. The construction selects a particular real pair

$$P_N(e) = (m_{0,N}(e), \varepsilon_{0,N}^{\mathrm{v}}(e))$$

on $R_{\mathrm{top}}(e)$, independently of twists and external sources. Use this very pair, without retuning, on every $R \in \mathcal{T}(e)$, with the same total lattice mass

$$\boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e), \qquad |m_{0,N}(e)| \le \tfrac{1}{8}(e\sqrt{\eta})^{3/4-8\varepsilon} < \tfrac{1}{8}.$$

For the finitely many meshes below the top construction's threshold one may set $m_{0,N} = 0$; the assertions about pressure concern $N \ge N_{\min}(R, e)$.

There are $C_6, \kappa_{\mathrm{strip}} > 0$, depending only on $\rho$, such that

$$\varepsilon_5(x) = C_6 \log \frac{1}{1 - \exp(-\kappa_{\mathrm{strip}} x)}, \qquad x > 0,$$

has the following property. Let $\mathcal{D}_m(p)$ be the 192-dimensional Fourier symbol of $D_0 + m$, and put

$$p_{\mathrm{ch}}^0(m) = \int_{[-\pi,\pi]^3} \log\det \mathcal{D}_m(p) \frac{dp}{(2\pi)^3}, \qquad p_F = p_{\mathrm{ch}}^0(\bar{m}\eta) + \varepsilon_{0,N}^{\mathrm{v}}(e).$$

For every $R \in \mathcal{T}(e)$ and $N \geq N_{\min}(R, e)$,

$$|\log z_{r,n} - \log z_{r,n}^{\text{free}} - \varepsilon_{0,N}^{\text{v}}(e) n^3| \leq 3e_*^{1/4-9\varepsilon},$$
$$|\log z_{r,n} - p_F n^3| \leq \varepsilon_5(\bar{m}R) + 3e_*^{1/4-9\varepsilon}.$$

In particular, for $n_* = R^*/\eta$ and $N \geq N_{\text{ref}}(e) := N_{\min}(R^*, e)$,

$$\left|L^3 \log z_{r,n_*} - \log z_{r,Ln_*}\right| \leq C_{\text{mat}}, \qquad C_{\text{mat}} = (L^3+1)(\varepsilon_5(1/L)+1).$$

Both factors here use exactly $\boldsymbol{\mu}_N$ and $q$. The bound is independent of $e, N$. On the reference torus

$$a_* \in \{h/L, h/L^2\}, \quad r_* \in \{r_K, Lr_K\}, \quad L^{-3} < \bar{m}a_* \leq L^{-1}.$$

The terminal physical scales are independent of $N$ above threshold. The total lattice mass can have either sign. The construction makes the final mass shift and vacuum term zero on the top torus; it does not impose zero terminal shifts separately on every smaller torus.

Proof work. Construct the exact block integration with the $1/48$ incoming chamber measure at the first step. Its local activity must be produced by the convergent fluctuation logarithm, not by a formal expansion. Prove extraction, shortest-chain field adjustment, scaling and reblocking estimates in the actual coefficient norms, retaining the external connection as an independent variable. Prove that odd-cover projection commutes with the local step: the full signed spinor kernel, rather than its sign alone, must be independent of a representative of the cover.

The chosen pair $P_N(e)$ is the pair constructed by the mixed-boundary fixed point on $R_{\text{top}}$: zero initial activity, zero terminal mass shift and zero terminal vacuum term. This identifies the sequence used in the next card; the theorem does not assert that every sequence satisfying the displayed size bound has these properties. Transfer the whole trajectory to smaller members of $\mathcal{T}(e)$ by the cover comparison. Iterate the exact photon change of variables, including its Jacobian and orthonormal curvature rows, to identify the terminal density for bounded Borel coarse observables. Estimate its complement and twist average; compare the free determinant to its pressure. This last step yields the displayed finite-volume estimates.

Proof tools. The regulator and the uniform kernel theorem, finite weighted Berezin completion, positive inverse-root locality, connected polymer logarithms, and cubic polymer geometry. All field-domain inclusions, activity norms and numerical choices are internal proof tasks here. Their verification is essential, but importing an unnamed collection of renormalization assumptions would defeat this statement.

Source. Current manuscript, Sections 11–16 and 20–22: pp. 103–178 and 197–227. The selected pair is TeX 16791–16888; the exact common-pressure conclusion is Theorem 22.8, TeX 21877–21925.

Scale and stop arithmetic: TeX 16254–16466. The iterated Borel density identity is Theorem 22.2, TeX 20868–20935; its tower is proved in Section 22.1 rather than inferred from one step. Ordered choices and positive ceilings are proved in Section 24, pp. 244–247.

## 12. Normalized products for the same tuned bare sequence

### Q4. Normalized products for the same tuned bare sequence

Model-specific. Overview Target.

Objects and hypotheses. Fix the regulator, $\rho, \bar{m}, L, M, \varepsilon, e_G$ and the particular bare sequence $P_N(e)$ selected by the common construction theorem. Thus the mass used below is exactly $\boldsymbol{\mu}_N = \bar{m}L^{-N} + m_{0,N}(e)$ chosen on $R_{\mathrm{top}}(e)$; it is not a new choice depending on the volume or on the probe.

Let $\beta \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ be real, put $a = d^*\beta$, $f = da$, and

$$b(f) = -6\,\mathrm{Tr}\big[(\bar{m} + \gamma\cdot\partial)^{-1} i\gamma\cdot a\big]^4.$$

This is a real trace of a fourth power of a Schatten-four operator. Positivity of $b(f)$ is not needed for the product estimate. It is the additional probe hypothesis needed later to conclude non-Gaussianity.

For $\eta = L^{-N}$, use the centred difference $\delta_{\eta,\mu}u(x) = (u(x+\eta e_\mu/2) - u(x-\eta e_\mu/2))/\eta$, and set

$$a'_{\eta,\nu} = -\sum_{\mu\neq\nu} \delta_{\eta,\mu}\beta_{\mu\nu}, \qquad f^\eta_{\mu\nu} = \delta_{\eta,\mu}a'_{\eta,\nu} - \delta_{\eta,\nu}a'_{\eta,\mu}.$$

Define the bond cochain

$$\widehat{a}_\eta(c, c+e_\nu) = \eta^{1/2}(J_\eta a'_{\eta,\nu})(\eta c + \eta e_\nu/2).$$

When used on a torus, periodize this covering cochain by summing its period translates. Then $X_\eta = F^\sharp_\eta(f^\eta) = \langle d\widehat{a}_\eta, F\rangle$, and put $v_\eta = \|d\widehat{a}_\eta\|^2_{\ell^2(\mathbb{Z}^3)}$.

Reference source and normalization. Let $R^* = hMr_K/L$, $n_* = R^*/\eta$, $\ell_c = R^*/(2L)$ and $N_* = 8L^3$. Choose a translate by a vector in $\eta\mathbb{Z}^3$ placing the buffered support of $\beta$ strictly inside the first open cell of side $\ell_c$. Such a placement is possible when $D_\beta + 4 + 3\eta \le \ell_c$, where $D_\beta$ is a support-cube side. Reflect this source into the $(2L)^3$ cells of the reference torus; write its cochains as $\widehat{a}_c$. Their curvatures satisfy $\langle d\widehat{a}_c, d\widehat{a}_{c'}\rangle = v_\eta \mathbf{1}_{\{c=c'\}}$.

For $t = (t_c)$, define

$$A_t = \frac{1}{e}\sum_c t_c\widehat{a}_c, \qquad \Phi_\eta(t) = \frac{\displaystyle\int \frac{d\theta}{(2\pi)^3}\int \det(D_q[A(u) + H_\theta + A_t] + \boldsymbol{\mu}_N)\, d\gamma_{n_*}(u)}{\displaystyle\int \frac{d\theta}{(2\pi)^3}\int \det(D_q[A(u) + H_\theta] + \boldsymbol{\mu}_N)\, d\gamma_{n_*}(u)}.$$

The exact-curvature Gaussian, transverse lift and harmonic twist are defined in the preceding construction card. The denominator is positive and all normalizing factors are included. The free one-cell ratio is

$$N_{0,\eta}(t) = \frac{\det(D_q[(t/e)\widehat{a}_\eta] + \bar{m}\eta)}{\det(D_q[0] + \bar{m}\eta)} \quad \text{on the antiperiodic torus } C(n_*), \qquad c_m = N^{(m)}_{0,\eta}(0), \quad m = 2, 4.$$

Its source phase is $\exp(it\eta J_\eta a'_\eta)$, so $c_m$ is independent of $e$ apart from the reference side $R^*(e)$.

Claim. There are $e_\sharp(\beta) > 0$, $C < \infty$, a mesh threshold $N_\beta(e) < \infty$, and $\sigma > 0$ such that for $0 < e \le \min(e_G, e_\sharp(\beta))$, $N \ge N_\beta(e)$, the normalized reference products obey

$$\Theta_m = \left|\left[\prod_{c=1}^{N_*}(\partial^m_{t_c} - c_m)\right]\Phi_\eta(0)\right|^{1/N_*},$$

$$\Theta_4 \le Ce^{\mathfrak{e}}, \qquad (1+|c_2|)\Theta_2 \le Ce^{\mathfrak{e}}, \qquad \mathfrak{e} = \tfrac{1}{2}(1/4 - 9\varepsilon) - 12\varepsilon,$$

$$\sigma = \mathfrak{e} - \frac{2(L^3-1)}{8L^3(L-1)^3} > 0, \qquad |c_2| + |c_4|^{1/2} \le C,$$

$$|c_4 - 3c_2^2 - b(f)| \le C(e + \eta^{1/4}).$$

Constants are independent of mesh and placement, and $\|f^\eta - f\|_{H^s} \le \eta^2\|\beta\|_{H^{s+4}}$, $v_\eta \to \|f\|_2^2$. The threshold includes $N_{\mathrm{ref}}(e)$ and the support-placement condition.

Write $\lambda(e) = R^*(e)/L$, $H_2(x;v) = x^2 - v$, and $H_4(x;v) = x^4 - 6vx^2 + 3v^2$. On every all-flux cutoff torus of physical side

$$R = 8hL^T, \qquad T \geq T^* + 1, \qquad R^* = hL^{T^*},$$

using that same $\boldsymbol{\mu}_N$, there are $u_2, u_4 \geq 0$, independent of $T$, with

$$u_4 \leq Ce^\sigma, \qquad (1 + |c_2|)u_2 \leq Ce^\sigma,$$

such that every finite set $Y \subset \lambda(e)\mathbb{Z}^3$, distinct modulo $R\mathbb{Z}^3$, satisfies

$$\left| \mathbb{E}^{\mathrm{af}}_{N,T} \prod_{y \in Y} \left[ e^{-m} H_m(F^\sharp_\eta(\tau_y f^\eta); v_\eta) - c_m \right] \right| \leq u_m^{|Y|}, \qquad m = 2, 4.$$

The notation $e^{-m}$ is an inverse power of the charge. In particular, uniformly in these tori,

$$\left| \kappa_4(F^\sharp_\eta(f)) - e^4 b(f) \right| \leq Ce^{4+\sigma} + C(\eta^2 + e^4\eta^{1/4}).$$

The result gives both the estimate on a single cumulant and the stronger distinct-translate product estimate. The latter is needed to pass the conclusion to almost every ergodic component of a limiting law.

Proof work. Differentiate the exact terminal determinant telescope with external sources. Remove the local constants and separate the one-cell free factor from connected multi-cell errors. After projection to fourth jets, write the normalized function as a product of free jets times error-set factors integrated against a positive small-field measure of mass at most one, plus a bounded complement. The finite algebraic covering estimate B6 then annihilates every error family that fails to cover a cell. Its smallness bounds must be verified for the terminal activity produced by the selected $P_N(e)$.

For the passage to all fluxes and arbitrary volume, combine the common matter defect with the photon determinant and flux-tail bounds. In particular

$$P_n(w) \leq \exp\left( -\frac{2\pi^2 |w|^2}{q^2 n} \right), \qquad q^2 n_* = e^2 R^*.$$

The rectangular chessboard transfer loses the explicitly displayed power $2(L^3 - 1)/(8L^3(L - 1)^3)$ of $e$. Finally compare the free fourth response with the continuum Schatten-four trace and replace $f^\eta$ by $f$ using Gaussian domination. A pressure bound alone does not imply these normalized-product estimates.

Proof tools. The regulator; the specific bare sequence constructed in the common tuning theorem; its two-reference-volume matter bound; all-flux Gaussian domination and rectangular chessboard positivity; the free coefficient and normalized finite-vector response theorems. No conclusion is claimed for arbitrary bare sequences merely satisfying a size bound.

Source. Current manuscript, Sections 18–19 and 23–25, pp. 181–196 and 228–255. The normalized reference response is `thm:interacting-reference-error`, TeX 23465–23556; positive probe-dependent ceilings are verified in Sections 23.8 and 24. The all-volume output is `prop:transfer-UV-inputs`, TeX 24814–24969. Reference product transfer and its exact exponent: TeX 24563–24813. Free ratio and coefficients: TeX 17327–17348, 17614–17634, 17900–17905. The source and smooth-test identities are TeX 18345–18598.

13. Application: construct the continuum nine-mirror law

## A1. Application: construct the continuum nine-mirror law

Model-specific. Overview Target.

Statement. Fix the chamber cutoff D0, a physical charge $e > 0$, a reference mass $0 < \bar{m} \leq 1$ and real numbers $m_{0,N}(e)$ independent of the outer volume $T$. Use the same numbers at every outer volume. Suppose the finite laws have the counting Gaussian bound and every fixed-spacing volume-limit plaquette law is invariant and reflection positive in the three coordinate and six diagonal mirror directions. Then volume-first limits of the physical step fields exist in $H_{-4}$, the negative oscillator Sobolev space defined in B3. Every such limit is a Borel law $\mu$ of real tempered two-forms with

$$dF = 0 \quad \mu\text{-a.s.}, \qquad \mathbb{E}_\mu e^{F(f)} \leq e^{\|f\|_2^2/2} \quad (f \text{ real Schwartz}).$$

It is invariant under all translations and signed coordinate permutations, and is invariant and reflection positive at every height in the nine mirror families. Its reflection form is the character form, with the exterior-square action defined in B1. Every chosen volume-first law can be realized by a joint cutoff sequence along which all fixed finite-family moments and cumulants converge.

Inputs in the QED3 application. Q1 supplies the finite Gaussian bound and the same-law lattice RP. B3 supplies compactness and identifies each fixed-spacing field limit as the step image of a lattice limit. The chosen sequence from Q3 ensures that this is the same family used for the response estimates. The limit passage itself does not require a renormalization estimate.

Proof outline. Use discrete Bianchi and cell averaging to pass closedness. Approximate translations by mesh translations. For a mirror height in $\mathbb{Z}[L^{-1}]$, a compact test with positive support margin induces a plaquette test whose entire plaquettes lie on the permitted lattice side once the margin exceeds $2\eta$. Apply RP to the recovered lattice law before sending the mesh to zero. Translation invariance then moves the mirror to every real height. Extend by character density to positive-side $L^2$ observables.

Internal proof details. M10 and M22.

Source. Manuscript §6, pp. 53–57, Proposition 6.1.

### A2. Application: assemble the OS axioms

Model-specific. Overview Target.

Statement. Let $\mu$ be a translation-invariant law of real tempered two-forms on $\mathbb{R}^3$, with $\mathbb{E}_\mu e^{F(f)} \le e^{\|f\|_2^2/2}$ and character reflection positivity at the nine coordinate and diagonal planes. No closedness assumption is needed here. Its moment distributions have Gaussian regularity, permutation symmetry, fixed-plane tensor reflection positivity and Euclidean covariance away from the big diagonal $\Delta_n = \{x_i = x_j \text{ for some } i \neq j\}$. For almost every translation-ergodic component of the same parent law, these statements hold together with tensor clustering.

Assembly. Translation invariance and characteristic RP first give invariance under the nine reflections, by the reflection-invariance argument in B1. Apply B2 to inherit the Gaussian bound and the nine mirror forms and to obtain mixing on one full-measure component set. Apply B1 to the parent and to each of these components. The short kernel and supported-tensor argument B4 supplies the remaining OS statements. The exterior-square coefficient representation has planar weights of absolute value at most one, so it satisfies B1's representation hypothesis.

Optional full-Schwartz conclusion in QED3. The law from A1 is also closed. The separate QED3 theorem M17 therefore gives Euclidean covariance on every Schwartz test, invariance of the law, and all-plane RP. The quantitative three-dimensional estimate and the tensor argument M18 then give E0′–E3 for the parent and E0′–E4 for the same components, exactly as in the canonical manuscript. This strengthening is not a hypothesis of B1.

Internal proof details. B4 is an assembly note rather than a principal theorem box. M14 and M22 give the component and limiting-law inputs. M17–M18 record the stronger QED3 specialization.

Source. General rotation manuscript, Theorem A; QED3 manuscript §7.5, Theorem 8.6 and Proposition 27.1.

## A3. Application: a positive fourth cumulant in those same components

Model-specific. Overview Target.

Statement. Fix $0 < \rho < 1$, $0 < \bar{m} \le 1$ and the common tuned bare mass sequence constructed in Q3. Choose a real compact smooth two-form $\beta$ with $f = dd^*\beta$ and

$$b(f) = -6\,\mathrm{tr}[(S_{\bar{m}} i\gamma \cdot d^*\beta)^4] > 0, \qquad S_{\bar{m}} = (\bar{m} + \gamma \cdot \partial)^{-1}.$$

The continuum probe theorem B7 supplies such a choice. There are $C_f < \infty$, $\sigma > 0$ and a positive admitted charge ceiling $e_f$, independent of the parent limit and its component, such that for every $0 < e \le e_f$ and every volume-first limit $\mu$ of the corresponding chamber laws D0, on one full-measure set of translation-ergodic components of this *same* parent law,

$$|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \le C_f e^{4+\sigma}.$$

After reducing that ceiling so $C_f e^\sigma \le b(f)/2$, this gives $\kappa_4^{\mu_\xi}(f) \ge e^4 b(f)/2 > 0$. The component set may be intersected with the OS set from A2.

Inputs. Q4 supplies estimates for every finite set of distinct translated cell probes, not only a parent fourth moment. B3 supplies moment convergence; B2 supplies conditional centering and the Gaussian bound on the same components. B5 is the general conditional-cumulant criterion. B7 supplies the sign of the continuum coefficient.

Proof outline. Along a joint sequence realizing the fixed parent law, take a common subsequence of the bounded reference coefficients. Replace the cutoff probes by their common $L^2$ limit in every fixed translated product, using Gaussian moments. Apply B5 with $g = e$, $X = F(f)$ and $v = \|f\|_2^2$. The inherited Gaussian inequality gives $\mathbb{E}[X \mid \mathcal{I}] = 0$. Use the free coefficient comparison and the bounds on both Wick degrees to obtain the stated error. Make the final probe-dependent charge restriction only after its uniform error constant has been fixed.

Internal proof details. M19, T02–T04 and T17–T18.

Source. Manuscript §§25–26, pp. 247–257.

## Appendix A. Detailed proof tasks and supporting statements

The following pages expand the principal results. Each states its own inputs and gives the source location of the argument.

### B4. Tensor tests and the OS assembly

Interface. Overview Target.

Kernel extension. Let $E$ be a finite-dimensional real orthogonal representation of $O(d)$, and let $\lambda$ be a law on $\mathcal{S}'(\mathbb{R}^d; E)$ satisfying $\mathbb{E}e^{F(f)} \le e^{C\|f\|_2^2}$ for real tests. Gaussian tails and Hölder give

$$|\mathbb{E}\prod_{j=1}^{n} F(f_j)| \le (K_C\sqrt{n})^n \prod_j \|f_j\|_2.$$

The Schwartz kernel theorem therefore gives a unique tempered tensor distribution $S_n$, with $S_0 = 1$. Reality and permutation symmetry follow from the random-field products. This is the routine kernel extension used in the assembly.

Reflection positivity. Fix $r(x) = (-x_1, x_2, \dots, x_d)$, acting on coefficients by the given representation. Assume translation invariance and character reflection positivity at $x_1 = 0$. The reflection-invariance argument in the rotation paper then makes reflection a unitary operator on $L^2(\lambda)$. The Gaussian moment estimate extends random tensor evaluations $Q_\Psi$ continuously to Schwartz tensor tests with values in $L^2(\lambda)$. A compact Fourier tensor expansion preserves support inside the positive open half-space. Growing cutoffs and a small inward translation handle the closed half-space. Thus the character reflection form applies to $\sum_{k=0}^{M} Q_{\Psi_k}$ and gives

$$\sum_{k,l=0}^{M} S_{k+l}(\overline{r\cdot\Psi_k}\otimes\Psi_l) \ge 0, \qquad \operatorname{supp}\Psi_k \subset \{x_1 \ge 0\}^k.$$

The degree-zero entry is a scalar and is conjugated in the first block.

Covariance and clustering. B1 supplies covariance on the complement of the big diagonal $\Delta_n = \{x_i = x_j \text{ for some } i \ne j\}$. This is the covariance domain in the general OS application. If, in addition, ordinary $L^2$ correlations mix, apply mixing to $Q_\Psi$ and $Q_\Phi$ to obtain the tensor cluster identity

$$S_{k+l}(\Psi \otimes \tau_a^{\otimes l}\Phi) \longrightarrow S_k(\Psi)S_l(\Phi) \quad (|a| \to \infty).$$

The primary OS assembly uses one fixed reflection plane and off-diagonal covariance. It does not infer invariance of the law or characteristic-functional RP at every plane.

QED3 strengthening. The separate closed-two-form rotation result M17 supplies full moment covariance and all-plane law-level RP in QED3. Together with the quantitative Hermite estimate M09, the same assembly then gives the stronger full-Schwartz version of the manuscript's first-page theorem. Ordinary closedness is used only in that additional QED3 argument.

Source. Schwartz kernel theorem; QED3 manuscript §5.3 and §27, especially TeX 25188–25325.

These tensor and support arguments are componentwise; their qualitative conclusion applies to finite $E$ and arbitrary fixed $d$. The numerical constants of the three-dimensional two-form estimate are not generalized.

### Z01. Assemble the field-strength existence theorem

MODEL-SPECIFIC. Overview Target.

Objects and hypotheses. Fix $0 < \rho < 1$ and $0 < \bar{m} \le 1$, standard chamber weights, and the constants and charge intervals admitted by T16. The tuned volume-independent bare mass is $m_{0,N}(e)$ from T01. On cubic tori take periodic gauge fields and fermions antiperiodic in all three coordinate directions, with

$$\eta = L^{-N}, \quad R = 8hL^T, \quad h = L^{\lfloor \log_L \bar{m}^{-1} \rfloor}, \quad q = e\sqrt{\eta}, \quad \boldsymbol{\mu}_N = \bar{m}\eta + m_{0,N}(e).$$

Let real compact smooth $\beta$ satisfy $f = dd^*\beta$ and $b(f) = -6\,\mathrm{tr}[(S_{\bar{m}} i\gamma \cdot d^*\beta)^4] > 0$, where $S_{\bar{m}} = (\bar{m} + \gamma \cdot \partial)^{-1}$.

Claim. For every admitted sufficiently small fixed $e > 0$ volume-first limits exist. Every such parent law has closed tempered support, Gaussian domination and tensor OS conditions E0′–E3. In almost every translation-ergodic component of this *same* law, E0′–E4 hold on all tensor Schwartz tests, and

$$|\kappa_4(f) - e^4 b(f)| \le C_f e^{4+\sigma}, \qquad \kappa_4(f) \ge \tfrac{1}{2} e^4 b(f) > 0.$$

The constants and positive charge ceiling can depend on the fixed probe as stated; the error is uniform in parent limit and component.

Proof work. Instantiate the conditional finite estimates only after the compatible constant choices. Use M22 for the parent law and M14 for one full-measure component set. Apply M17 and M13 to each of those components, then M18 for the full tensor statement. Along a joint sequence realizing the chosen volume-first law, extract one common subsequence of the four bounded reference coefficients. Pass all fixed finite translated products to that law, use M19 and T18, and intersect its full-measure set with the OS set. T02 supplies a positive probe. Choose the probe ceiling so its positive leading coefficient dominates the uniform error.

Inputs. M01, M13, M14, M17–M19, M22 and T01–T02, T16–T18. Earlier nodes state the exact primitive and reference inputs behind these final conclusions. No uniqueness, full cutoff convergence, charged-field construction or pole-mass identification is part of this target.

Source. Manuscript Theorem 1.1, pp. 1–2, and its complete assembly on pp. 259–261.

**M01. The chamber law is defined at every real bare mass**

MODEL-SPECIFIC. Overview Target.

Objects and hypotheses. Let $\Gamma \subset \mathbb{Z}^3$ have rank three and $\min_{0\neq\gamma\in\Gamma} |\gamma|_\infty \geq 4$, $q > 0$, $0 < \rho < 1$, and $w_i > 0$. Set $C_0 = \{0 < x_3 < x_2 < x_1 < 1/2\}$ and choose $v_0 \in C_0$. The four walls have inward normals $e_3, (e_2 - e_3)/\sqrt{2}, (e_1 - e_2)/\sqrt{2}, -e_1$ and reflections $s_i$. For $W = \mathbb{Z}^3 \rtimes O_{\mathrm{cub}}$, the fermion sites are $gv_0$, their four neighbours are $gs_iv_0$, and $\pi(k + rv_0) = k$. Use the Hermitian Clifford triple $\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu)$. For a type-$i$ bond $xx'$ let

$$(\mathcal{D}_A\psi)(x) = M\psi(x) - \sum_{i=0}^{3} \frac{w_i}{2}(\rho - \gamma_{(x'-x)/|x'-x|})e^{iqA(\pi x,\pi x')}\psi(x'), \qquad M \in \mathbb{R}.$$

Spinors obey $\psi(x + g) = \chi(g)\psi(x)$ for a group homomorphism $\chi : \Gamma \to \{1, -1\}$. Set $A(x, x) = 0$ when a chamber hop has equal photon projections. Photon configurations are $(A, z)$, with real link cochain $A$ and closed integer plaquette cochain $z$, modulo $(A, z) \sim (A + 2\pi k/q, z - dk)$. Their curvature is $u = dA + 2\pi z/q$. Give them Maxwell weight $e^{-\|u\|^2/2}$, counting the integer cochains and integrating link phases over one fundamental cell. The fermion integral is normalized to give $\det \mathcal{D}_A$.

Claim. This defines a finite measure with strictly positive total mass for every real $M$; its normalized curvature law is supported on $du = 0$. Prove determinant nonnegativity and strict normalization separately. Prove the lattice, gauge and charge-conjugation identities with the two-form orientation signs, preserving or transporting the character as appropriate. The special weights for the continuum construction are $(6, 6\sqrt{2}, 6\sqrt{2}, 6)$.

Proof work. Verify simple transitivity of the chamber action and the four-neighbour graph, including the eight bonds above a photon link. Use the paired spin blocks for determinant nonnegativity. Exhibit a nonzero extreme Laurent coefficient independently of $M$, so the determinant does not vanish identically. Integrability of the all-flux Maxwell sum and this strictness establish normalization even when the total lattice mass $M - \rho\sum_i w_i/2$ is negative.

Inputs. Finite cochain algebra, Clifford matrices and finite Berezin integration. The geometry and operator are defined here; no continuum law or renormalization estimate is used.

Source. Manuscript pp. 5–11, Propositions 2.1–2.3.

**M02. Reflection positivity on an adapted finite torus**

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M01

Objects and hypotheses. Use the finite functional $Z[G]$ specified in M01. Fix one mirror $\ell(x) = c$, where $\ell = x_j$ or $x_j \pm x_k$, with $c \in \frac{1}{2}\mathbb{Z}$ for coordinate mirrors and $c \in \mathbb{Z}$ for diagonal mirrors. Let the torus be adapted to this mirror: a fundamental slab has height period $p$, the second reflecting wall is $\ell = c + p/2$, and the fermion character on the normal closing vector is $-1$. The period group splits as $\Gamma = \Gamma_\parallel \oplus \mathbb{Z}v$, where $\ell = \langle n_\ell, \cdot \rangle$, $v = pn_\ell/|n_\ell|^2 \in \mathbb{Z}^3$ and $\ell(\Gamma_\parallel) = 0$. A positive integrand is local and even in Grassmann generators and has support in $c < \ell < c + p/2$. Reflection $\Theta$ includes conjugation and reversal of Grassmann order. Bounded scalar curvature functions are included. With upward unit normal $n$ its generator action is $\Theta\psi_\alpha(x) = -\sum_\beta \bar{\psi}_\beta(\theta x)(\gamma_n)_{\beta\alpha}$ and $\Theta\bar{\psi}_\alpha(x) = -\sum_\beta (\gamma_n)_{\alpha\beta}\psi_\beta(\theta x)$; scalar coefficients undergo tensor pullback and complex conjugation. For an insertion lifted to a box $B$, require for each Grassmann coefficient

$$|a_\Psi(\widehat{A})| \le C \exp\{(1/2 - \varepsilon) \sum_{\sigma \subset B} (d\widehat{A})_\sigma^2\}, \qquad C < \infty, \quad 0 < \varepsilon \le 1/2.$$

Here the local lift has $d\widehat{A} = u$ and the same link phases; the coefficient transforms by the inverse charge of its monomial, so its evaluated insertion is gauge invariant.

Claim. Construct the reflected Gram representation

$$Z[(\Theta G')G] = \langle \mathcal{S}G', \mathcal{S}G \rangle, \qquad Z[(\Theta G)G] \ge 0,$$

with its actual positive boundary measure and finite cut-mode sum. Reflection covariance also holds. The support class is the open *torus half*, not the infinite positive half-space of a periodic extension.

Proof work. Diagonalize each crossing hopping into coefficients $w_i(1 \pm \rho)/2 > 0$ and expand its finite Grassmann subsets. Keep the Berezin signs at both walls and the antiperiodic closing character. Split the Maxwell incidence constraints at the cut, impose closedness by Fourier multipliers, and perform the Poisson sum and cut-link gauge average. Show positivity of the resulting Gauss multiplier. Complete arbitrary upper data into the actual half-slab before using the Gram identity; prove absolute convergence coefficient by coefficient for the stated growth class.

Inputs. M01 supplies the explicit graph and finite functional. The proof must verify the crossing and boundary incidence identities for that graph; these are not inferred from normalization.

Source. Manuscript pp. 11–16, Proposition 2.4 and Theorem 2.5.

## M03. Local slab kernels, adjoints and closing traces

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M01, M02

Objects and hypotheses. On a cylinder with a finite transverse period cell fix a mirror $\ell = c$. Let $\mathcal{M}^j(c)$ be its contained $j$-cells and $\mathcal{X}^j(c)$ the cells cut in their relative interiors. The coboundary restricted to cut faces is $d_c$; its upper-face part is $d_c^+$. Set $r_q = 2\pi/q$. A boundary variable is $\xi = (A_0, z_0, \vartheta, k, I)$: mirror links $A_0 \in [0, r_q)$, mirror integer plaquettes $z_0$, one angle $\vartheta_Q \in [0, 2\pi)$ per cut cube, one integer $k_\sigma$ per cut plaquette, and a subset $I$ of the finite ordered crossing modes. Their monomials $\zeta_I$ and positive weights $\omega_I$ are supplied by the crossing expansion in M02. With $\mathfrak{q}(I)$ the signed mode charge on cut links, put

$$\kappa_\sigma = r_q^{-1}(d_c^*\vartheta - 2\pi k)_\sigma, \quad D_c(k, I) = r_q^{|\mathcal{X}^1(c)|}\mathbf{1}_{d_c^* k + \mathfrak{q}(I) = 0},$$

$$d\nu_c(\xi) = dA_0 \prod_{\sigma \in \mathcal{M}^2(c)} e^{-u_\sigma^2/2} \prod_{Q \in \mathcal{X}^3(c)} \frac{d\vartheta_Q}{2\pi} \Big( \prod_{\sigma \in \mathcal{X}^2(c)} \frac{q}{\sqrt{2\pi}} e^{-\kappa_\sigma^2/2} \Big) D_c(k, I).$$

Integer and subset indices carry counting measure; set $\mathcal{H}(c) = L^2(Y_c, \nu_c)$. A closed-slab integrand has even fermion support strictly between its two walls, is invariant under $(A, z) \mapsto (A + r_q k, z - dk)$, and its coefficients satisfy $|a_\Psi| \le C \exp\{(1/2 - \varepsilon) \sum_{\text{interior plaquettes}} u_\sigma^2\}$, $\varepsilon > 0$. Local products use disjoint lifted growth boxes and their period translates.

Claim. Construct scalar kernels between these boundary spaces as follows. Integrate interior Maxwell weights and closedness constraints, and integrate $\zeta_I G e^{-S} \Theta \zeta_{I'}$ over interior fermions. Multiply by $\sqrt{\omega_I \omega_{I'}}$ and by the upper/lower boundary phases $e_c = \exp\{i\langle\vartheta, d_c^+ z\rangle - i\langle\kappa, d_c^+ A\rangle\}$ and their reflected conjugates. The positive mode weights belong to this kernel, not to $\nu_c$. Prove $\nu_c$ is finite and the resulting operators $\mathcal{S}[G] : \mathcal{H}(c) \to \mathcal{H}(c')$ are Hilbert–Schmidt. Gluing slabs gives operator composition; reflection gives the adjoint; a compatible lattice translation gives the corresponding unitary conjugation. For a minimal positive closing-height vector $g$, at least two consecutive slabs, and admissible insertions $G_1, \dots, G_r$, prove the closing trace

$$Z_T[G_1 \cdots G_r] = \operatorname{Tr}_{\mathcal{H}(c_0)}(V_g^{-1} U_f^{(1+\chi(g))/2} \mathcal{S}[G_r] \cdots \mathcal{S}[G_1]), \qquad r \ge 2.$$

Here $V_g$ is translation of boundary data and $U_f$ multiplies a mode subset by its fermion parity. Its factor disappears for antiperiodic closure.

Proof work. Prove the common lift is representation independent. The Maxwell majorant must be integrable on both boundary copies; it proves the Hilbert–Schmidt assertion before any trace inequality is used. Match each Grassmann top coefficient under gluing, compute the reflected order and closing parity, and justify the all-flux boundary direct sum. The abstract composition and adjoint calculus is reusable after these local identities have been supplied.

Inputs. M01–M02: finite functional and cut factorization. The three geometric inputs are lift injectivity, disjoint growth boxes, and compatible closing character. A decay estimate alone supplies none of them.

Source. Manuscript pp. 17–23, Theorem 3.1.

**M04. Chessboard comparison with an extensive pressure error**

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M01, M03

Objects and hypotheses. Let a finite torus functional have slab operators with the gluing, adjoint and closing-trace identities of M03. Fold a local insertion over a complete reflected tiling. Keep charge, mass and hopping weights fixed. Let $C(n)$ be a coordinate cube and $D(n)$ the equal- volume diagonal-adapted torus with periods $(n/2)(1,-1,0)$, $n(1,1,0)$ and $ne_3$, antiperiodic on these periods. Use eventual sides divisible by eight, with an unbounded set of eventual divisors. The cells have wall forms $x_1-x_2, x_1+x_2, x_3$, offsets in $\mathbb{Z},\mathbb{Z},\mathbb{Z}+1/2$, and even width $h$ with $4h$ an eventual divisor. A cell $Q$ is an intersection of three successive open height intervals. For a real local trigonometric polynomial $O$ supported strictly inside $Q$, let $N_n = n^3/|Q|$ and $V_n$ be its full reflected fold. Define $\mu^t_{D(n)}$ by the positive scalar tilt $e^{tV_n}$.

Claim. Prove the Schatten chessboard inequality and a pressure error $\Delta(n)\to 0$ with

$$|\log Z_{C(n)} - \log Z_{D(n)}| \le n^3\Delta(n), \qquad \mu_{C(n)}(O) \le \mu^t_{D(n)}(O) + |Q|\Delta(n)/t \quad (t>0).$$

The error is extensive before division by $tN_n$. It does not give a bounded ratio of partition functions or equality of local laws.

Proof work. A full fold is a trace of an even Schatten power. Apply trace Hölder successively in the three admissible frames, retaining closing vectors, characters and embedded-box conditions. Establish the pressure limit by the positive $Y^*Y$ representation and separate convexity in the three side lengths. For the local comparison use Jensen on the original source, the full-fold inequality, and convexity of $\log Z_D[e^{tV_n}]$. Volume must tend to infinity at fixed $t$.

Inputs. M03 and strict finite normalization from M01. The exact lattice straightening and admissible tilings are model verification within this task; Schatten Hölder is general.

Source. Manuscript pp. 23–26 and 30–31, Proposition 3.2, Theorem 3.3, Propositions 3.4 and 3.6.

### M05. All-flux Gaussian domination from a slice trace

Model-specific. Overview Target.

Proof dependencies. M01, M03

Objects and hypotheses. On a finite coordinate torus use the Maxwell curvature $u = dA + 2\pi z/q$ and normalized determinant-weighted all-flux law of M01. For a real plaquette cochain $\beta$, let $Z(\beta)$ be its unnormalized partition integral with Maxwell weight $\exp(-\|u-\beta\|^2/2)$ in place of $\exp(-\|u\|^2/2)$. All fluxes and winding classes are retained. Time may be any of the three coordinate directions.

Claim. For every real $\beta$,

$$0 < Z(\beta) \le Z(0), \qquad \mathbb{E}e^{\langle \beta, u\rangle} \le e^{\|\beta\|^2/2}.$$

The constant is one in the variance convention above. In particular there is a uniform Gaussian tail for each plaquette.

Proof work. Construct the coordinate-slice trace from M03, with magnetic flux, electric winding, compact gauge average and antiperiodic closure. First unfold the real quotient and prove that the full flux/winding slice operators are Hilbert–Schmidt after summation. This justifies the cyclic trace formula on its actual boundary Hilbert space. Remove electric unitary factors by trace Hölder in one time direction, leaving a repeated magnetic slice and zero electric shift. Choose a different coordinate as time; the remaining shift is then electric and can be removed. Completing the Maxwell square gives the normalized exponential bound. Its complex-source consequence comes by taking the real part of the source.

Inputs. M01 supplies the finite functional and M03 the closing trace identities. The source-dependent all-flux factorization is part of what this task proves. This task does not infer a zero-flux conditional bound from the all-flux inequality; that later bound needs its own slice proof.

Source. Manuscript pp. 26–30, Theorem 3.5.

### M06. Uniform insertion bounds in the tilted comparison laws

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M03, M04, M05

Objects and hypotheses. Let $G$ be an even gauge-invariant local integrand with the sub-Maxwell coefficient growth specified in M03. This includes the heat-regularized kernels below. Polynomial specializations have $G(\widehat{A}) = \sum_a b_a e^{i\langle \delta_a, \widehat{A}\rangle} \Psi_a$ with finite support and $d^*\delta_a = q\mathfrak{q}(\Psi_a)$, where $\mathfrak{q}$ counts $\psi$ minus $\bar{\psi}$ at each photon site. The support radius is fixed. Use the adapted tori $D_i$ in M04. An insertion at a cut is the boundary-mode matrix obtained by integrating this polynomial between the cut monomials of M03. Its local operator acts on the quotient photon variables and the finite crossing Fock space; the electric characters are retained in its representation. The real source $O = \sum_\beta a_\beta e^{i\langle \beta, u\rangle}$ has coefficient norm $|O|_1 = \sum_\beta |a_\beta|$ and tilt $t \geq 0$. For original diagonal cell width $h$ and support radius $r$, choose a coordinate width $h'' \in h\mathbb{Z}$ and an integer ratio $k \geq 1$ with $h'' \geq 2r + 6h$, $kh \geq 16h''$, $kh \in h''\mathbb{Z}$, and $4h'', 2kh$ eventual divisors of the volume sequence. The coarse diagonal cell width is $kh$, not $k$. Assume $G$ has a coordinate bound $C_G$: on every rectangular coordinate torus with sides at least four, for every nonempty disjointly embedded family of translated or signed-coordinate-reflected boxes $g_a B$ and every real plaquette cochain $\beta$,

$$\left| Z_T[e^{i\langle \beta, u\rangle} \prod_{a=1}^{r} G^{g_a}] \right| \leq C_G^r Z_T[1],$$

provided the following free-column condition holds. Every insertion box, together with its period translates, must embed disjointly. In each coordinate direction select a plaquette column such that no insertion box or period translate contains a plaquette of that column.

Claim. Under this coordinate-bound hypothesis prove the amplification estimate

$$|\mu_i^t(G)| \leq C_G \exp\{2t|O|_1 k^3 + |Q|k^3 \Delta_i\}.$$

Verify the hypothesis for heat-regularized bounded local Weyl operators $X$ inserted in the cut-mode kernel, with bound $\|X\|$ (or $2^{|S|/2}\|X\|$ for the variance-modified kernel on a link set $S$), and for bounded bosonic insertions with their supremum norm. Here the operator norm is the supremum over quotient realizations *and* electric characters. Arbitrary gauge-invariant polynomials are treated subsequently in M07. Its support-dependent constants are uniform in insertion position and volume at fixed physical lattice parameters. Prove separately the analogous unbounded local exponential estimate that gives tightness of the tilted laws.

Proof work. Put the local Weyl algebra in normal form and separate its translation characters by finite averaging. Prove its norm assertion by the measurable quotient section and direct-integral electric shifts. Use the free column in the coordinate trace. Apply the two chessboards to amplify a one-cell insertion into the coefficient bound. For exponentials use their positive bosonic value and a second chessboard plus M05; a bounded-operator estimate does not cover this unbounded case.

Inputs. M03–M05 and the explicit local insertion algebra. Nonempty disjoint families, enlarged lift boxes and the free column are part of the statement to verify.

Source. Manuscript pp. 33–40, Propositions 4.1–4.2 and Theorem 4.3.

### M07. Two layers and elimination of the outer fields

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M01, M06

Objects and hypotheses. Work with the same finite tilted functional and fixed local supports as M06. Fermion generators on a coordinate link plane have invertible matching hoppings, since $0 < \rho < 1$. Test polynomials form $\mathcal{P}$: $P(\widehat{A}) = \sum b_a e^{i\langle \delta_a, \widehat{A}\rangle} \Psi_a$ has finite support and $d^*\delta_a = q\mathfrak{q}(\Psi_a)$. At $c = j + 1/2$ let $\mathcal{P}_1(c)$ read only photon links in the two planes $j, j+1$ and between them, and fermions at the two ends of bonds crossing $c$: the lower plane's last thin layer and the upper plane's first thin layer. Keep the finite heat-regularization parameters explicit. The photon integrals are unfolded over all fluxes and the finite phase polynomials extend entirely to complex links; the local Maxwell majorants allow the stated imaginary contour shifts.

Claim. For each $P \in \mathcal{P}$ construct $Y_P \in \mathcal{P}_1(c)$, independent of volume and tilt, such that for each fixed lattice isometry $g$,

$$|\mu_i^t(P^g) - \mu_i^t((Y_P)^g)| \le C_{P,g}\, t, \qquad 0 \le t \le 1, \quad i \ge i_0(P, g).$$

For every $Y \in \mathcal{P}_1(c)$ prove $\lim_{t\downarrow 0} \limsup_{i\to\infty} |\mu_i^t(\Theta_c Y) - \overline{\mu_i^t(Y)}| = 0$. First fix the finite heat error, then take volume to infinity and tilt to zero, and only then remove the heat error. As an intermediate output, extend M06's insertion estimate to every fixed $P \in \mathcal{P}$, with a constant $C_P$ independent of volume and insertion position and the same exponential factor. This extension uses the thin-layer recursion, rather than assuming a coordinate bound for every polynomial at the outset.

Proof work. Use finite Berezin integration by parts to write the fermion equations of motion. Their nilpotent recursion moves the fermion fields to the two crossing layers, retaining the finite normal-form identity $Z_T[\Phi P] = \sum_a b_a Z_T[\Phi \varphi_{\beta_a} Y_a]$ for every bounded bosonic factor $\Phi$: decompose each exponent as $\delta = d^*\beta + qj$, with $j$ joining the charged endpoints in these layers, and $\varphi_\beta = e^{i\langle \beta, u\rangle}$. Represent the $Y_a$ by the heat-regularized insertions of M06; absorb the curvature characters in its coordinate bound and pass the bounds through the finite recursion. Obtain near-Hermiticity from the resulting unital almost-contractive functional, keeping the two heat regularizations distinct. Perform the all-flux contour shift orbit by orbit, keeping the endpoint Maxwell estimate and source error. The outer-plane exponents cancel, leaving a smaller fixed polynomial. Iterate only finitely many times, taking a common volume threshold for all support boxes. Transport identities explicitly; the diagonal torus is not assumed to have the missing coordinate symmetry.

Inputs. M01 gives the exact hopping and integration identities; M06 bounds the local operations. The contour and layer identities are separate model verifications, not consequences of RP.

Source. Manuscript pp. 40–47, Proposition 4.4 and Theorem 4.5, equations (4.37)–(4.52).

### M08. Diagonal positivity of the original volume-limit law

Model-specific. Overview Target.

Proof dependencies. M02, M04, M05, M06, M07

Objects and hypotheses. Let $\Omega$ be the closed subspace of real plaquette arrays on $\mathbb{Z}^3$ satisfying $du = 0$, with product topology. At fixed $q, \rho, w_i, M$ let cubic sides $n_i \to \infty$ have arbitrarily large eventual divisors and be eventually divisible by eight. Assume the fold bound of M04, insertion tightness of M06, outer-field identities of M07 and half-torus wall positivity of M02.

Claim. Every cubic weak limit on $\Omega$ is translation/cubic/sign invariant, Gaussian dominated, and reflection positive at $x_j = c$ for $c \in \frac{1}{2}\mathbb{Z}$ and $x_j \pm x_k = c$ for $c \in \mathbb{Z}$. For a local positive-side observable $A$, RP means $\nu((\Theta A)A) \geq 0$, with tensor pullback and conjugation. It is the original cubic law $\nu$ that has this property.

Proof work. For each real local trigonometric polynomial $O$, take subordinate adapted weak limits at fixed $t$, then let $t \downarrow 0$. Each fixed cylinder eventually fits inside the positive torus half before passing its reflected form to a limit. M07 gives coordinate-link reflections and even translations of the comparison state. These translations place any local $O$ in the permitted cell. Thus $\nu(O) \leq \nu_O(O)$ for some state $\nu_O$ in a compact convex class of diagonal-positive laws. Separate this class by local trigonometric polynomials to conclude $\nu$ belongs to it. The comparison state may depend on $O$ throughout.

Inputs. M02, M04, M06 and M07. M05 gives tightness and domination of the original laws. Fixing parameters before volume limits is essential; this statement contains no joint-cutoff estimate.

Source. Manuscript pp. 47–49, Lemma 4.6 and Theorem 4.7.

## M09. Gaussian domination, moments and analytic generating functions

INTERFACE. Overview Target.

Objects and hypotheses. On a probability space $(\Omega, \mathcal{F}, \lambda)$, let $X$ be a real linear random map on a dense subspace of a real Hilbert space $\mathcal{K}$, satisfying $\mathbb{E}e^{X(f)} \le e^{\|f\|_{\mathcal{K}}^2/2}$ for every real $f$ there. Write $F(f) = X(f)$ and $\|f\|_2 = \|f\|_{\mathcal{K}}$ below. Examples are finite lattice cochains with their counting norm and Schwartz two-forms with the componentwise $L^2$ norm. Extend evaluation complex linearly. In a field-law application the sigma-algebra is generated by these evaluations.

Claim. The map extends consistently to $\mathcal{K}$ with values in every finite $L^p(\lambda)$; it is centered, has variance at most $\|f\|_2^2$, and

$$\lambda(|F(f)| > u) \le 2e^{-u^2/(2\|f\|_2^2)}, \qquad \mathbb{E}\prod_{j=1}^{n} |F(f_j)| \le c_n \prod_j \|f_j\|_2, \quad c_n = 2\,2^{n/2}\Gamma(1+n/2)$$

for real tests. Complex tests cost at most $2^{n/2}$ more. Finite- family generating functions are entire. Their logarithms are uniformly analytic on $\sum_j |z_j|\|f_j\|_2 < 2/5$; moments determine the law.

Proof work. Apply the bound to $tf$ for both signs, differentiate at zero using the exponential majorant, optimize the exponential tail, and integrate it. Hölder gives products and continuity in every $L^p$. Prove a uniform zero-free disk by the centered exponential remainder, before using Cauchy's formula on logarithms. Characteristic functions of finite evaluations give determination of the whole evaluation law.

Inputs. Only the displayed Gaussian inequality and the specified Hilbert space. Closedness, reflection positivity, a lattice, and a mass parameter are absent from the statement.

Source. Manuscript pp. 49–51, Proposition 5.1 and §5.2.

**M10. Compactness and recovery of a fixed-spacing lattice law**

INTERFACE. Overview Target.

Proof dependencies. M05, M09

Objects and hypotheses. Let $0 < \eta \le 1$, let $n \ge 4$ be an integer, and put $R = \eta n \ge 1$. Let $u$ be a closed plaquette field on the cubic torus of side $n$, with $\mathbb{E}e^{\langle \beta, u\rangle} \le e^{\|\beta\|_{\ell^2}^2/2}$ on the torus. At spacing $\eta$ put $F^\sharp = \eta^{-3/2}u(\sigma)$ on the cube $Q_\eta(\sigma)$ of side $\eta$ centered at $\eta x(\sigma)$, where $x(\sigma)$ is the unit-lattice plaquette centre, separately in each orientation. Thus $f_{[\eta]}(\sigma) = \eta^{-3/2}\int_{Q_\eta(\sigma)} f$ and $F^\sharp(f) = \sum_\sigma f_{[\eta]}(\sigma)u(\sigma)$. For finite tori use $f^R(x) = \sum_{k\in\mathbb{Z}^3} f(x + Rk)$ in this identity. Consider families with $n \to \infty$ at each fixed $\eta$, followed by $\eta \to 0$; a realizing joint sequence also has $R \to \infty$. Let $h_\kappa$ be the two-form Hermite basis of $1 + |x|^2 - \Delta$ with eigenvalues $\lambda_\kappa = 2|k|_1 + 4$ and define $\|F\|_{-s}^2 = \sum \lambda_\kappa^{-2s}|F(h_\kappa)|^2$.

Claim. The step-field laws satisfy $\mathbb{E}\|F^\sharp\|_{-3}^2 < 61$ and are tight in $H_{-4}$. Fixed-spacing volume limits and subsequent spacing limits exist. Every fixed-spacing limit $\mu^\eta$ is *exactly* the step image of a product-topology lattice limit $\nu^\eta$ along its defining volume sequence. Along any realizing joint sequence, all finite-family moments and cumulants converge.

Proof work. Establish cell-average contraction and, with $Q_R = [-R/2, R/2)^3$, the periodization estimate $\|f^R\|_{L^2(Q_R)} \le \|f\|_2 + 26R^{-2}\||x|^2 f\|_{L^2(\mathbb{R}^3\setminus Q_R)}$. Hermite summation gives the common expected $H_{-3}$ norm; its balls are compact in $H_{-4}$. For the converse, choose bumps $b^\sigma$ inside single cells with $(b^\sigma)_{[\eta]} = 1_\sigma$. The inverse coordinate map $J_\eta F = (F(b^\sigma))_\sigma$ is continuous. Prove polynomial growth and $H_{-3}$ support of its limit by tails and Fatou. Recover $F$ from the Borel step map using the closed hyperplanes $F(f - \sum f_{[\eta]}b^\sigma) = 0$ for one countable dense family. Uniform exponential integrability gives analytic generating-function, moment and cumulant convergence.

Inputs. The finite Gaussian inequality (M05 in the application) and the specified cell map. The step map on the infinite product space is Borel; continuity is not assumed.

Source. Manuscript pp. 51–56, §§5.4, 6.1–6.4.

## M11. One-plane reflection-positive reconstruction

INTERFACE. Overview Target.

Objects and hypotheses. Let a translation-invariant probability law on a real tempered-field space be invariant and RP at $r(x_1, x') = (-x_1, x')$. Its positive-side sigma-algebra is generated by compact tests in $x_1 > 0$. Put $U(a)A(F) = A(\tau_{-a}F)$ and $\Theta A = \overline{A(rF)}$. Assume evaluations generate the sigma-algebra and the translation action is strongly continuous on $L^2$; for these field laws this follows by density of bounded characters.

Claim. The quotient completion of $b(A, B) = \mathbb{E}[(\Theta A)B]$ has a contractive class map and commuting generators $H \geq 0$, $P_i = P_i^*$ with

$$[U(te_1 + y)A] = e^{-tH} e^{iy\cdot P}[A] \quad (t \geq 0, y \perp e_1).$$

There is a joint spectral resolution on $[0, \infty) \times e_1^\perp$, and $\Omega = [1]$ is fixed.

Proof work. Character density first extends RP to all positive-side $L^2$. Repeated Cauchy–Schwarz in $\|[U(t)A]\|^2 = b(A, U(2t)A)$ proves time contractivity and descent through null vectors. Strong continuity, positive square roots and spectral calculus define $H$. Parallel translations are isometries with inverse; prove their commutation with the semigroup before using a joint spectral measure.

Inputs. Translation invariance and one invariant RP plane. Gaussian domination, closedness and rotation symmetry are not needed.

Source. Manuscript pp. 57–58, Proposition 7.1.

### M12. Two mirrors force the sharp energy–momentum cone

INTERFACE. Overview Target.

Proof dependencies. M11

Objects and hypotheses. Let a translation-invariant field law be invariant and RP at $x_1 = 0$ and $x_1 + x_2 = 0$, with the geometric tensor reflection action. Define $H, P_2$ from the coordinate reflection quotient as in M11: $[U(te_1 + xe_2)A] = e^{-tH+ixP_2}[A]$. The second mirror has its own quotient Hilbert space and generators.

Claim. On the entire coordinate quotient the joint spectral measure obeys $E \geq |p_2|$, with constant one. Equivalently $\mathbf{1}_{\{E<|p_2|\}}(H, P_2) = 0$.

Proof work. For $A$ supported in $0 < x_1 < D$, $|x_2| < D$, give one correlation two Fourier–Laplace representations, one with positive measure $\rho(E, p)$ and one with a mixed finite measure of variation at most $\|A\|_2^2$. For every $h \in L^1(0, \infty)$ compare them to obtain

$$\int e^{-2DE} |\widehat{h}(E - ip)|^2 \, d\rho \leq \|A\|_2^2 \sup_{u \geq 0} |\widehat{h}(u)| \sup_{v \in \mathbb{R}} |\widehat{h}(iv)|.$$

Use the actual Laplace transforms $e^{-n} z^{1-4isn/\pi}(1 + z)^{-3}$, $s = \pm 1$, and verify their $L^1$ inverse transforms and boundary values. Their two suprema have bounded product but the modulus grows on every forbidden sector. Exhaust the sector, then use dense compact characters to annihilate its spectral projection on the whole space.

Inputs. M11 applies in both directions. No continuous rotation covariance or identification of the two quotient spaces is assumed. This sharp cone precedes the angular argument.

Source. Manuscript pp. 58–60, Proposition 7.2; the supplied rotation example also states the two-plane comparison.

### M13. Invariant projections and clustering in every direction

INTERFACE. Overview Target.

Proof dependencies. M11, M12

Objects and hypotheses. Let a translation-invariant tempered-field law be invariant and RP at the coordinate and diagonal planes $x_j = 0$, $x_j \pm x_k = 0$. Let $P_j$ project in ordinary $L^2$ onto invariants of translations along $e_j$, and $P$ onto invariants of all translations. Coordinate quotients have the nonnegative $H_j$ of M11 and the cones $H_j \geq |P_i^{\mathrm{mom}}|$ from M12.

Claim. $P_1 = P_2 = P_3 = P$. The quotient map is an isometry from $\operatorname{ran} P_j$ onto $\ker H_j$. For all ordinary $L^2$ observables,

$$\langle A, U(a)B\rangle \longrightarrow \langle A, PB\rangle \quad (|a| \to \infty).$$

For an ergodic law this is factorization and $\ker H_j = \mathbb{C}\Omega_j$.

Proof work. Move an invariant observable into either half-space by approximating with local characters and translating the approximants. The linear geometric reflection is a positive involution there and hence the identity. Time averaging identifies its quotient range with zero energy. The sharp cone kills all transverse momenta at zero energy, so invariance in one coordinate implies invariance in every direction. For opposite half-space local observables the spectral decay bound is uniform in the transverse displacement. Extend by $L^2$ approximation, and select a coordinate tending to infinity along each subsequence of large translation vectors.

Inputs. M11–M12 and local character density. Rotation invariance, Gaussian domination and a mass gap are absent. Ergodicity alone would not imply this mixing conclusion.

Source. Manuscript pp. 60–61, §§7.3–7.4, Proposition 7.3.

### M14. The same Gaussian and mirror hypotheses in almost every component

INTERFACE. Overview Target.

Proof dependencies. M09, M13

Objects and hypotheses. Let $\lambda$ be a translation-invariant law of real tempered two-forms on the evaluation standard Borel space. Assume $\mathbb{E}e^{F(f)} \le e^{\|f\|_2^2/2}$ for all real Schwartz $f$, and invariance and RP at the nine coordinate/diagonal mirror directions. Let $\mathcal{I}$ be the all-translation invariant sigma-algebra and $\lambda = \int \lambda_\xi \, d\nu(\xi)$ its conditional-law decomposition. If closedness is desired assume $dF = 0$ almost surely in the parent law.

Claim. There is one measurable full-measure set of translation-ergodic components having the same Gaussian constant for *all* real Schwartz tests, all nine invariant RP mirror forms at every height, cubic invariance and, when the parent is closed, closedness. These are the component laws to which the rotation and clustering criteria must be applied.

Proof work. Use a countable determining character algebra to construct and identify simultaneous ergodic conditional laws. An invariant nonnegative weight has a positive-side representative and is fixed by geometric reflection; apply RP to its square root times a test observable. Condition these weighted forms to pass positivity and invariance. Transport conditional laws to the other mirror frames and use uniqueness of the ergodic decomposition, then intersect the finitely many full-measure sets. For the Gaussian bound, average $e^{F(f)}$ over expanding translation cubes, estimate every integer moment of that average by the decay of translated $L^2$ overlaps, and deduce the essential bound on its invariant conditional expectation. A countable dense family and Fatou extend it to every test on one component set. Pass the closedness event by conditioning.

Inputs. M09 and the invariant-reflection argument in M13. An exponential bound or symmetry of a mixture is not a bound or symmetry of each component without this argument.

Source. Manuscript pp. 61–62, §7.5.

## M15. Bounded field insertions and holomorphic planar kernels

INTERFACE. Overview Target.

Proof dependencies. M09, M11, M12

Objects and hypotheses. Let a real two-form law on $\mathbb{R}^3$ be translation invariant, Gaussian dominated with constant $1/2$, and invariant and RP at $x_1 = 0$. Its coordinate quotient is $(\mathcal{H}, H, P_2, \Omega)$; assume the sharp joint cone $H \geq |P_2|$. For a slab test $f$ define $B_\ell(f)[A] = [F(f)U(\ell e_1)A]$, initially on bounded positive-side observables, with $\operatorname{supp} f \subset \{0 \leq x_1 \leq \ell\}$. Set $W = \{z \in \mathbb{C}^2 : \Re z_0 > |\Im z_1|\}$ and $d_W(z) = \Re z_0 - |\Im z_1|$.

Claim. $B_\ell(f)$ is bounded by $\sqrt{2\mathrm{e}}\|f\|_2$, independently of observable degrees. After localizing only the $(x_1, x_2)$ plane, one obtains insertion operators for transverse profiles $h \in L^2(\mathbb{R}; \Lambda^2\mathbb{C}^3)$. For $\xi, \eta \in W$ they are

$$A_h(\xi, \eta) = \operatorname*{w-lim}_{\epsilon \downarrow 0} T(\xi - \epsilon e_0) B_{2\epsilon}(\tau_{\epsilon e_1}(j_\epsilon h)) T(\eta - \epsilon e_0), \quad T(z) = e^{-z_0 H + i z_1 P_2},$$

where $j_\epsilon$ is the normalized planar disk indicator and $e_0 = (1, 0)$ in the coordinates of $W$. Their norm is at most $C\|h\|_2 / \min(d_W(\xi), d_W(\eta))$. For $N \geq 2$ and $Z = (z_1, \ldots, z_N)$ with successive differences in $W$, construct $G_N(Z; h_1, \ldots, h_N)$, holomorphic in $Z$, with

$$|G_N| \leq C^N \prod_j \|h_j\|_2 \big(\min_{j<N} d_W(z_{j+1} - z_j)\big)^{-N}.$$

On strictly time-ordered compact planar tests,

$$\mathbb{E} \prod_j F(\alpha_j h_j) = \int \prod_j \alpha_j(z_j) G_N(Z; h)\, dZ.$$

Construct vectors $\Psi(Z; h)$ for ordered positive-side points ($z_1 \in W$ and all successive gaps in $W$), with inner product $G_{a+b}(r\bar{Z}_{\mathrm{rev}}, Z'; Jh_{\mathrm{rev}}, h')$; $r$ reverses planar time, $Jh = \overline{r \cdot h}$ acts on tensor indices, and the first tuple is reversed. The empty tuple is the vacuum and the centered one-point function is zero. This is the continued Gram identity consumed by the angular argument.

Proof work. Double a source slab and apply Gaussian domination before taking a Cauchy derivative; prove null-vector descent for the insertion. The sharp cone supplies bounded complex translations. Holomorphic averaging replaces the shrinking localization radius by the fixed neighbouring gap. Prove weak point-insertion limits are unique, establish their shift/adjoint identities and norm holomorphy, and show that splitting an inner gap does not change a product. Integrate these operators to recover genuine ordered moments and their Gram vectors. No pointwise random field is introduced.

Inputs. M09, M11 and M12, each with the assumptions restated above. The transverse variable remains smeared throughout.

Source. Manuscript pp. 62–68, Propositions 8.1–8.3.

### M16. No contact term for bounded closed-form correlations

INTERFACE. Overview Target.

Objects and hypotheses. Let $N \geq 1$ and let $T$ be a complex $N$-linear form on $L^2(\mathbb{R}^3; \Lambda^2\mathbb{C}^3)$ with $|T(f_1, \ldots, f_N)| \leq C_N \prod \|f_j\|_2$. It is closed in each distributional two-form slot: replacing that slot by $d^*\zeta$, with compact smooth three-form $\zeta$, gives zero. Suppose $T$ vanishes on all compact tests whose projections to the $(x_1, x_2)$ plane are pairwise disjoint.

Claim. $T = 0$ on all $L^2$ tests. The same result holds over every open planar set, and can therefore be used after other entries have been fixed. This is a distributional statement; the correlation kernel is not assumed to be an $L^2$ function.

Proof work. Use the Hodge identification with vector tests. In the bilinear case, vanishing on separated planar supports makes planar multipliers move between slots. Represent the resulting bounded functional by an $L^2$ vector. Its closedness and the closedness after multiplication force the vector to be transverse-constant in the remaining direction, hence zero by $L^2$ integrability. At higher orders, separate groups of slots by induction over open planar sets. Partition the plane into small squares: mixed-square contributions vanish, while the equal-square sum tends to zero by absolute continuity of the $L^2$ integral. Upgrade by density.

Inputs. Bounded multilinearity and slotwise closedness alone; no law, Gaussian bound, reflection form or analyticity is required. This lemma must be reapplied after separated-support angular identities have been established.

Source. Manuscript pp. 71–72, Proposition 8.4.

### M17. Rotational covariance from discrete mirrors: the standalone criterion

INTERFACE. Overview Target.

Proof dependencies. M09, M11, M12, M15, M16, M23, M24

Objects and hypotheses. Let $\mu$ be a Borel law of real tempered two-forms on $\mathbb{R}^3$. For real Schwartz tests $f$ assume (i) $dF = 0$ almost surely; (ii) $\mathbb{E}e^{F(f)} \le e^{\|f\|_2^2/2}$; (iii) invariance under translations and signed coordinate permutations; and (iv) RP at $x_j = 0$ and $x_j \pm x_k = 0$. For $g(x) = Qx + a$ the active test action is $(g \cdot f)(x) = (\Lambda^2 Q) f(g^{-1}x)$. RP means positivity of $\mathbb{E}[\overline{A(rF)}A(F)]$ for every positive-side character sum $A = \sum z_a e^{iF(f_a)}$.

Claim. The law is invariant under every Euclidean isometry and RP at every affine plane. The criterion contains no chamber operator, charge, mass, ergodicity assumption or cumulant estimate.

Proof work. M23 supplies the finite angular window and two-point base case. M24 supplies positive-kernel rigidity and identification of the continued density with actual separated-support moments. Apply M16 to remove contact terms. Entire moment determinacy gives invariance of the law; conjugating the coordinate reflection form gives positivity at every affine plane.

Inputs. M09, M11–M12, M15–M16 and M23–M24; their law hypotheses are exactly those written here. The criterion is in three dimensions and its proof uses the polygon argument.

Source. Manuscript pp. 62–77, Theorem 8.6 and its preceding estimates. The higher-angular analytic and positive-kernel subtasks are separately listed on the chart as M23–M24.

### M18. From a Euclidean field law to the full tensor OS axioms

INTERFACE. Overview Target.

Proof dependencies. M09, M13, M17

Objects and hypotheses. Let $\lambda$ be a law of real tempered two-forms on $\mathbb{R}^3$ with $\mathbb{E}e^{F(f)} \le e^{\|f\|_2^2/2}$ for every real Schwartz $f$, invariant under every Euclidean isometry and RP at every origin plane. Define $\mathcal{S}_n = \mathcal{S}(\mathbb{R}^{3n}; (\Lambda^2\mathbb{C}^3)^{\otimes n})$ and $S_n(f_1 \otimes \cdots \otimes f_n) = \mathbb{E}\prod F(f_j)$, $S_0 = 1$. For $g(x) = Qx + a$ define $(g \cdot \Psi)(x_1, \dots, x_n) = (\Lambda^2 Q)^{\otimes n}\Psi(g^{-1}x_1, \dots, g^{-1}x_n)$. For the clustering clause also assume $\langle A, U(a)B\rangle \to \overline{\mathbb{E}A}\,\mathbb{E}B$ for all ordinary $A, B \in L^2(\lambda)$ as $|a| \to \infty$.

Claim. Extend $S_n$ to all $\mathcal{S}_n$ and prove exactly tensor regularity, Euclidean covariance, reflection positivity, permutation symmetry, and, under the extra mixing hypothesis, clustering. In particular

$$|S_n(\Psi)| \le 2\,66^n\sqrt{n!}\,q_{6n}(\Psi), \quad q_m(\Psi) = \max_{|\alpha|\le m}\sup_y (1+|y|^2)^{m/2}|\partial^\alpha\Psi(y)|,$$

with the maximum also over tensor components. Covariance means $S_n(g \cdot \Psi) = S_n(\Psi)$, and permutation symmetry includes the arguments and their tensor indices. For a fixed reflection $r$ and a common closed half-space $H$ bounded by its plane, take $\operatorname{supp}\Psi_k \subset H^k$ for $1 \le k \le m$ and $\Psi_0 \in \mathbb{C}$. The RP form is $\sum_{k,l} S_{k+l}(\overline{r \cdot \Psi_k} \otimes \Psi_l) \ge 0$ for that finite family, and degree zero reflection conjugates its scalar. Under mixing the cluster formula is $S_{k+l}(\Psi \otimes \tau_a^{\otimes l}\Phi) \to S_k(\Psi)S_l(\Phi)$ as $|a| \to \infty$, where $\tau_a f(x) = f(x-a)$.

Proof work. Expand in the two-form Hermite basis and prove the absolute coefficient bound using two oscillator powers in each slot. Construct the random evaluation $Q_\Psi$ by an absolutely convergent $L^2$ series. Establish its product, permutation and tensor-action rules first pointwise or on products, then by the established norms. Give a support-preserving Fourier tensor expansion, compact cutoff and inward translation to show $Q_\Psi$ belongs to the positive-side $L^2$ space even for plane-touching supports. Apply the law-level reflection form to $\sum Q_{\Psi_k}$. Apply mixing with $A = \overline{Q_\Psi}$ to obtain the un-conjugated tensor cluster identity.

Inputs. M09 for moments; M17 for covariance/all-plane RP; M13 for the extra mixing conclusion in ergodic laws. Closedness and non-Gaussianity are separate outputs of the overall construction, not assumptions of this conversion.

Source. Manuscript pp. 51–52 and 258–259, Proposition 27.1.

### M19. Distinct translated products bound invariant conditional expectations

INTERFACE. Overview Target.

Objects and hypotheses. Let a probability space carry a measure-preserving action $U(y)$ of a full-rank lattice $\Lambda \subset \mathbb{R}^3$. Let real $G$ belong to every finite $L^p$. Suppose that for every finite set of distinct lattice points $Y$,

$$\left| \mathbb{E} \prod_{y \in Y} U(y) G \right| \le u^{|Y|}, \qquad u \ge 0.$$

Let $P_\Lambda$ denote conditional expectation onto its invariant sigma-algebra. If the action extends to all translations, write $\mathcal{I}$ for the smaller all-translation invariant algebra.

Claim. $\|P_\Lambda G\|_\infty \le u$, and consequently $|\mathbb{E}[G \mid \mathcal{I}]| \le u$ almost surely. No ergodicity of the discrete lattice action is required.

Proof work. Average $G$ over growing finite lattice cubes. In each fixed even moment expand into tuples of sites. Distinct-index tuples are bounded by the assumption. Repeated-index tuples have vanishing relative count; their expectations are uniformly bounded for that fixed moment by Hölder. The mean ergodic limit and uniform higher moments pass the bound to every even moment of $P_\Lambda G$. Let the moment order tend to infinity. The tower property for conditional expectations gives the all-translation bound.

Inputs. Only a measure-preserving lattice action and the displayed distinct-product bound. In the application T17 supplies the bound for centered scaled Wick polynomials, while M10 passes it to the limit law. Positivity of a parent fourth cumulant is not an alternative input.

Source. Manuscript pp. 255–256, Lemma 26.1.

## M20. Polymer counting, reblocking and nonwrapping lifts

INTERFACE. Overview Target.

Objects and hypotheses. Let $L \geq 5$ be odd, $M \geq L^2$ a power of $L$, and let a cubic torus side $\ell$ have $\ell/(LM)$ an odd integer at least 13. A polymer $X$ is a nonempty face-connected union of $M$-cubes. Define $Md_M(X)$ as the infimum length of a connected net meeting their closed cubes. A net is a union of images of specified straight segments; length is the sum of their represented Euclidean lengths, with the infimum also over those representations. Let $\overline{X}^{\,L}$ be its closure in $LM$-cubes.

Claim. With $c_X = 54$, $\kappa_0 = 54 \log 72$, $K_0 = e^{\kappa_0}/36$, prove, with $|X|_M$ the number of cubes,

$$d_M(X) \leq |X|_M - 1, \quad |X|_M \leq c_X(d_M(X)+1), \quad d(p,q) \leq M(d_M(X)+2) \quad (p,q \in X),$$
$$\sum_{X \supset \square} e^{-\kappa_0 d_M(X)} \leq K_0, \qquad \sum_{X \subset Y} e^{-\kappa_0 d_M(X)} \leq K_0 |Y|_M,$$

where sums run over nonempty connected polymers and $Y$ is any cube union. For reblocking require also that $\ell/L$ has the same side conditions, and prove

$$Ld_M(Y) \leq d_M(X) \quad (\overline{X}^{\,L} = LY), \qquad \sum_{\overline{X}^{\,L} = LY} e^{-\kappa' d_M(X)} \leq c_X K_0 L^3 e^{-L(\kappa' - \kappa_0 - 1) d_M(Y)}.$$

Here $\kappa' \geq \kappa_0$. If $2M(d_M(X)+2) < \ell$, construct an injective adjacency-preserving covering lift $\widetilde{X}$ with the same net distance. If also $M(d_M(X)+4) < \ell$ and $M \geq 2$, then at every mesh $\eta = L^{-k} \leq 1$ there is a lattice box $W$ containing every bond meeting $\widetilde{X}$, with each side at most $M(d_M(X)+3)$. Projection is injective on $W$, and every torus bond meeting $X$ has exactly one lift in $W$. Construct $W$ by enlarging each coordinate interval of the closed lifted cubes by one mesh at either end.

Proof work. Construct a rooted cube tree from a near-minimal represented net, count its finite neighbours and tree codes, and integrate the exponential tail. Scale the represented segments before taking the infimum for reblocking. A path shorter than one period cannot identify distinct required lifts; check every adjacency and the enlarged bond box separately. All constants must remain independent of torus side.

Inputs. Cubic geometry only. This is an upstream geometric lemma used by localization, cover comparison and polymer logarithms; it has no field-theoretic input.

Source. Manuscript pp. 261–263, Appendix A and Proposition A.1.

### M21. Exponential locality of a positive inverse square root

INTERFACE. Overview Target.

Objects and hypotheses. Let $T$ be a finite real symmetric matrix with $T \geq t_0 I$, $t_0 > 0$. Its indices have a symmetric nonnegative support distance $d$ with $d(a,a) = 0$, and positive constants $g, C_T$ with triangle defect $\phi \geq 0$, meaning $d(a,c) \leq d(a,b) + d(b,c) + \phi$, and $|T(a,b)| \leq C_T e^{-g d(a,b)}$. Suppose

$$S > 0, \qquad S \geq e^{g\phi/2} \sup_a \sum_b (d(a,b) + \phi) e^{-g d(a,b)/2}, \quad \theta = \min\{g/2, t_0/(2C_T S)\}.$$

Claim. For every $x \geq 0$,

$$|(T+x)^{-1}(a,c)| \leq (t_0/2 + x)^{-1} e^{-\theta d(a,c)}, \qquad |T^{-1/2}(a,c)| \leq \sqrt{2/t_0}\, e^{-\theta d(a,c)}.$$

The constants are uniform in matrix size when the displayed data are uniform.

Proof work. Conjugate by the diagonal exponential weight centered at $c$. The distance defect and Schur test bound the perturbation by $t_0/2$. Invert with that preserved lower bound, then use the positive resolvent integral for $T^{-1/2}$. Justify the matrix-element integral with its positive integrable majorant.

Inputs. Finite positive matrix and a summable metric kernel bound. Photon fields, Grassmann variables and an RG scheme do not occur. S04 verifies these hypotheses for the axial photon quadratic form.

Source. Manuscript p. 95, the positive-resolvent argument in §10.4.

### M22. The continuum nine-mirror interface

INTERFACE. Overview Target.

Proof dependencies. M05, M08, M10

Objects and hypotheses. At each spacing $\eta = L^{-N}$ take all fixed-spacing volume limit laws of the step field in M10. Suppose their recovered unit-lattice laws satisfy M08 at that fixed charge and mass. A volume-first limit is any weak limit in $H_{-4}$ of these laws as $N \to \infty$. Use the exterior-square action on tests; a lattice plaquette includes all its corners when determining its positive-side support.

Claim. Every volume-first limit is a law of closed real tempered two-forms, is translation/cubic/sign invariant, satisfies the Gaussian bound with constant $1/2$, and is invariant and RP at every height in the three coordinate and six diagonal mirror families. It can be realized by a joint cutoff sequence along which all fixed finite- family moments and cumulants converge.

Proof work. Select the realizing joint sequence from the fixed-spacing volume limits. Prove the discrete Bianchi test annihilates the step field and differs from the continuous test by at most $\eta^2 \sum_j \|\partial_j^3 \xi\|_2/24$. Approximate arbitrary translations by mesh translations. At a mirror height in $\mathbb{Z}[L^{-1}]$, compact tests with margin $\delta > 2\eta$ induce plaquette tests strictly on the positive lattice side. Use M10's converse identification, then M08, before passing to the spacing limit. Translation invariance moves the mirror to every real height; character density extends the form to the whole positive-side $L^2$ space.

Inputs. M05, M08 and M10, with mass independent of outer volume at each fixed spacing. Compactness of arbitrary joint cutoffs alone does not establish this interface.

Source. Manuscript pp. 55–57, Proposition 6.1.

### M23. A finite angular window and the two-point base case

INTERFACE. Overview Target.

Proof dependencies. M09, M11, M12, M15, M16

Objects and hypotheses. Let a closed two-form law on $\mathbb{R}^3$ have Gaussian domination, translations, cubic invariance and the nine RP planes. Identify two-forms with axial vectors by $*f = (f_{23}, -f_{13}, f_{12})$. In the $(x_1, x_2)$ plane use polarization vectors $e_1 \pm ie_2, e_3$ with rotation weights $1, -1, 0$. Let $\lambda = (\lambda_1, \dots, \lambda_N)$ be the tuple of these polarizations and real transverse $L^2(\mathbb{R})$ profiles; each label represents their product as an axial-vector profile. Conjugation exchanges the two complex polarizations and fixes the real profiles. Let $w$ be the sum of the polarization weights. Ordered partially smeared moment kernels are those constructed in M15.

Claim. For ordered complex planar configurations, let $\Lambda_\beta$ be imaginary planar rotation and prove

$$G_N(\Lambda_\beta Z; \lambda) = \sum_{|w+4m| \le N} c_m(Z; \lambda) e^{-(w+4m)\beta}, \qquad \beta \in \mathbb{R}.$$

At two points only the required tensor weight survives. For a permutation $s$ and a coordinate quarter turn $R_\upsilon$, let $\mathcal{Z}(s, \upsilon)$ consist of tuples whose permuted successive gaps, after $R_{-\upsilon}$, lie in $W$; put $\mathcal{Z}_N = \bigcup_{s,\upsilon} \mathcal{Z}(s, \upsilon)$ and $\mathcal{O}_N = \mathcal{Z}_N \cap (\mathbb{R}^2)^N$. The coefficients are holomorphic on $\mathcal{Z}_N$, symmetric in points and labels, invariant under common translations, and satisfy

$$c_m(\Lambda_\beta Z; \lambda) = e^{-(w+4m)\beta} c_m(Z; \lambda), \qquad \overline{c_m(X; \lambda)} = c_{-m}(X; \bar{\lambda}).$$

Conjugating labels reverses their weights. On real configurations with $X, R_t X \in \mathcal{O}_N$ they satisfy $c_m(R_t X; \lambda) = e^{i(w+4m)t} c_m(X; \lambda)$. Extend them uniquely to real-analytic coefficients on all *real* distinct planar configurations with this same phase rule. Also $\sum_m c_m = G_N$ on the tube union. These conclusions precede full law covariance.

Proof work. For two points obtain the positive matrix spectral measure and the bound $\int e^{-tE} d\rho(E, p) \le Ct^{-2}$. The complex rotation orbit has exponential type two. Bianchi gives the endpoint ODE; its unwanted $r^2$ solution contradicts the inverse-square bound. Use M16 to extend the two-point identity to all tests. For all orders glue only intersecting convex ordered tubes, prove the finite window on narrow configurations, and extend via the constant-coefficient angular ODE and holomorphic uniqueness. Transport the coefficients to distinct configurations using their already-established analytic phase rule. A covariance-dependent larger tube is not an input.

Inputs. M09, M11–M12, M15–M16. This task proves the analytic and two-point inputs of M24; it does not already conclude that the law is rotationally invariant.

Source. Manuscript pp. 68–74, §§8.4–8.8 and Propositions 8.4–8.5.

### M24. Positive leading coefficients and angular rigidity

INTERFACE. Overview Target.

Proof dependencies. M15, M16, M23

Objects and hypotheses. Use M23's finite angular coefficients of a Gaussian-dominated, closed, cubic-invariant nine-mirror law. Define $\mathfrak{a}(0) = \mathfrak{a}(1) = 0$. For $N \geq 2$ set $\mathfrak{a}(N) = \max\{|m| : c_m(\cdot;\lambda) \not\equiv 0 \text{ for some } N\text{-tuple of labels } \lambda\}$, with maximum zero for an empty set. The finite window and $|w| \leq N$ give $\mathfrak{a}(N) \leq N/2$. Use M15's *actual* Hilbert vectors for continued reflected moments; positivity is not continued as a scalar inequality.

Claim. The indices satisfy

$$2\mathfrak{a}(a+b) \leq \mathfrak{a}(2a) + \mathfrak{a}(2b) \quad (a,b \geq 0), \qquad a \mid 2\mathfrak{a}(2a) \quad (a \geq 1), \qquad \mathfrak{a}(N) = 0 \quad \text{for all } N.$$

Identify the resulting continued kernels with actual moments on all separated supports. This gives the remaining proof inputs for the standalone criterion M17.

Proof work. Scale the Gram form by its largest growing exponential and pass to its positive leading kernel. Cauchy–Schwarz gives the first inequality and propagates zero diagonal entries. A regular $2a$-gon with one repeated label gives the divisibility selection rule on that configuration. The halving construction and analytic uniqueness propagate its zero to arbitrary configurations; this propagation is indispensable. For $\nu(a) = \mathfrak{a}(2a)$, integer convexity makes its increments eventually constant; the divisibility eliminates their intercept, and the two-point base case makes their slope zero. Finally use analytic displacement between separated groups to identify the continued density with actual moment integrals, then M16 to remove contact terms.

Inputs. M15, M16 and M23 with the same law and tensor conventions. M17 subsequently uses moment determinacy and conjugation to conclude full Euclidean invariance and all-plane RP.

Source. Manuscript pp. 74–77, §§8.8–8.11.

## S01. The chamber gap and the singular free inverse

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M01

Objects and hypotheses. Fix $0 < \rho < 1$. A chamber site is $x = (c, g)$, with $c \in a\mathbb{Z}^3$, $a > 0$, and $g$ a signed permutation matrix. Work on the covering lattice or a torus of side $R \in a\mathbb{N}$, with any unitary character in its three period directions. Pairings use $a^3/48$ per chamber site and the Euclidean norm on $\mathbb{C}^4$; $|\cdot|_2$ on a spin matrix denotes its induced Euclidean operator norm. The internal neighbors are $x^{(i)} = (c, gs_i)$, where $s_0$ reverses the third coordinate, $s_1$ exchanges coordinates $2, 3$, and $s_2$ exchanges $1, 2$. The external neighbor is $x^{(3)} = (c + age_1, gr_{e_1})$, with $r_{e_1}$ the first-coordinate reflection. Put

$$n_0 = e_3, \quad n_1 = (e_2 - e_3)/\sqrt{2}, \quad n_2 = (e_1 - e_2)/\sqrt{2}, \quad n_3 = -e_1, \qquad w = (6, 6\sqrt{2}, 6\sqrt{2}, 6).$$

Use the Hermitian Euclidean representation $\gamma_\mu = \sigma_3 \otimes \sigma_\mu$ and $\gamma_5 = \sigma_1 \otimes 1_2$, with Pauli matrices

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The four-component Dirac matrices satisfy the Clifford relations; $\gamma_5^2 = 1$ and $\gamma_5\gamma_\mu = -\gamma_\mu\gamma_5$. Set $\widehat{d}_i(x) = -gn_i$, $M_i(x) = \frac{1}{2}w_i(\rho - \gamma \cdot \widehat{d}_i(x))$, and

$$D_A f(x) = -a^{-1} \sum_{i=0}^{3} M_i(x)[U_A(x, x^{(i)})f(x^{(i)}) - f(x)].$$

Internal transports are one, and the external transport is $\exp(ie_a aA(c, c+age_1))$. Let $J\xi(c, g) = \xi(c)$, $J^* f(c) = 48^{-1} \sum_g f(c, g)$, $Q = 1 - JJ^*$. Let $\lambda_{\mathrm{int}} > 0$ be the gap of $\sum_{i=0}^{2} w_i(1 - R_i)$ on the mean-zero functions of $g$; the three generators above make its kernel precisely the constants.

Claim. The bond form $\mathcal{L}_A = \sum_i w_i(1 - U_A R_i)$ satisfies, for real $A$,

$$(f, \mathcal{L}_A f) \geq \lambda_{\mathrm{int}} \|Qf\|_2^2, \qquad a^{-2}(f, \mathcal{L}_A f) \geq c_* \|\nabla_A J^* f\|_2^2, \quad c_* = \frac{\min(6, \lambda_{\mathrm{int}})}{18}.$$

Here $\nabla_{A,\mu}\xi(c) = a^{-1}[u_A(c, c + ae_\mu)\xi(c + ae_\mu) - \xi(c)]$. For every finite set $E$ of photon sites the same two bounds hold with $\|Qf\|_2$ restricted to sites over $E$, the positive bond-energy sum restricted to chamber bonds whose two endpoints lie over $E$, and $\nabla_A$ restricted to photon bonds with both endpoints in $E$. Moreover $QD_A Q + t$ has inverse norm at most $(t + \rho\lambda_{\mathrm{int}}/(2a))^{-1}$ when its denominator is positive. For the zero-field symbol at $a = 1$, split $D_0(p) + m$ by $J, Q$, write $H_m = D_{hh}^{-1}$, and let $B_m = D_{ll} - D_{lh}H_m D_{hl}$, $0 \leq m \leq 1$. There are positive constants $\theta_0, c_e, \kappa_{\mathrm{strip}}, C_G, c_b$ and finite $C_\nu$, depending only on $\rho$, such that $H_m, B_m$ are $2\pi\mathbb{Z}^3$-periodic and holomorphic on a neighborhood of $\{|\operatorname{Im} p|_\infty \leq \theta_0\}$. Put $g_h = \rho\lambda_{\mathrm{int}}/2$, $c_{\mathrm{lh}} = \sqrt{3}(1 + \rho)$, $t(p) = \sum_\mu(1 - \cos p_\mu)$, $K_\rho = \rho + 2c_{\mathrm{lh}}^2/g_h$. On real $p \in [-\pi, \pi]^3$,

$$\|H_m(p)\| \leq (g_h + m)^{-1}, \qquad \|D_{lh}(p)\| + \|D_{hl}(p)\| \leq 2c_{\mathrm{lh}}(2t(p))^{1/2},$$

and

$$B_m = m + i\gamma \cdot \sin p + E_m, \quad \|E_m\| \leq K_\rho \sum_\mu (1 - \cos p_\mu),$$

$$\operatorname{Re} B_m \geq m + \rho c_* \sum_\mu (1 - \cos p_\mu),$$

$$|B_m v| \geq c_e(|p|^2 + m^2)^{1/2}|v|, \qquad \|\partial_p^\nu B_m^{-1}\| \leq C_\nu(\rho)(|p|^2 + m^2)^{-(1+|\nu|)/2}.$$

The inverse estimate persists in a strip of width $\kappa_{\mathrm{strip}} m$ when $m > 0$; at $m = 0$ it is asserted away from $p = 0$. The covering Green kernel splits into a label-independent part of size $C_G(1 + |c - c'|_\infty)^{-2}$ and a remainder of size $C_G(1 + |c - c'|_\infty)^{-3}$. A nearest-neighbor difference of the first part has

the latter bound. For $m > 0$, all have the further factor $\exp(-\kappa_{\mathrm{strip}} m|c - c'|_\infty)$. In mesh $a$, let $G(x, x') = a^{-2} G_{\mu a}(x/a, x'/a)$, $0 \le \mu a \le 1$. Then

$$|G(x, x')|_2 \le 2C_G(a + |c - c'|_\infty)^{-2},$$
$$|G(x^{(i)}, x') - G(x, x')|_2 \le 20 C_G a (a + |c - c'|_\infty)^{-3} \quad (0 \le i \le 3).$$

For distinct $x = (c, g)$, $\widetilde{x} = (\widetilde{c}, \widetilde{g})$, put $d_X(x, \widetilde{x}) = \max(|c - \widetilde{c}|_\infty, a)$. For $0 \le \alpha \le 1$ the full Hölder output is

$$|G(x, x') - G(\widetilde{x}, x')|_2 \le 6240 C_G d_X(x, \widetilde{x})^\alpha [(a + |c - c'|_\infty)^{-2-\alpha} + (a + |\widetilde{c} - c'|_\infty)^{-2-\alpha}].$$

Also prove the free coercivity and cube remainder estimates

$$\|D_0 f\|_2^2 \ge c_b a^{-2}(f, \mathcal{L}_0 f), \qquad \|f - P_E f\|_{2,E}^2 \le C_P \ell_E^2 a^{-2}(f, \mathcal{L}_0 f)_E.$$

Here $E$ is a lattice cube of side $\ell_E \ge a$, $P_E f$ is the constant mean over $E \times O_{\mathrm{cub}}$, and the subscript $E$ uses the local bond sum just defined. One may take $C_P = (8c_*)^{-1} + \lambda_{\mathrm{int}}^{-1}$. All constants are uniform in the mesh, size, character and allowed mass.

Proof work. Use the internal spectral gap and the eight external crossings in each photon direction. Eliminate the heavy component while retaining the real-part form. Combine the Dirac lower bound near zero with the real-part bound away from zero. Differentiate the inverse identity. The general Fourier subproblem is explicit: let $T$ be a $2\pi\mathbb{Z}^3$-periodic finite matrix symbol which is $C^N$ off $p = 0$ on the Brillouin torus and satisfies $\|\partial^\nu T(p)\| \le A|p|_2^{-q-|\nu|}$, $q < 3$, for $|\nu| \le N$, with an integer $N > 3 - q$. Prove Fourier decay $CA(1 + |n|_\infty)^{-(3-q)}$ by a small-ball split and integration by parts. Apply this to the block inverse, then shift contours for positive mass. Obtain the displayed mesh-scaled one-hop and Hölder outputs, including intracluster differences. Finally combine the fiber coercivity with scalar cube Poincaré.

Inputs. Chamber model and spin algebra; finite Fourier analysis. Exports: chamber coercivity, free singular symbol and free-kernel regularity. The Wilson compression $D_{ll}$ is not the Schur complement $B_m$.

Source. PDF pp. 79–83, §§9.1–9.3, Propositions 9.1–9.4, (9.5)–(9.22); setup repeated here from current TeX lines 6790–6886. Proofs and statements: TeX 6888–7261.

## S02. Blocked inverses and their local walk expansion

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S01

Objects and hypotheses. Let $L \geq 5$ be odd, $\eta = L^{-k}$, and use chamber spinors with measure $\eta^3/48$. The finite side is a positive integer multiple of $L$, with any unitary period character. The chamber Dirac stencil has four neighbors, three internal and one joining adjacent photon sites. Its inputs are the free coercivity, cube Poincaré and mesh-scaled Green bounds of S01, with constants independent of $\eta$. Unit blocks and $L$-blocks are centered odd cubes. Their covariant averages use the arithmetic mean of the six coordinate-order paths and its reverse phase; denote the induced projections by $\Pi_1(A), \Pi_L(A)$. Put

$$b = 1, \quad b_L = b/L, \quad b_- = b(1 - L^{-1}), \quad b_k = \frac{b(1 - L^{-1})}{1 - L^{-k}} \quad (k \geq 1).$$

For $k \geq 1$, set

$$\begin{aligned} F_y(A) &= D_A + \mu + \alpha_y \Pi_1(A) + \beta_y \Pi_L(A), \\ \alpha_y &= b_k i\gamma_5 y (b_k + i\gamma_5 y)^{-1}, \\ \beta_y &= b_k^2 b_L (b_k + b_L + i\gamma_5 y)^{-1} (b_k + i\gamma_5 y)^{-1}. \end{aligned}$$

Include $y = \infty$ by $\alpha_\infty = b_k, \beta_\infty = 0$. At $k = 0$, instead use $F_y = D_A + \mu + b_L \Pi_L(A) + i\gamma_5 y$, with real $y$. Here $0 \leq \mu \leq 1$. For unit blocks define the row norm

$$N_\gamma(T) = \sup_z \sum_{z'} e^{\gamma d(z,z')} \|\mathbf{1}_{\Delta_z} T \mathbf{1}_{\Delta_{z'}}\|_{\infty \to \infty}.$$

The sup norm is the supremum of the Euclidean spinor norm. $d$ is the periodic sup distance; kernels use the chamber column measure $\eta^3/48$. For a pair of distinct chamber sites put $d_X(x, x') = \max(d(c, c'), \eta)$ and

$$U_A(x, x') = \exp(ie_k \eta\, \tau A(c, c')), \qquad (\delta_{\alpha,A} v)(x, x') = \frac{U_A(x, x') v(x') - v(x)}{d_X(x, x')^\alpha}, \quad 0 < d_X(x, x') < 1.$$

Here $\tau A(c, c')$ is the mean of the six coordinate-order path sums. If $c = c'$, the transport is one. For pair-valued operators the left block indicator restricts the first point and the output sup runs over all such short pairs. At $k = 0$, $N_\gamma^{\ker}$ is the same weighted row sum with the localized operator norm replaced by the supremum of the normalized matrix kernel over the two clusters. The inverse domain consists of $A = A_0 + A_1$, $A_0$ real, $\|A_1\|_\infty < e_k^{-1+\varepsilon}$, where on every $11M$ cube $A_0 = B_0 + d\lambda$ with $B_0, \lambda$ real and $\|B_0\|_\infty < e_k^{-1+\varepsilon}$.

Claim. Fix $0 < \alpha < 1$. There are $C_I, C_5, C_{00}, \gamma_5', a_z > 0$, depending only on $L, \rho, \alpha$, with $C_5, C_{00} \geq 1$, such that the zero-connection $L^2$ inverse norm is at most $LC_I$ for positive levels. If $f$ is supported on the unit block $\Delta_{z'}$, $v = F_y(0)^{-1} f$, $x$ lies over $\Delta_z$, and $0 < d_X(x, \widetilde{x}) \leq 1$, then

$$|v(x)|_2 + d_X(x, \widetilde{x})^{-\alpha} |v(x) - v(\widetilde{x})|_2 \leq C_5 \exp(-\gamma_5' d(z, z')) \|f\|_\infty.$$

At zero level,

$$|F_y(0)^{-1}(x, x')|_2 \leq C_{00}(1 + |y|)^{-1} \exp(-a_z d(c, c')).$$

Choose

$$\gamma = \tfrac{1}{2} \min(\gamma_5', a_z) > 0, \qquad a' = \max(C_5, C_{00}) \sum_{n \in \mathbb{Z}^3} \exp(-\gamma |n|_\infty).$$

Use a rate with $\gamma L \leq 1/6$, reducing it and redefining $a'$ by this sum if needed. In particular $a'$ depends only on $L, \rho, \alpha$, not on $M$, level, side, character, mass or $y$. These estimates give $N_\gamma(F_y(0)^{-1}) \leq a' w_k(y)$ and $N_\gamma(\delta_{\alpha,0} F_y(0)^{-1}) \leq a'$; at zero use $N_\gamma^{\ker}$ for the kernel bound. For $e_k \|B\|_\infty \leq \delta$, $3L\delta \leq 1$, $2C_V a' \delta \leq 1$, the perturbed inverse obeys

$$N_\gamma(F_y(B)^{-1}) \leq 2a' w_k(y), \qquad N_\gamma(\delta_{\alpha,B} F_y(B)^{-1}) \leq 18a',$$

where $C_V = 2\exp(1)(42+6b)$, $w_k = 1$ at positive levels and $w_0(y) = (1+|y|)^{-1}$. If $M > 4L$, fold local $3M$ cubes into tori of side $6M$ and patch their inverses with $\sum h_\square^2 = 1$. For $q(p) = 54\pi C_V a' p/M \le 1/2$, length-$n$ walk terms obey

$$\sup_z \sum_{z'} e^{\gamma d(z,z')} \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} S_\omega \mathbf{1}_{\Delta_{z'}}\| \le 16 a' p w_k(y) q(p)^n,$$

$$\sup_z \sum_{z'} e^{\gamma d(z,z')} \sum_{|\omega|=n} p^{n+1} \|\mathbf{1}_{\Delta_z} \delta_{\alpha,A} S_\omega \mathbf{1}_{\Delta_{z'}}\|_{\infty\to\infty} \le 152 a' p q(p)^n.$$

Their sum is the exact inverse, analytic on the stated local-gauge domain, and has row bounds $32a' w_k(y)$ and $304a'$. Require $6Le_0^\varepsilon \le 1$, $4C_V a' e_0^\varepsilon \le 1$ for all $e_k \le e_0$.

Proof work. Prove the averaging-penalty real-part inequalities before invoking the free coercivity. Conjugate by an exponential distance weight. Use the local free Green equation to convert $L^2$ to sup and Hölder control without mesh losses. Handle the level-zero large-$y$ estimate separately. The complex perturbation is a Neumann series. Verify both local inverse identities on the interior of the folded cube. Bound $[F_y, h_\square]$ by $\pi C_V/M$, then count at most eight initial cutoffs and 27 choices per extra walk step.

Inputs. Chamber coercivity and free-kernel regularity; nested gauge averages. Exports: exact analytic inverse, local inverses, weighted walk sums. The quotient uses $d_X = \max(|c-c'|_\infty, \eta)$, with $0 < d_X < 1$; at $k = 0$ no such pair exists.

Source. PDF pp. 83–89, §§9.4–9.6, Propositions 9.5–9.7, (9.23)–(9.41), especially (9.28). TeX 7263–7808.

## S03. The photon minimizer and uniform differentiated kernels

MODEL-SPECIFIC. Overview Target.

Objects and hypotheses. Let $L \geq 5$ be odd, $\eta = L^{-k} = N^{-1}$, and take periodic cochains on $\eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with pairing $\eta^3 \sum$, where $\ell$ is an integer multiple of $L$. All differences are mesh-normalized; write $d$ for the coboundary, $\delta = d^T$, $\Delta_{\mathrm{sc}} = \delta d$ on scalars. For a bond coefficient $B_\mu(c)$, define

$$\partial_\nu B_\mu(c) = \eta^{-1}[B_\mu(c+\eta e_\nu) - B_\mu(c)], \qquad (\delta_\alpha F)(c,\widetilde{c}) = \frac{F(\widetilde{c}) - F(c)}{d(c,\widetilde{c})^\alpha}, \quad 0 < d(c,\widetilde{c}) \leq 1,$$

where $d(c,\widetilde{c})$ is periodic sup distance. The quotient on a kernel acts in its fine row point, with row direction and column index fixed. Its bounds below are componentwise; the $L^2$ operator norm uses the cochain pairing. For unit blocks $\Delta_z = \{|c - z|_\infty < 1/2\}$, define

$$(Q_k\lambda)(z) = \eta^3 \sum_{c\in\Delta_z} \lambda(c),$$

$$(Q_k A)_\mu(z) = \eta^3 \sum_{c\in\Delta_z} \sum_{j=0}^{N-1} \eta A_\mu(c + j\eta e_\mu),$$

$$(Q_k^{(2)} F)_{\mu\nu}(z) = \eta^3 \sum_{c\in\Delta_z} \sum_{i,j=0}^{N-1} \eta^2 F_{\mu\nu}(c + i\eta e_\mu + j\eta e_\nu).$$

Let $R_k$ be orthogonal projection onto $\Delta_{\mathrm{sc}}(\ker Q_k)$. Define $H_k^{\mathrm{ph}} A$ by minimizing $\|da\|_2^2$ subject to $Q_k a = A$, $R_k \delta a = 0$; put $\Delta_k = (dH_k^{\mathrm{ph}})^T dH_k^{\mathrm{ph}}$.

Claim. First verify that the bond average is onto, $Q_k d = dQ_k$, $dQ_k = Q_k^{(2)} d$, and $\|Q_k^{(2)} F\|_2 \leq \|F\|_2$. The minimizer exists uniquely, is real linear, preserves constant and closed fields, and $Q_k H_k^{\mathrm{ph}} = 1$, $H_0^{\mathrm{ph}} = 1$. Its Euler characterization is

$$Q_k a = A, \qquad \delta da \in \operatorname{ran} Q_k^T, \qquad \Delta_{\mathrm{sc}} \delta a \in \operatorname{ran} Q_k^T,$$

with the second range on bonds and the third on scalars. For every fixed $0 < \alpha < 1$, there are positive $C_H(\alpha), \gamma_H, C_\Delta, C_\Delta^{(2)}$, depending only on the fixed averaging construction and $\alpha$, independently of $k, \ell$, such that

$$|H_k^{\mathrm{ph}}(b,b')|, \quad |\partial H_k^{\mathrm{ph}}(b,b')|, \quad |\delta_\alpha \partial H_k^{\mathrm{ph}}((b,\widetilde{b});b')| \leq C_H(\alpha) e^{-\gamma_H d(b,b')},$$

where $b, \widetilde{b}$ have the same direction and initial separation at most one. Also

$$|\Delta_k(b,b')| \leq C_\Delta e^{-\gamma_H d(b,b')}, \qquad \|\Delta_k\|_{2\to 2} \leq C_\Delta^{(2)}.$$

No restriction on the harmonic component is imposed.

Proof work. Minimize the curvature first and identify all minimizers using the periodic discrete Hodge decomposition. A unique scalar in $\ker Q_k$ imposes the Landau condition. Solve the Euler system in the alias classes $q_m = p + 2\pi m$. With

$$u(s) = N^{-1} \sum_{j=-(N-1)/2}^{(N-1)/2} e^{isj/N}, \qquad s_N(s) = 2N \sin(s/(2N)),$$

use $u(s) s_N(s) = 2\sin(s/2)$. Separate the zero alias and cancel its apparent pole as in (10.11). Lower-bound the three denominators on the real cube, then keep a complex strip of width independent of $N$. Prove the summable alias estimate (10.15), including its first derivative and $\alpha$-moment. Identify the periodic kernel before shifting contours; periodize the covering estimate.

Inputs. Finite-dimensional minimization; periodic Fourier/Hodge algebra. Exports: photon minimizer and uniform differentiated kernels. The fixed-strip alias calculation is a verification for these averages; a different averaging map is not covered automatically.

Source. PDF pp. 89–93, §§10.1 and 10.3, Proposition 10.1, (10.1)–(10.5), (10.9)–(10.17). TeX 7819–7946, 8002–8219.

## S04. Axial coordinates and a local positive inverse square root

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M21, S03

Objects and hypotheses. On a unit torus of side $\ell$, partition into odd $L$-blocks. The one-step bond average $\mathbf{Q}$ averages straight length-$L$ paths; $\tau A(y,z)$ is the mean of the six coordinate-order path sums from block centre $y$ to $z$. The photon maps are $H_k^{\mathrm{ph}}$ and $\Delta_k = (dH_k^{\mathrm{ph}})^T dH_k^{\mathrm{ph}}$, with $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$, the plaquette-average contraction, and exponential kernels for $H_k^{\mathrm{ph}}$, its derivatives and $\Delta_k$.

Claim. On the axial slice,

$$\mathbf{Q}Z = \tau Z = 0 \quad \Longrightarrow \quad \|Z\|_\infty \le \tfrac{3}{2}L^2\|dZ\|_\infty, \qquad \|Z\|_2^2 \le 9L^7\|dZ\|_2^2.$$

Construct an orthonormal coordinate basis $\{\Upsilon\}$ from reflection eigenvectors of $\ker\tau$ on interior block bonds and point vectors at noncentral face bonds. Its dimension is $(2-2L^{-3})\ell^3$. There is a real bijection $C$ onto the axial slice: leave these entries unchanged and fill a central face bond by

$$(Cz)(b_c) = -\sum_{\substack{b \text{ on the face} \\ b\neq b_c}} z(b) - \sum_{\text{interior } b} \frac{n_c(b)}{L} z(b),$$

where $n_c(b)$ counts the defining straight paths. It satisfies $\|Cz\|_\infty \le 3L^{9/2}\sup_\Upsilon |z(\Upsilon)|$, $\|z\|_2 \le \|Cz\|_2 \le (5L^3)^{1/2}\|z\|_2$, and $\sum_b |(C\Upsilon)(b)| \le 14L^{3/2}$. Every point of $\operatorname{supp} C\Upsilon$ is within $L$ of every point of $\operatorname{supp}\Upsilon$. The positive form $\widetilde{\Delta}_k = C^T\Delta_k C$ has a uniform gap and an exponential kernel. Therefore

$$C_k^{1/2} = \widetilde{\Delta}_k^{-1/2}, \qquad |C_k^{1/2}(\Upsilon,\Upsilon')| \le C_P^{\mathrm{ph}} e^{-\gamma_P d(\Upsilon,\Upsilon')}.$$

Use of the general matrix input M21. Apply M21 to $T = \widetilde{\Delta}_k$. The verification to supply here is

$$T \ge t_0 I, \quad t_0 = c_P^{-1}, \quad c_P = \max(9L^7, 5L^3 C_\Delta^{(2)}),$$
$$|T(\Upsilon,\Upsilon')| \le C_{\mathrm{Dt}} \exp[-\gamma_H d(\Upsilon,\Upsilon')], \quad C_{\mathrm{Dt}} = (14L^{3/2})^2 C_\Delta \exp(2\gamma_H L).$$

The support distance has triangle defect $L$; its row sums are bounded by $c_{\mathrm{bas}}(g) = 2L^3\exp(2gL)(1+6/g)^3$. Thus M21 may use

$$S_P = \exp(\gamma_H L/2)(L+2/\gamma_H)c_{\mathrm{bas}}(\gamma_H/4), \quad \gamma_P = \min\{\gamma_H/2, (2c_P C_{\mathrm{Dt}} S_P)^{-1}\}.$$

Its conclusion gives the claimed root decay, with $C_P^{\mathrm{ph}} = \max\{C_H(\alpha), C_{\mathrm{Dt}}, \sqrt{2c_P}, 4\exp(2)\}$. The general positive-matrix theorem is proved in M21, not repeated here.

Further photon output. Let $A_k^{\min}(A')$ minimize the $\Delta_k$ form under $\mathbf{Q}A = A'_L, \tau A = 0$, where $A'_L = L^{-1/2}A'(\cdot/L)$. For $\mathrm{Sq}_k = CC_k^{1/2}$,

$$\langle A_k^{\min}(A') + \mathrm{Sq}_k\zeta,\ \Delta_k(A_k^{\min}(A') + \mathrm{Sq}_k\zeta)\rangle = \|dH_{k+1}^{\mathrm{ph}}A'\|_2^2 + \|\zeta\|_2^2,$$
$$H_k^{\mathrm{ph}}A_k^{\min}(A') = (H_{k+1}^{\mathrm{ph}}A')_L - d\omega_k, \quad Q_{k+1}\omega_k = 0.$$

The standard product Gaussian restricted to $|\zeta(\Upsilon)| \le p_0$ has probability $\exp(-\ell^3\delta\varepsilon^0)$, where $0 \le \delta\varepsilon^0 \le 12e^{-p_0^2/2}$ for $p_0 \ge 1$.

Proof work. Use path strips for interior bonds and a face mean for face bonds. Count plaquette multiplicities for the $L^2$ estimate. Count $\ker\tau$ and solve the central-bond constraints. The gap of $\widetilde{\Delta}_k$ follows from axial coercivity and the plaquette-average contraction. Apply the displayed matrix lemma, retaining the support-distance defect $L$. Orthogonality to the fluctuation slice gives the exact Gaussian split. Compare consecutive constrained minima to identify $\omega_k$. Count the independent coordinate tails.

Inputs. Photon minimizer and differentiated kernels; M21's positive inverse-square-root theorem with the inputs verified above. This gives axial coordinates, inverse-square-root decay and the exact one-step Gaussian split. The iterated Gaussian tower and its Jacobian require a separate proof.

Source. PDF pp. 90–91, 93–96, §§10.2,10.4–10.5, (10.6)–(10.8), (10.18)–(10.29). The general matrix lemma is on p. 95, (10.25). TeX 7947–8001, 8220–8536.

## S05. Weakening, local gauges and fluctuation domain margins

Model-specific. Overview Target.

Proof dependencies. M20, S02, S03, S04
Objects and hypotheses. Let $L \geq 5$ be odd, $\eta = L^{-k}$, $M = L^m \geq L^2$, $\ell/(LM)$ odd and at least 13. For bond coefficients $B_\mu(c)$, put

$$\partial_\nu B_\mu(c) = \eta^{-1}[B_\mu(c+\eta e_\nu) - B_\mu(c)],$$
$$\|B\|_1 = \sup_{c,\mu} |B_\mu(c)|, \qquad \|B\|_2 = \sup_{c,\mu,\nu} |\partial_\nu B_\mu(c)|,$$
$$\|B\|_3 = \sup_{\mu,\nu} \sup_{0<d(c,\widetilde{c})\leq 1} \frac{|\partial_\nu B_\mu(\widetilde{c}) - \partial_\nu B_\mu(c)|}{d(c,\widetilde{c})^\alpha}, \quad 0<\alpha<1.$$

$d$ is periodic sup distance. A local norm uses only bonds and difference cells contained in its set. A differentiated kernel uses these operations in its fine row point; its column and directions stay fixed. An $M$-polymer is a face-connected union of partition cubes. Let $Md_M(X)$ be the infimum length of a represented Euclidean net meeting its closed cubes. Assume its size/counting/lifting estimates, in particular $|X|_M \leq 54(d_M(X)+1)$. Photon parameter cubes instead have half-side $M$ or $LM$ and overlap; they are linked when sharing a lattice site. The exact photon kernels $H_k^{\mathrm{ph}}$, its first and short Hölder first derivatives and $C_k^{1/2}$ have common exponential bound $C_P^{\mathrm{ph}} e^{-\gamma_P d}$. The exact fermion inverse is given by local walk terms $S_{y,\omega}$ with length-$n$ weighted row bounds $16a'pw_k(y)q(p)^n$, and Hölder bounds $152a'pq(p)^n$, where $q(p) = 54\pi C_V a' p/M$. Here $a', C_V, \gamma > 0$ are the constants in S02's explicit zero-row and walk estimates, depending only on $L, \rho, \alpha$. Use the row norm

$$N_\gamma(T) = \sup_z \sum_{z'} \exp(\gamma d(z,z'))\|\mathbf{1}_{\Delta_z} T \mathbf{1}_{\Delta_{z'}}\|_{\infty\to\infty}.$$

Its sup norms are Euclidean spinor norms. For pair outputs the first point is restricted, and $\delta_{\alpha,A} v = (U_A(x,x')v(x') - v(x))/d_X(x,x')^\alpha$ on $0 < d_X = \max(d(c,c'),\eta) < 1$, with the six-path transport and transport one within a cluster.
Claim. Construct polynomial photon weakening by multiplying the exact kernel entries by parameter products along ordered shortest paths. Smooth only the reconstruction row, using the partition of unity and its first-difference Lipschitz bounds (10.35). For $0 < \alpha_0 \leq 1/4$ and

$$2\alpha_0(402)\log M \leq \gamma_P M,$$

the reconstruction, its two differentiated kernels, and the root family have bounds

$$116 C_P^{\mathrm{ph}} M^{402\alpha_0} e^{-\gamma_P d/2}$$

on $|s| \leq M^{\alpha_0}$. For fermions put $X_\omega = \bigcup_{i=0}^n \square_i^{\sim 1}$, $s_\omega = \prod_{\square \subset X_\omega} s_\square$. Use right, left and symmetrized walks. If $\alpha_0 = 1/54$, $q(M^{27\alpha_0}) \leq 1/2$ and $q(e^{27\kappa'}) \leq 1/2$, then

$$N_\gamma(S_y^\#(s,A)) \leq 32a' M^{27\alpha_0} w_k(y),$$
$$N_\gamma(\delta_{\alpha,A} S_y^\#(s,A)) \leq 304 a' M^{27\alpha_0},$$
$$\|\mathbf{1}_{\Delta_z} S_y^\#(X,A) \mathbf{1}_{\Delta_{z'}}\| \leq 32a' e^{27\kappa'} w_k(y) e^{-\gamma d(z,z') - \kappa' d_M(X)}.$$

At $k=0$, $w_0(y) = (1+|y|)^{-1}$, with normalized kernel interpretation. At full strength all families recover the exact operators. They are continuous on closed parameter polydiscs, analytic inside and in the connection.

When parameters vanish outside an active union, the families restrict to its linked components and read only that component's data. A nonzero fermion row at $(c,g)$ requires every partition cube meeting $K_c = \{|c'-c|_\infty \leq 2L\}$ to be active. Keep these locality assertions as part of the output.

Domain output. Write $\theta_i = e_k^{-3/4+i\varepsilon}$, $i = 1, 2, 3$. The small complex field norms are sup, first difference and short Hölder first difference with thresholds $\theta_i$. In $\widetilde{R}_k$ also allow a real field with these bounds in independent real local gauges on cubes of side $(2L+1)M$; in $\widetilde{R}_k^-$ leave the margin $e_k^{\varepsilon}\theta_i$ in the global complex summand. For every real coarse $A$, a local real gauge gives all three norms of $H_k^{\mathrm{ph}}A$ bounded by $C_B\|dA\|_\infty$, with $C_B = C_B(L, M, \alpha)$ independent of size, mesh and mean. Let $\mathrm{Dom}_k = \{A \text{ real} : H_k^{\mathrm{ph}}A \in \frac{1}{4}\widetilde{R}_k\}$. For $p_{0,k} = (-\log e_k)^{p_0}$, $p_k = (-\log e_k)^p$, require

$$p_{0,k} \le e_k^{-\varepsilon/4}, \quad 2C_w^2 c_\gamma^2 (3L^{9/2}) e_k^{\varepsilon/4} \le 1, \quad 8C_B p_k \le \theta_3,$$
$$8e_k^{3/4-7\varepsilon/2} \le 1, \qquad L^{3/4} \ge 8/3.$$

Here $C_w e^{-\gamma_w d}$ bounds the weakened photon kernels, with fixed $C_w > 0, \gamma_w > 0$. The constant $c_\gamma > 0$ bounds both

$$\sup_{\Upsilon} \sum_{\Upsilon'} \exp[-\gamma_w d(\Upsilon, \Upsilon')], \qquad \sup_{b} \sum_{b'} \exp[-\gamma_w d(b, b')],$$

over coordinate supports and unit bonds, independently of $k, \ell$. Then $\|dA\|_\infty \le p_k$ implies $A \in \mathrm{Dom}_k$. Let $\mathrm{Box}_k = \{\zeta : |\zeta(\Upsilon)| \le p_{0,k} \text{ for every coordinate } \Upsilon\}$. Every $\zeta$ in this box gives

$$Z(s, \zeta) = H_k^{\mathrm{ph}}(s) C C_k^{1/2}(s)\zeta, \qquad \max_i \|Z\|_i \le \tfrac{1}{2} e_k^{-\varepsilon/2}.$$

For $A \in \widetilde{R}_k^-$, $A + tZ \in \widetilde{R}_k$ on $|t| \le \frac{1}{2} e_k^{-3/4+9\varepsilon/2}$. For $e_{k+1} = \sqrt{L} e_k$, the exact photon block preserves Dom under $A_k^{\min}(A') + \mathrm{Sq}_k\zeta$.

Proof work. Count path parameters and absorb their polynomial growth by half the exponential decay; differentiate the row smoothing explicitly. Use $|X_\omega|_M, d_M(X_\omega) \le 27(n+1)$ in the two weighted walk sums. Prove component restriction and the stronger row-vanishing margin directly. For the local photon gauge use path sums, bound seam bonds by $O(\ell^2\|dA\|)$, and absorb this by the remote kernel decay. Sum the two photon kernels through the coordinate map $C$; scale the outgoing real local representative and retain the one-quarter domain margin.

Inputs. Walk sums; photon kernels, axial coordinates and root decay; polymer geometry. Exports: analytic weakening with component locality and fluctuation domain margins. A weakened root is not asserted positive or a square root at general complex parameter values.

Source. PDF pp. 97–101, §§10.6–10.8, Propositions 10.2–10.4, (10.30)–(10.44); pp. 102–104, §11.1, Proposition 11.1, (11.1)–(11.7). TeX 8537–9009, 9091–9248.

## S06. Fine-to-coarse resolvents and local fermion maps

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S02, S05

Objects and hypotheses. At mesh $\eta = L^{-k}$, $k \geq 1$, average the 48 chamber labels by $J^*$, then the nested covariant unit blocks by $Q_k(A)$. Write

$$\mathcal{Q} = Q_k(A)J^*, \quad \mathcal{Q}^\sharp = JQ_k(-A)^T, \quad P_L = Q(-A)^T Q(A), \qquad \mathcal{Q}\mathcal{Q}^\sharp = 1.$$

The maps use normalized scalar phases, so $\mathcal{Q}^\sharp\mathcal{Q} = \Pi_1$, $\mathcal{Q}^\sharp P_L \mathcal{Q} = \Pi_L$; the symbol $\sharp$ is not a Hermitian adjoint for complex $A$. Take $A$ in the inverse domain: $A = A_0 + A_1$, $A_0$ real, $\|A_1\|_\infty < e_k^{-1+\varepsilon}$, and on every $11M$ cube $A_0 = B_0 + d\lambda$ for real $B_0, \lambda$ with $\|B_0\|_\infty < e_k^{-1+\varepsilon}$. The total connections from the activity domain are used only under the explicit inclusion recorded in S09. Use $\mathsf{C} = i\gamma_2$ in the Hermitian Euclidean representation; thus $\bar{H}_k(A) = \mathsf{C}^{-1}H_k(-A)\mathsf{C}$. Let $S_k = (D_A + \mu + b_k\Pi_1)^{-1}$, $0 < \mu \leq 1$, and put

$$H_k = b_k S_k \mathcal{Q}^\sharp, \quad \mathfrak{H}_k = b_k \mathcal{Q} S_k, \quad D_k = b_k - b_k^2 \mathcal{Q} S_k \mathcal{Q}^\sharp, \quad M_k = b_k - D_k.$$

The fine inverse $S_y$ exists with the penalized resolvent parameters of (9.23); set

$$B_{k,y} = (b_k + i\gamma_5 y)^{-1}(1 - P_L) + (b_k + b_L + i\gamma_5 y)^{-1}P_L.$$

Claim. For real $y$, the coarse operator is invertible and

$$(D_k + b_L P_L + i\gamma_5 y)^{-1} = B_{k,y} + b_k^2 B_{k,y}\mathcal{Q}S_y\mathcal{Q}^\sharp B_{k,y} =: \Gamma_{k,y},$$
$$S_y = S_k + H_k \Gamma_{k,y} \mathfrak{H}_k.$$

This algebra proves coarse invertibility from the fine inverses. With weakened right/left/symmetrized fine families, define

$$H_k(s) = b_k S_\infty(s)\mathcal{Q}^\sharp, \quad \Gamma_k(s) = B_{k,0} + b_k^2 B_{k,0}\mathcal{Q}S_0^{\mathrm{sy}}(s)\mathcal{Q}^\sharp B_{k,0}, \quad M_k(X) = b_k^2 \mathcal{Q} S_\infty^{\mathrm{sy}}(X)\mathcal{Q}^\sharp.$$

For these bounds $a', \gamma > 0$ are S02's defined zero-row constants, with $\gamma L \leq 1/6$; the fine input is exactly the weighted walk and polymer inequalities of S05 at the fixed positive rate $\kappa'$. All sup norms use Euclidean spinor norms; normalized chamber kernels use column measure $\eta^3/48$. Specifically $H_k$ and its short covariant quotient satisfy

$$\|H_k(s)\|_{\infty\to\infty} \leq 32\exp(1) b_k a' M^{27\alpha_0}, \qquad \|\delta_{\alpha,A} H_k(s)\|_{\infty\to\infty} \leq 304\exp(1) b_k a' M^{27\alpha_0}.$$

The quotient is $(U_A(x,x')v(x') - v(x))/\max(d(c,c'),\eta)^\alpha$ on $0 < \max(d(c,c'),\eta) < 1$, with the six-path phase and transport one within a cluster. The separate outputs for $\Gamma_k$ and $M_k$ are

$$|\Gamma_k(s)(z,z')| \leq C_\Gamma M^{27\alpha_0} e^{-\gamma d(z,z')},$$
$$|M_k(X)(z,z')| \leq 32\exp(1) b_k^2 a' e^{27\kappa'} e^{-\gamma d(z,z') - \kappa' d_M(X)},$$
$$C_\Gamma = 2(1 + 2\exp(1))/b_- + 128\exp(1)(1 + 2\exp(1))^2 a'.$$

The assumed fine-family analyticity, component restriction and local read-dependence pass to these families. At $k = 0$ use instead $\Gamma_0 = (D_A + \mu + b_L J P_L J^*)^{-1}$ on all 192 chamber components. Its weakened normalized kernel has the same decay; $H_0(s)$ is the product of parameters of cubes meeting $K_c = \{|c' - c|_\infty \leq 2L\}$, bounded by $M^{8\alpha_0}$. At one it is the identity. There is no $M_0$.

Proof work. Use both resolvent identities for $Z = b_k^2 \mathcal{Q}^\sharp B_{k,y}\mathcal{Q}$ and multiply the proposed inverse on both sides. Bound the averaging phases by their nested path lengths. Each $B$ has rows and columns in one $L$-block; compose their row sums and retain the distance shift $2L$. Nested blocks and the fine component restriction give locality. Transposition exchanges right and left walk expansions, explaining the symmetrized covariance.

Inputs. Exact and weakened fine inverse estimates, including their support and component statements; normalized gauge averages. Exports: coarse resolvent, field maps, local Dirac kernels. $\mathfrak{H}_k$ is the right resolvent map; the barred field map is separately $\bar{H}_k(A) = \mathsf{C}^{-1}H_k(-A)\mathsf{C}$.

Source. PDF pp. 104–107, §§11.2–11.3, Propositions 11.2–11.3, (11.8)–(11.19). TeX 9249–9513.

## S07. Weighted Berezin integration and exact block completion

INTERFACE. Overview Target.

Objects and hypotheses. Let $I$ be a finite ordered mode set with positive weights $\nu(t)$, and independent generators $W(t), \bar{W}(t)$. An operator kernel acts using the column weight $\nu(t')$; its quadratic form is $\sum_{t,t'} \nu(t)\nu(t')\bar{W}(t)O(t,t')W(t')$. Define $\mathrm{D}W$ as extraction of the coefficient of $\prod_t W(t)\bar{W}(t)$, multiplied by $\prod_t \nu(t)^{-1}$. For invertible $C$, put

$$\mathrm{d}\mu_C(W) = (\det C)\,\mathrm{D}W \exp[-\langle \bar{W}, C^{-1}W\rangle_\nu].$$

For any finite families of Grassmann generators give an ordered monomial the product of its generator weights, and a polynomial the sum of the moduli of its coefficients times those weights. This is equivalently the absolute antisymmetric-kernel norm with the index measures and the factors $1/n!$. Claim. The norm is submultiplicative. With $w_W = \max_t \nu(t)^{-1/2}$,

$$\left\| \int \mathrm{d}\mu_I(W) F \right\|_w \le \|F\|_{w,w_W},$$

$$\int \mathrm{d}\mu_C(W) F(\bar{W}, W) = \int \mathrm{d}\mu_I(W) F(\bar{W}, CW),$$

$$\int \mathrm{d}\mu_C(W) W(t)\bar{W}(t') = C(t,t'), \qquad \int \mathrm{D}W\, e^{-\langle \bar{W}, OW\rangle_\nu} = \det O.$$

The last formula does not require $O$ invertible. Disjoint mode sets factor, with external generators retained as coefficients. For linear odd fields $v_1, \dots, v_{2r}$, prove the Pfaffian Wick formula; odd moments vanish. If a linear substitution sends family $\varphi_i$ to $\sum_j O_{ij}\theta_j$, its norm is at most one whenever

$$\sum_j \left( \sup_t \sum_s |O_{ij}(t,s)| \nu'_j(s) \right) w'_j \le w_i$$

for each occurring input row.

For two finite spinor spaces with their specified pairings, let $K = D + b_L A^\sharp A$ be invertible, $\Gamma = K^{-1}$, and set

$$\Psi^c = b_L \Gamma A^\sharp \Phi, \quad \bar{\Psi}^c = b_L \Gamma^T A^T \bar{\Phi}, \quad D' = b_L - b_L^2 A \Gamma A^\sharp.$$

Put $s = b_L A^\sharp \Phi$, $\bar{s} = b_L A^T \bar{\Phi}$. Then completion of the block quadratic form gives

$$\int \mathrm{D}\Psi\, \exp\{-\langle \bar{\Psi}, K\Psi\rangle + \langle \bar{\Psi}, s\rangle + \langle \bar{s}, \Psi\rangle - b_L\langle \bar{\Phi}, \Phi\rangle\} F(\bar{\Psi}, \Psi)$$

$$= \det K\, e^{-\langle \bar{\Phi}, D'\Phi\rangle} \int \mathrm{d}\mu_I(W) F(\bar{\Psi}^c + \bar{W}, \Psi^c + \Gamma W).$$

Here $A^\sharp$ is the specified algebraic reverse map. In the chamber application it is the transpose of the averaging map at the reversed connection. No Hermitian conjugation of Grassmann fields is used.
Proof work. Compute each paired monomial in the identity Gaussian. Its modulus is exactly covered by $w_W$. A linear change of the unbarred variables multiplies the top coefficient by the operator determinant. Left Grassmann differentiation and integration by parts fix the Wick signs. Bound images of generators for the substitution estimate. Translate by the displayed means, keeping odd factors in their original order, and cancel both cross terms.
Inputs. Finite exterior algebra and determinants only. Exports: weighted moments, substitutions, exact Gaussian block identity. In the application $\nu = 1/48$ at level zero, so $w_W = \sqrt{48}$ and the identity two-point kernel is $48\delta_{tt'}$. At positive levels $\nu = 1$.
Source. PDF pp. 107–110, §§11.4–11.5, Proposition 11.4, (11.20)–(11.29). TeX 9514–9754.

## S08. A local determinant logarithm and its small-field gain

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M20, S05, S06, S07

Objects and hypotheses. On a torus partitioned into $M$-cubes, a polymer $X$ is face-connected and $Md_M(X)$ is its minimal represented-net length. Assume $|X|_M \le 54(d_M(X)+1)$ and the nonwrapping lift/box criteria. Put $\eta = L^{-k}$, $0<\alpha<1$, $b=1$, $b_L = b/L$, $b_k = b(1-L^{-1})/(1-L^{-k})$ for $k \ge 1$. Let $C_w, \gamma_w, \kappa, \kappa_0$ be fixed positive constants. The fine-block norm below is the operator norm for the supremum of the Euclidean spinor norm. Normalized fine chamber kernels use column measure $\eta^3/48$. At positive levels the unit-spinor operators satisfy

$$\delta Z_k(A) = \det(D_k + b_L P_L), \quad B_{k,y} = (b_k + i\gamma_5 y)^{-1}(1-P_L) + (b_k + b_L + i\gamma_5 y)^{-1} P_L.$$

Here $P_L$ is a covariant block projection, $\mathcal{Q}\mathcal{Q}^\sharp = 1$, and the fine polymer inverses obey

$$\|\mathbf{1}_{\Delta_z} S_{k,y}(X,A)\mathbf{1}_{\Delta_{z'}}\| \le C_w e^{-\gamma_w d(z,z') - \kappa_w d_M(X)}, \qquad \kappa_w = \kappa + \kappa_0 + 3.$$

At zero use $\delta Z_0 = \det F_0$ on all chamber modes, with measure $1/48$, and the normalized kernel bound has the further factor $(1+y)^{-1}$. Assume analyticity, gauge and step covariance, the transpose/reality identities, $0<\mu\le 1$, and the proved inverse domain consisting of locally real-gauge-small fields.

Claim. Define

$$\widetilde{E}_k(X,A) = -ib_k^2 \int_0^\infty \mathrm{Tr}[\gamma_5 B_{k,y} \mathcal{Q} S_{k,y}(X,A) \mathcal{Q}^\sharp B_{k,y}]\,\mathrm{d}y \qquad (k \ge 1),$$

$$\widetilde{E}_0(X,A) = \int_1^\infty \mathrm{Tr}[F_0 S_{0,y}(X,A)]\,\frac{\mathrm{d}y}{y} - i\int_0^1 \mathrm{Tr}[\gamma_5 S_{0,y}(X,A)]\,\mathrm{d}y.$$

The chamber trace includes its $1/48$ measure. Let $E_k^d$ be the average of $\widetilde{E}_k(A)$, $\widetilde{E}_k(-A)$, $\overline{\widetilde{E}_k(\bar{A})}$, $\overline{\widetilde{E}_k(-\bar{A})}$; set $E_k^{\det}(X,A) = E_k^d(X,A) - E_k^d(X,0)$. These are absolutely convergent, local analytic functions and

$$|\widetilde{E}_k(X,A)|,\ |E_k^d(X,A)| \le C_d M^3 e^{-(\kappa+1)d_M(X)},$$

$$\delta Z_k(A) = \exp\{c_k^d \ell^3 + \sum_X \widetilde{E}_k(X,A)\}, \qquad \frac{\delta Z_k(A)}{\delta Z_k(0)} = \exp\{\sum_X E_k^{\det}(X,A)\}.$$

Here $c_0^d = 0$, $c_k^d = 4[(1-L^{-3})\log b_k + L^{-3}\log(b_k+b_L)]$, and

$$C_d = 16(54)C_w \max\{\tfrac{\pi}{2} b \exp(1)(1+2\exp(1)), 2 + 58\exp(1) + b\exp(1)\}.$$

Each $E_k^d$ reads only bonds meeting $X$, is real-gauge invariant and even, and satisfies the analytic reality identity.

For the small-field assertion define the norms explicitly by

$$\|B\|_1 = \sup_{c,\mu} |B_\mu(c)|, \qquad \|B\|_2 = \sup_{c,\mu,\nu} |\partial_\nu B_\mu(c)|,$$

$$\|B\|_3 = \sup_{\mu,\nu} \sup_{0<d(c,c')\le 1} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c,c')^\alpha},$$

$$\partial_\nu B_\mu(c) = \eta^{-1}[B_\mu(c+\eta e_\nu) - B_\mu(c)].$$

The distance is periodic sup distance; local norms use only bonds and difference cells inside their stated set. For $A$ with local-real-gauge plus global complex bounds $(e_k^{-3/4+\varepsilon}, e_k^{-3/4+2\varepsilon}, e_k^{-3/4+3\varepsilon})$ on its three norms, and $\|C\|_\infty < e_k^{-1+2\varepsilon}$, $\|\partial C\|_\infty < e_k^{-1+3\varepsilon}$, prove

$$|E_k^d(X, A+C) - E_k^d(X,C)| \le 8 C_d M^3 e_k^{1/4} e^{-\kappa d_M(X)}.$$

Require $v = -\log e_k \ge 9$, $e_k^\varepsilon \le 1/4$, $8M(v/4+4)e_k^\varepsilon \le 1$, $\ell \ge Mv^2$, and $e_k^{1/4} + e_k^\varepsilon < 1$.

Proof work. At real $A$, use the real eigenvalues of the Hermitian matrix $K\gamma_5$ to integrate scalar logarithms; retain the ranks of $P_L, 1-P_L$ and the integral phases. Two $B$ factors give $(b_k^2+y^2)^{-1}$. At zero keep the split integration intervals and the $(1+y)^{-1}$ factor. Analytic continuation and continuous $2\pi i\mathbb{Z}$-valued differences justify the symmetrized quotient without a global logarithm branch. For the gain, large polymers pay $e_k^{1/4}$ from their decay. Lift the remaining polymers into a nonwrapping product box, subtract a real path-sum gauge, extend the sup-small local field by zero and apply Cauchy's difference formula on a disk of radius $\frac{1}{3}e_k^{-1/4}$, with the external slot fixed.
Inputs. Local fine/coarse resolvent kernels; weighted trace conventions; polymer lifts and geometry.
Exports: local determinant primitives and their $e_k^{1/4}$ difference gain.
Source. PDF pp. 110–113, §§11.6–11.8, Propositions 11.5–11.6, (11.30)–(11.38). TeX 9755–10116.

## S09. Scalar extraction and exact quadratic normalization

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M01, M20, S07

Objects and hypotheses. Define the activity space $\mathcal{A}_k^{x,\mathbb{C}}$, with real subspace $\mathcal{A}_k^x$, used explicitly in S11, S13, S15 and S18. Fix odd $L \geq 5$, $\eta = L^{-k}$, $M = L^m \geq L^2$, $\ell/(LM)$ odd and at least 13, $0 < \varepsilon \leq 1/200$, $0 < \alpha < 1$, and $0 < e_k < 1$. An $M$-polymer is small if $d_M(X) \leq L$, with an injective covering lift. Put $\mathrm{Vol}(X) = M^3|X|_M$. An activity $E(X; A, C)$ is a balanced polynomial in independent ordinary and pair Grassmann legs, with holomorphic coefficients reading only bonds meeting $X$. Ordinary legs are over $X$; an allowed pair has all six coordinate-order paths in $X$, or is an external nearest-neighbor pair with an endpoint over $X$. Its separation is $d_X = \max(|c - c'|_\infty, \eta)$, and the pair range is $0 < d_X \leq M$; intracluster pairs are included. Index measures are $\eta^3/48$ and $(\eta^3/48)^2$, while generator weights are $h_1 = e_k^{-1/4}$, $h_2 = e_k^{-1/4+\varepsilon}$. Ordinary kernels use the antiperiodic convention in each chamber argument; pair indices are simultaneous period classes of their two endpoints. The coefficient norm of the separately antisymmetric kernels is

$$\|E(X; A, C)\|_{h_k} = \sum_{n,m} \frac{h_1^n h_2^m}{n!m!} \int |E_{nm}(X; A, C; \xi; \zeta)| \, \mathrm{d}\xi \, \mathrm{d}\zeta.$$

To specify its domains, put

$$\begin{aligned} \partial_\nu B_\mu(c) &= \eta^{-1}[B_\mu(c + \eta e_\nu) - B_\mu(c)], \\ \|B\|_1 &= \sup |B_\mu(c)|, \quad \|B\|_2 = \sup |\partial_\nu B_\mu(c)|, \\ \|B\|_3 &= \sup_{\mu,\nu} \sup_{0<d(c,c')\leq 1} \frac{|\partial_\nu B_\mu(c') - \partial_\nu B_\mu(c)|}{d(c, c')^\alpha}, \qquad \theta_i = e_k^{-3/4+i\varepsilon}. \end{aligned}$$

The distance is periodic sup distance; a local norm uses only its contained bonds and difference cells. $R_k = \{B : \|B\|_i < \theta_i,\ i = 1, 2, 3\}$. A field belongs to $\widetilde{R}_k$ if it is $A_0 + A_1$, $A_0$ real, $A_1 \in R_k$, and on every enlarged cube of side $(2L + 1)M$, $A_0 = A_\square + d\lambda_\square$ with real $A_\square, \lambda_\square$ and $\|A_\square\|_{i,\square^\dagger} < \theta_i$. The external domain is

$$\mathfrak{X}_k = \{C : \|C\|_1 < e_k^{-1+2\varepsilon},\ \|C\|_2 < e_k^{-1+3\varepsilon}\}.$$

The activity norm is precisely

$$\|E\|_k = \sup_X \exp(\kappa d_M(X)) \sup_{(A,C)\in\widetilde{R}_k\times\mathfrak{X}_k} \|E(X; A, C)\|_{h_k}.$$

Assume the polymer counting bound $\sum_{X\supset\square} e^{-\kappa_0 d_M(X)} \leq K_0$, $\kappa \geq \kappa_0$.

Here is the complete symmetry package. Use $\gamma_\mu = \sigma_3 \otimes \sigma_\mu$, $\gamma_5 = \sigma_1 \otimes 1_2$, $\mathsf{C} = i\gamma_2$, and $\varkappa = \gamma_5\mathsf{C}$. For a spin lift $S$ generated by the four $\gamma$-matrices and a translation $a \in M\mathbb{Z}^3$, let $\widetilde{g}(c, g) = (r(S)c + a, r(S)g)$, where $S\gamma_\mu S^{-1} = r(S)_{\mu\mu}\gamma_\mu$. Define

$$a_{\widetilde{g}}\psi(x) = S\psi(\widetilde{g}^{-1}x), \qquad a_{\widetilde{g}}\bar{\psi}(x) = (S^{-1})^T\bar{\psi}(\widetilde{g}^{-1}x).$$

For pair generators transform both endpoints and use the same spin matrices. $G_\lambda$ multiplies an unbarred ordinary or pair leg by $\exp(ie_k\lambda(c))$, with $c$ its first photon point, and a barred leg by its inverse. Define on ordinary and pair legs

$$\mathrm{Cg}\,\psi = \mathsf{C}\bar{\psi}, \quad \mathrm{Cg}\,\bar{\psi} = -\mathsf{C}\psi, \qquad \mathbf{K}\psi = \varkappa\psi, \quad \mathbf{K}\bar{\psi} = \varkappa\bar{\psi},$$

with $\mathbf{K}$ antilinear. The complex activity class requires

$$\begin{aligned} a_{\widetilde{g}}E(\widetilde{g}X; \widetilde{g}A, \widetilde{g}C) &= E(X; A, C), \\ \mathrm{Cg}\,E(X; -A, -C) &= E(X; A, C), \\ G_{\lambda+\lambda'}E(X; A^1, C^1) &= E(X; A, C) \end{aligned}$$

whenever the two pairs are in the domain and $A^1 = A - d\lambda,\ C^1 = C - d\lambda'$ on bonds at distance at most one from $X$, for real $\lambda, \lambda'$. Its real subspace additionally requires

$$\mathbf{K}E(X; -\overline{A}, -\overline{C}) = E(X; A, C).$$

All equations act on independent generators before evaluation. For the total-connection use in S06 and later cards, the explicit conditions

$$e_k^{\varepsilon} \le \tfrac{1}{4}, \qquad e_k^{1/4} + e_k^{\varepsilon} \le 1, \qquad 4e_k^{1/4+2\varepsilon} + 2e_k^{3\varepsilon} \le 1$$

give $\widetilde{R}_k + \mathfrak{X}_k \subset \widetilde{G}_k$, where $\widetilde{G}_k$ has a global complex bound $e_k^{-1+\varepsilon}$ and local real-gauge sup bounds $e_k^{-1+\varepsilon}$ on every $11M$ cube. They also give $e_k \| \operatorname{Im}(A + C)\|_\infty \le 1$ and $e_k \|d(A + C)\|_\infty \le 1$.

Claim. Write $E^{[2,0]} = \int \bar{\psi}(x)^T K_X(x, x')\psi(x')\,\mathrm{d}x\,\mathrm{d}x'$. On small $X$, define

$$a_0 = \frac{E_{00}(X; 0, 0)}{\operatorname{Vol}(X)}, \qquad a_2 = \frac{1}{\operatorname{Vol}(X)} \int_{\widetilde{X}\times\widetilde{X}} K_X(0, 0; x, x').$$

The sums $\varepsilon_v(E) = -\sum_{X \supset \square,\ \text{small}} a_0$ and $M(E) = -\sum a_2$ are independent of the cube. In fact $M(E) = m(E)1_4$, with complex-linear maps real on real activities, and

$$|\varepsilon_v(E)| \le K_0 M^{-3} \|E\|_k, \qquad |m(E)| \le K_0 M^{-3} e_k^{1/2} \|E\|_k.$$

Subtract $a_0 \operatorname{Vol}(X)$ and $\int_X \bar{\psi}^T a_2 \psi$ only on small polymers, to obtain $RE$. Then $\|RE\|_k \le 2\|E\|_k$ and

$$\sum_X E(X) = -\varepsilon_v(E)\ell^3 - m(E)\langle\bar{\psi}, \psi\rangle + \sum_X RE(X).$$

There is an activity adjustment, preserving evaluated values, which makes the entire ordinary quadratic kernel vanish at $(A, C) = (0, 0)$ on each small polymer. Construct it by averaging over shortest face-chains between the cubes of $x, \widetilde{x}$, retaining the initial chamber label at intermediate cube centres, and telescoping

$$\psi(p_i) = U_{A+C}(p_i, p_{i+1})\psi(p_{i+1}) - d_X(p_i, p_{i+1})^{\alpha}\chi(p_i, p_{i+1}).$$

Average over $\widetilde{x}$ with normalized measure $\operatorname{Vol}(X)^{-1}\mathrm{d}\widetilde{x}$; use inverse transports for barred legs. This defines a homomorphism $\Theta$. Set $E^{\mathrm{nat}} = E - E^{[2,0]} + \Theta E^{[2,0]}$. If $90(54)LMe_k^{\varepsilon} \le 1$ and $e_k^{\varepsilon} \le 1/4$, then its pointwise norm is at most $10\|E(X)\|$. For $(RE)^{\mathrm{nat}}$, the vacuum and full ordinary quadratic kernel are both zero at zero connection on small $X$. All these maps preserve the activity symmetries and commute with setting the external slot to zero.

Proof work. Derive matrix covariance from unique antisymmetric kernels. The summed extraction commutes with all four matrices $\gamma_1, \gamma_2, \gamma_3, \gamma_5$, whose commutant is scalar. Use the quadratic weight $h_1^{-2} = e_k^{1/2}$ and polymer counting. Reorder finite sums for the exact extraction identity. For adjustment keep the chain length and phase products: the path length and sum of $d_X^{\alpha}$ are at most $3(54)LM$. Telescoping proves equality only after the prescribed field evaluation. At zero connection the remaining ordinary part is a common spatial mean, annihilated by the zero integrated kernel. Use $h_2/h_1 = e_k^{\varepsilon}$ for the norm bound.

Inputs. Weighted Grassmann substitutions; explicit activity symmetries; polymer counting/lifts. Exports: scalar counterterms and a zero ordinary quadratic kernel. Individual polymer matrices $a_2$ need not be scalar.

Source. PDF pp. 114–118, 122–124, §§12.1–12.3,12.5, Proposition 12.1, (12.1)–(12.17), (12.34)–(12.39). TeX 10142–10565, 10905–11070.

## S10. A bounded left inverse using interior and boundary legs

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S01, S02, S05, S06, S07

Objects and hypotheses. At positive level $k$, let $\eta = L^{-k}$, $0 < \mu \le 1$, $b_k \ge b_- > 0$. On chamber spinors with measure $\eta^3/48$, $\mathcal{Q} = Q_k(A)J^*$ is the normalized nested block average, $\mathcal{Q}^\sharp = JQ_k(-A)^T$, $\mathcal{Q}\mathcal{Q}^\sharp = 1$. The chamber stencil is $D_A = -\eta^{-1}\sum_i M_i(x)(U_A R_i - 1)$, with three internal and one external neighbor, $M_i = \frac{1}{2}w_i(\rho - \gamma\cdot\widehat{d}_i)$. Its geometric cancellation is $\sum_i w_i\widehat{d}_i = 0$; the external matrices have norm at most 6. Let

$$F_\infty = D_A + \mu + b_k\mathcal{Q}^\sharp\mathcal{Q}, \quad H_k = b_k F_\infty^{-1}\mathcal{Q}^\sharp, \quad T_k = \mathcal{Q}(b_k^{-1}D_A + 1 + \mu/b_k).$$

The inverse is assumed only where $T_k H_k = 1$ is used. For a union $Y$ of partition $M$-cubes, split the row of $T_k$ at a unit site $y \in Y$ into $T_Y^{(1)}$, retaining ordinary differences whose neighbor stays in $Y$, and

$$(T_Y^{(2)}\chi)(y) = -b_k^{-1}\eta^{\alpha-1}\int_{\Delta_y} \mathbf{1}[c_3 \notin Y]\phi_A(y,c)M_3(x)\chi(x,x^{(3)})\,\mathrm{d}x.$$

The nested phase has modulus at most $\exp(3\mathfrak{i}/2)$, where $\mathfrak{i} = e_k\|\operatorname{Im} A\|_\infty$; put $q_d = e_k\|dA\|_\infty$.

Claim. $T_k H_k = 1$. On evaluated boundary legs $\chi = \delta_{\alpha,A}\psi$, $T_Y^{(1)}\psi + T_Y^{(2)}\chi = T_k\psi$ on $Y$. Only ordinary legs over $Y$ and external boundary pairs occur; coefficients read bonds meeting $Y$. Uniformly in the mesh and $Y$,

$$\|T_Y^{(1)}\|, \|T_k\| \le e^{3\mathfrak{i}/2}[1 + b_-^{-1}(54Lq_d e^{6q_d} + 12 + 6e^{\mathfrak{i}} + \mu)],$$
$$\|T_Y^{(2)}\| \le 6b_-^{-1}e^{3\mathfrak{i}/2}\eta^\alpha.$$

These are Euclidean spinor sup norms; the scalar coefficient row sum is at most twice this bound. If $\mathfrak{i}, q_d \le 1$, all ordinary/barred rows are bounded by

$$C_T = 2\exp(3/2)[1 + b_-^{-1}(54L\exp(6) + 12\exp(1) + 1)].$$

They have the required lattice, local real-gauge, charge and reality covariance. The substitution $\Psi \mapsto T_Y^{(1)}\psi + T_Y^{(2)}\chi$, with its barred counterpart, preserves support and decreases coefficient norm if the unit-generator weight $w_3$ satisfies $C_T(w_1 + w_2) \le w_3$. On evaluating $\psi = H_k\Phi$ and its covariant pair field, the substitution returns $\Phi$ on $Y$. At zero level these maps are identity on all chamber components.

Proof work. The left inverse is algebraic, but its uniform row estimate requires the cancellation of the apparent $\eta^{-1}$ loss. Reindex internal hops and internal external hops, using $M_i(x^{(i)}) - M_i(x) = w_i\gamma\cdot\widehat{d}_i(x)$. Boundary site counts are $48\eta^{-2}$, compared with $48\eta^{-3}$ total sites in a unit block. For the remaining nested-phase difference, classify a site by the first common block scale $j$. There are at most $\eta^{-3}L^{-(j-1)}$ sites, while the closed-path phase error is at most $6L^{2(j-k)}q_d\exp(6q_d)$. Sum these factors before taking a bound. The boundary Hölder row gains $\eta^\alpha$ from its face count. Verify the homomorphism and covariance on each independent generator.

Inputs. Exact field map and nested gauge averages; chamber cancellation; weighted substitution theorem. Exports: support-preserving reconstruction and its explicit uniform row constant. The activity keeps pair generators independent until evaluation.

Source. PDF pp. 118–122, 139–140, §§12.4,13.4, Proposition 12.2, (12.18)–(12.33), (13.23)–(13.25). TeX 10566–10904, 12768–12866.

## S11. Scaling and contraction of a normalized local activity

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M20, S07, S09

Objects and hypotheses. The activity input is exactly $\mathcal{A}_k^{x,\mathbb{C}}$ defined in S09, with its complete kernel support, measures, analytic domains and three covariance equations; for reality use the additional antilinear equation defining $\mathcal{A}_k^x$. The local description below uses this fixed package. Let $L$ be odd, $M = L^m \geq L^2$, $\eta = L^{-k}$, and both $\ell, \ell/L$ satisfy the admissible side condition $\ell/(LM)$ odd and at least 13. A polymer is small when $d_M(X) \leq L$. Assume its nonwrapping lift, size/counting bounds with constants $c_X = 54, \kappa_0, K_0$, and

$$\sum_{\overline{X}^L = LY} e^{-\kappa' d_M(X)} \leq c_{\mathrm{rb}} L^3 e^{-L(\kappa' - \kappa_0 - 1) d_M(Y)}, \quad c_{\mathrm{rb}} = c_X K_0, \quad \kappa' \geq \kappa_0.$$

An activity has ordinary legs over $X$ and pair legs either with all six rectilinear paths in $X$, or external boundary pairs with an endpoint there. Weights are $h_1 = e_k^{-1/4}, h_2 = e_k^{-1/4+\varepsilon}$, and its norm is the supremum of the weighted coefficient norm multiplied by $e^{\kappa d_M(X)}$. Assume balance, holomorphy, local read dependence, independent local real gauges of its two slots, and charge conjugation, so the vacuum is even under $(A, C) \mapsto (-A, -C)$. For the stated preservation conclusions also retain the lattice and reality actions of the activity. The first slot has three thresholds $\theta_i = e_k^{-3/4+i\varepsilon}$, allowing local real gauges plus a global complex part; the external slot has thresholds $d_1 = e_k^{-1+2\varepsilon}$, $d_2 = e_k^{-1+3\varepsilon}$. Use

$$0 < \varepsilon \leq \bar{\varepsilon} = 1/200, \quad 1 - \varepsilon < \alpha < 1, \quad L^{1/8 - 3\bar{\varepsilon}/2} \geq 2400, \quad e_k^{\varepsilon} \leq 1/4,$$
$$160 L^{1+3\varepsilon/2} M e_k^{\varepsilon} \leq 1, \quad 9 M L^{\varepsilon} e_k^{2\varepsilon} \leq 1, \quad 90 c_X L M e_k^{\varepsilon} \leq 1,$$
$$\kappa \geq 7\kappa_0 + 7, \qquad L(\kappa - 6\kappa_0 - 4) - 1 \geq \kappa.$$

For an input $E$, let $E_{\mathrm{n}} = (RE)^{\mathrm{nat}}$: its small-polymer vacuum and ordinary quadratic kernel are zero at zero connection; pointwise it has norm at most 20 times the original norm, unchanged on large polymers.

Claim. Set $e_{k+1} = \sqrt{L} e_k$, $A_L = L^{-1/2} A(\cdot/L)$, and

$$S_L \psi(x) = L^{-1} \psi'(x/L), \quad S_L \chi(x, x') = L^{-1-\alpha} \chi'(x/L, x'/L), \quad (\mathcal{L}F)(Y) = \sum_{\overline{X}^L = LY} S_L F(X; A_L, C_L).$$

Here $\overline{X}^L$ is the union of $LM$-cubes containing $X$. On bidegree $(n, m)$,

$$\|(S_L F)^{[n,m]}\|_{h_{k+1}} = L^{-p(n,m)} \|F^{[n,m]}\|_{h_k}, \quad p(n,m) = n + (1+\alpha)m + (n+m)/8 - m\varepsilon/2.$$

The scaled pair range is $M/L$, including intracluster pairs. The linear normalized map preserves the activity symmetries and the zero-external-slot restriction, and

$$\|\mathcal{L} E_{\mathrm{n}}\|_{k+1} \leq c_{\mathrm{fl}} L^{-1/4+3\varepsilon/2} \|E\|_k, \qquad c_{\mathrm{fl}} = 61 c_{\mathrm{rb}}.$$

Relative to $w_k^E = e_k^{1/4-7\varepsilon}$, its factor is $c_{\mathrm{fl}} L^{-3/8+11\varepsilon/2}$; it is at most $1/16$ if $c_{\mathrm{fl}} L^{-3/8+6\bar{\varepsilon}} \leq 1/16$.

Proof work. First prove the necessary local gauge preparation on a small polymer, rather than assuming an analytic disk. On every read bond write

$$A_L - d\lambda_* = A^b + K_1, \qquad C_L = C^b + K_2$$

with $\lambda_*$ real and $K_i$ constant. The global $A^b$ threshold ratios are at most $(64, 137, 1382) L^{-15/8+3\varepsilon/2}$; both $C^b$ ratios are less than $L^{-7/4}$. Thus $(tA^b, tC^b)$ lies in the analytic domain for $|t| \leq L^{7/4}$. Use two cutoffs: radius $2LM$ for the Taylor remainder and $\min(e_k^{-\varepsilon}, \ell/5)$ for the linear part. Remove only the real symmetric linear part by a real quadratic gauge. Keep the bound $2\min(b_2, \ell b_3)$ on the other linear coefficients. Continue real constant-gauge identities holomorphically to complex

constants; the kernel comparison costs at most $\exp(2(n+m))$. Cauchy then gives $2L^{-7/2}N(X)$ for the vacuum and $2\exp(4)L^{-4}N(X)$ for the ordinary quadratic term. Other degrees gain $L^{-13/4+3\varepsilon/2}$. Reblock with the displayed geometry bound; large polymers supply the extra $\exp(-L)$. Finally prove that these formulas on covering generators commute with quotient maps under the margins (12.70),(12.74).

Inputs. Scalar extraction and quadratic normalization; weighted algebra; polymer reblocking/lifting. Exports: contraction with explicit relative factor, scaled domains and lifted local formulas. Integrated quadratic cancellation alone does not give the quadratic Cauchy gain.

Source. PDF pp. 124–133, §§12.6–12.9, Propositions 12.3–12.5, (12.40)–(12.74). TeX 11071–12017.

## S12. The exact density identity and the first chamber current

Model-specific. Overview Target.

Proof dependencies. S04, S05, S06, S07, S09

Objects and hypotheses. Incoming unit spinors have four components and counting measure at $k \geq 1$; at $k = 0$ there are 192 components per photon site and measure $1/48$. The outgoing $L$-spaced space has four components and measure $L^3$. Put $\mathcal{A}_k = Q(A)$ at positive levels and $\mathcal{A}_0 = Q(A)J^*$ at zero, with algebraic reverse $\mathcal{A}_k^\sharp = \mathcal{A}_k(-A)^T$. Let $\mathcal{D}_k = D_k$ for $k \geq 1$, $\mathcal{D}_0 = D_A + \mu$, and

$$K_k = \mathcal{D}_k + b_L \mathcal{A}_k^\sharp \mathcal{A}_k, \quad \Gamma_k = K_k^{-1}, \quad \delta Z_k = \det K_k.$$

Fix odd $L \geq 5$, $M = L^m \geq L^2$, and $\ell/(LM)$ odd and at least 13, with the outgoing side also admissible. The normalized density has exponent

$$-\langle \bar{\Psi}, \mathcal{D}_k(A + C)\Psi \rangle + I_k[m, E](A, C; \Psi) - \varepsilon_v n_k + \sum_X P_k(X, C), \qquad n_k = \ell^3,$$

multiplied by $\det \mathcal{D}_k(0)^{-1}$. Here $I_k$ evaluates $-m\langle \bar{\psi}, \psi \rangle + \sum E(X)$ at $\psi = H_k(A+C)\Psi$ and its barred and covariant pair fields. The density's first argument is a real coarse photon field reconstructed by $H_k^{\mathrm{ph}}$. Use the activity class $\mathcal{A}_k^{x,\mathbb{C}}$ defined by the complete domains and generator covariance equations in S09; the scalar data $m, \varepsilon_v$ may be complex in this exact identity. Assume the exact photon split $H_k^{\mathrm{ph}} A_k^{\min}(A') = (H_{k+1}^{\mathrm{ph}} A')_L - d\omega_k$, $\mathrm{Sq}_k = C C_k^{1/2}$, its box law, all domain inclusions, and the fine/coarse resolvent and field-map definitions. In particular $H_{k+1}^{\mathrm{ph}} A'$ has one-quarter local real-gauge bounds with thresholds $e_{k+1}^{-3/4+i\varepsilon}$, while $C'$ has thresholds $e_{k+1}^{-1+2\varepsilon}, e_{k+1}^{-1+3\varepsilon}$ on its sup and first derivative. The admitted box backgrounds $B, B + Z, B + Z - d\omega_k$ and their negatives lie in the proved open inverse domain. Use $0 < \mu \leq 1/L$, $e_{k+1} = \sqrt{L} e_k$, and zero input activity at $k = 0$.

Claim. For real $B$ and $\mu > 0$, prove invertibility of $D_B + \mu, D_k(B), K_k(B)$, with

$$D_k(B)^{-1} = b_k^{-1} + \mathcal{Q}_k(B)(D_B + \mu)^{-1} \mathcal{Q}_k^\sharp(B), \qquad \mathrm{Re}\, D_k^{-1} \geq b_k^{-1}.$$

Schur elimination and scaling identify the outgoing quadratic density, the induced mean field and

$$\det D_{k+1}(B') = b^{4n_{k+1}} \frac{\det \mathcal{D}_k(B'_L)}{\delta Z_k(B'_L)}.$$

Let $A = (H_{k+1}^{\mathrm{ph}} A')_L$, $C = C'_L$, $B = A + C$, $Z = H_k^{\mathrm{ph}} \mathrm{Sq}_k \zeta$, $m' = m + m(E)$, $E^r = RE$. With the block means $\Psi^c, \bar{\Psi}^c$, put $\Psi^f = \Psi^c + \Gamma_k(B)W$, $\bar{\Psi}^f = \bar{\Psi}^c + \bar{W}$. The transformed density is exactly

$$\begin{aligned} \det D_{k+1}(0)^{-1} \exp\{ &- \langle \bar{\Psi}', D_{k+1}(B')\Psi' \rangle - Lm' \langle \bar{\psi}', \psi' \rangle \\ &- L^3(\varepsilon_v + \varepsilon_v(E) + \delta\varepsilon_k^0) n_{k+1} + \sum_X E^r(X; A, C; \psi^{0,\#}) + \sum_X P_k(X, C) \} \\ &\cdot \frac{\delta Z_k(B)}{\delta Z_k(0)} \Xi_k, \end{aligned}$$

where $B' = H_{k+1}^{\mathrm{ph}} A' + C'$, $\psi^{0,\#}$ denotes the scaled outgoing fields, and

$$\begin{aligned} \Xi_k &= \int \mathrm{d}\mu_I(W)\, \mathrm{d}\hat{\mu}_k(\zeta)\, e^{V_k}, \\ V_k &= \mathrm{Act}[m', E^r](A + Z, C; \Psi^f) - \mathrm{Act}[m', E^r](A, C; \Psi^f) \\ &\quad + I[m', E^r](A, C; \Psi^f) - I[m', E^r](A, C; \Psi^c). \end{aligned}$$

Here Act includes the kinetic quadratic term. The photon law $\hat{\mu}_k$ is normalized on the coordinate box; its removed probability is exactly $\exp(-n_k \delta\varepsilon_k^0)$. The block transform has normalization $b^{-4n_{k+1}}$; its outgoing marginal is the unnormalized box integral of the incoming marginal.

First-step model verification. At $k=0$, $H_0=1$. On a cube define

$$V_0(Z,\square)=\tfrac{1}{2}\int_{\square}[\bar\psi(x)^T M_3(x)(e^{iqZ(c,c_3)}-1)(\psi(x)+\chi(x,x^{(3)}))\\ +(\bar\psi(x)+\bar\chi(x,x^{(3)}))^T M_3(x^{(3)})(e^{-iqZ(c,c_3)}-1)\psi(x)]\,\mathrm{d}x.$$

On evaluating pair differences, its sum is exactly $-\langle\bar f,(D_{B+Z}-D_B)f\rangle_{\mathrm{ch}}$. It has the four required symmetries and norm at most

$$12(1+\rho)M^3w_1(w_1+w_2)\sup_{b\text{ meeting }\square}|e^{iqZ(b)}-1|.$$

If $q|Z|\le\frac{1}{2}q^{1-\varepsilon/2}$, its $h_0/2$ norm is at most $12M^3q^{1/2-\varepsilon/2}$.

Proof work. Use real accretivity and the $\gamma_5$ identity for real invertibility. Apply Schur algebra and Sylvester's determinant identity with outgoing dimension $4n_{k+1}$. Remove $\omega_k$ by reciprocal barred/unbarred phases preserving each Berezin top coefficient. Complete the block Gaussian, factor the mean interaction and keep all determinant and box normalizations. For the current reindex by the external-neighbor involution; the two halves agree. Their coefficient before chamber summation is $1/96$.

Inputs. Exact photon split and domain preservation; block field identities; weighted Gaussian completion; scalar extraction. Exports: exact normalized density and first current. At zero the determinants have size $192n_0$ and the identity Wick kernel is $48\delta$; no heavy modes are removed.

Source. PDF pp. 133–142, §§13.1–13.3,13.5, Propositions 13.1–13.4, (13.1)–(13.22), (13.26)–(13.33). TeX 12018–12767, 12867–13082.

## S13. Finite localization before the connected expansion

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M20, S05, S06, S07, S09, S10

Objects and hypotheses. The incoming activity is in S09's precisely defined class $\mathcal{A}_k^{x,\mathbb{C}}$, or its real subspace, with all the generator transformations and domain definitions stated in S09. Use a finite torus partitioned into $M$-cubes. Polymers are face-connected; their net length $d_M$ satisfies $|X|_M \le c_X(d_M(X)+1)$, $\sum_{X\supset\square} e^{-\kappa_0 d_M(X)} \le K_0$, with $c_X = 54$. Photon parameter cubes have half-side $M, LM$; fermion parameters belong to partition $M$-cubes. All have radius $R = M^{\alpha_0}$. Let $P_0(X)$ be parameter cubes contained in $X$, and $Y'(X,P)$ the union of partition cubes meeting $X$ or the parameter cubes in $P_0(X)\cup P$. For a parameter-analytic family define

$$T_P G(X) = \int_{[0,1]^P} \partial_P G(s,X)\,\mathrm{d}s_P, \quad s=1 \text{ on } P_0(X),\ s=0 \text{ outside } P_0(X)\cup P,$$
$$\operatorname{Loc} G(Y) = \sum_{Y'(X,P)=Y} T_P G(X).$$

Assume $R - 1 \ge \exp((2L+1)^3\kappa + \log(6c_X))$.

General analytic claim. If $\|G(s,X)\| \le B_0 e^{-\kappa' d_M(X)}$, $\kappa_0 \le \kappa' \le \kappa$, and $T_P G = 0$ for disconnected $Y'$, then

$$\sum_Y \operatorname{Loc} G(Y) = \sum_X G(1,X), \qquad \|\operatorname{Loc} G(Y)\| \le 4c_X K_0 B_0 e^{-(\kappa'-\kappa_0-1)d_M(Y)}.$$

The disconnected cancellation is a hypothesis to verify, not a consequence of this norm bound.

RG objects and hypotheses. The analytic photon and fermion families have exponential kernel/short-Hölder bounds $C_w e^{-\gamma_w d}$, local polymer decay $e^{-(\kappa+\kappa_0+3)d_M(X)}$, and are exactly restricted to active components when parameters outside are zero. A fermion $H_k$ row at $(c,g)$ vanishes unless every cube meeting $\{|c'-c|_\infty \le 2L\}$ is active. Local Dirac terms are $M_k(X)$. The reconstruction rows have coefficient sums at most $C_T$ and create only external boundary pair legs. Let $e = e_k$, $h = (e^{-1/4}, e^{-1/4+\varepsilon})$, $w_W = 1$ at positive levels and $\sqrt{48}$ at zero. Input data obey $|m| \le 4e^{3/4-8\varepsilon}$, $\|E\|_k \le 4e^{1/4-7\varepsilon}$, with $E = 0$ at zero. The background slots have the local-real-gauge and complex thresholds $e^{-3/4+i\varepsilon}$, $i = 1,2,3$, the global margin $e^{\varepsilon}\theta_i$, and external thresholds $e^{-1+2\varepsilon}, e^{-1+3\varepsilon}$. For $\zeta$ in its coordinate box set

$$Z(s') = H_k^{\mathrm{ph}}(s') C C_k^{1/2}(s')\zeta, \quad |Z|_* = \max(\|Z\|, \|\partial Z\|, \|\delta_\alpha \partial Z\|) \le \tfrac{1}{2}e^{-\varepsilon/2},$$
$$B = A + C, \quad J^{\#} = H_k^{\#}(s'', B+Z) - H_k^{\#}(s'', B), \quad W^{\#} = H_k^{\#}(s'', B)(\Gamma_k(s'', B)W, \bar{W}),$$
$$r_1 = \tfrac{1}{2}e^{-3/4+9\varepsilon/2}, \quad r_2 = e^{-3/4+7\varepsilon}, \quad r_4 = e^{-1/4+3\varepsilon}.$$

Here # includes barred and covariant pair fields. Assume $2 \le r_2, r_4 \le r_1$ and all backgrounds on these contours stay inside the stated open domains. Use $0 < 1-\alpha < \varepsilon \le 1/200$, $L \ge 4^8$, $M > 4L$ and $\kappa \ge 7\kappa_0 + 7$. The fixed positive constants are $C_w, \gamma_w$ in S05–S06's explicit kernel estimates,

$$c_C = 3L^{9/2}, \qquad c_{\mathrm{fl}} = 61 c_X K_0, \qquad C_T = 2\exp(3/2)[1 + b_-^{-1}(54L\exp(6) + 12\exp(1) + 1)].$$

Choose $c_\gamma \ge 1$ bounding the two exponential sums over unit bonds and photon coordinate supports,

$$\sup_b \sum_{b'} \exp[-\gamma_w d(b,b')] \le c_\gamma, \qquad \sup_{\Upsilon} \sum_{\Upsilon'} \exp[-\gamma_w d(\Upsilon,\Upsilon')] \le c_\gamma.$$

They depend only on the fixed construction parameters, not on the level, side, allowed mass or charge. The stated domain assumption is the explicit S05 conclusion $A + tZ \in \widetilde{R}_k$ and $A + C + tZ \in \widetilde{G}_k$ for $|t| \le r_1$, with the domains defined in S09 and S06. It uses S09's three displayed total-connection inequalities.

Claim for the finite RG expressions. Write $\mathcal{C}_r f = (2\pi i)^{-1}\int_{|t|=r} f(t)/[t(t-1)]\,\mathrm{d}t$, $m' = m + m(E)$, $E^r = RE$. Define

$$\begin{aligned}
G_1(X) &= \mathcal{C}_{r_1}\langle\bar\Psi, M_k(X, B+tZ)\Psi\rangle,\\
G_2(X) &= \mathcal{C}_{r_2}E^r(X; A+tZ, C, \psi^\# + tJ^\#\Psi),\\
G_3(\square) &= \mathcal{C}_{r_2}(-m')\langle\bar\psi + t\bar J\bar\Psi, \psi + tJ\Psi\rangle_\square,\\
E_{\le 3}(Y) &= \mathrm{sub}_{Y,B}\sum_{i\le 3}\mathrm{Loc}\,G_i(Y),\\
G^W_{\le 3}(X) &= E_{\le 3}(X; \psi^\# + W^\#),\\
G_4(X) &= \mathcal{C}_{r_4}E^r(X; A, C, \psi^\# + tW^\#),\\
G_5(\square) &= \mathcal{C}_{r_4}(-m')\langle\bar\psi + t\bar W^c, \psi + tW^c\rangle_\square,\\
F(Y) &= \mathrm{Loc}(G^W_{\le 3} + G_4 + G_5)(Y).
\end{aligned}$$

Arguments suppressed in the last four lines are unchanged. At zero $G_1$ is the explicit chamber current, $G_2 = G_3 = G_4 = 0$, and no auxiliary $\Psi$ or reconstruction occurs. The reconstruction $\mathrm{sub}_{Y,B}$ replaces $\Psi$ by the bounded interior/boundary row. Both localizations have disconnected cancellation and exact sum identities. Their coefficients read only bonds meeting $Y$ and photon coordinates supported there; $F$ has fluctuation indices only in $Y$. The remaining legs have the required ordinary and admissible pair support. Moreover

$$\begin{aligned}
\|E_{\le 3}(Y)\|_{h/2} &\le 8c_X K_0 e^{1/4-5\varepsilon}\exp[-(\kappa-\kappa_0-1)d_M(Y)],\\
\|F(Y)\|_{h/4;w_W} &\le H_{\mathrm{cl}}\exp[-(\kappa-2\kappa_0-2)d_M(Y)],\\
H_{\mathrm{cl}} &= 36c_X^2K_0^2e^{1/4-5\varepsilon}.
\end{aligned}$$

$F$ is a fixed term plus a complex-linear function of $(m, E)$. The quantitative admission conditions used here are

$$c_{\mathrm{fl}}e^{\varepsilon},\ 3Me^{\varepsilon},\ 2c_C C_w^2 c_\gamma^2 e^{\varepsilon/4},\ 16c_X c_\gamma C_w C_T^2 M^3 e^{\varepsilon/2},\ 96M^3e^{1/4},$$
$$32C_wC_Te^{3\varepsilon/2},\ 88M^3e^{1/4-6\varepsilon},\ 2e^{5\varepsilon/2}\ \le 1, \qquad 112C_w^2c_\gamma e^{2\varepsilon}\le 3.$$

Proof work. Prove finite locality before using any connected expansion. At indicator parameter families, follow each composed row through its active component; the stronger $H$-row vanishing controls pair endpoints just outside $X$. A disconnected component contains a parameter absent from the expression, whose two finite-difference terms cancel. For the general lemma use repeated one-variable FTC, Cauchy derivative bounds and $d_M(Y) - d_M(X) \le (2L+1)^3|P|$. For numerical bounds use

$$|H^\#|_{1,\infty}\le 4C_w, \quad |J^\#|_{1,\infty}\le 16C_we^{3/4-9\varepsilon/2}, \quad w_W|H^\#\widetilde\Gamma|_{1,\infty}\le 28C_w^2c_\gamma.$$

Give auxiliary fields weight $C_Th_1$, then pass to $h/4$ in the second localization. Apply the general bound twice.

Inputs. Weakened primitive bounds with component restriction and row vanishing; domain margins; weighted substitutions; current and reconstruction; polymer geometry. Exports: a finite local fluctuation activity and its explicit norm bound. The weakened operators are not asserted to have finite range.

Source. PDF pp. 142–149, §§14.1–14.3, Proposition 14.1, Lemma 14.2, Propositions 14.3–14.4, (14.1)–(14.23). TeX 13083–13788.

## S14. A quantitative connected-polymer logarithm

INTERFACE. Overview Target.

Proof dependencies. M20, S07

Objects and hypotheses. Let $\mathcal{P}$ be the collection of nonempty face-connected $M$-polymers. Every $X_i$ and $Y_i$ below belongs to $\mathcal{P}$. Use represented-net length $d_M$, size bound $|X|_M \le c_X(d_M(X)+1)$, and rooted counting bounds

$$\sum_{X\supset\square} e^{-\kappa_0 d_M(X)} \le K_0, \qquad \sum_{X\subset Y} e^{-\kappa_0 d_M(X)} \le K_0|Y|_M.$$

Two polymers are incompatible when their union is face-connected, including equality. In a finite Grassmann algebra let $V(X)$ be even, with old-generator weights $w$ and fluctuation weight $w_W = \max \nu^{-1/2}$. It has fluctuation modes only in $X$ and depends on real box coordinates only through coordinates supported in $X$. The photon law is a product probability law. Compatible polymers have disjoint fluctuation and coordinate supports. Suppose

$$\|V(X)\|_{w;w_W} \le He^{-a_2 d_M(X)}, \quad a_2 = \kappa - 2\kappa_0 - 2, \quad \kappa \ge 7\kappa_0 + 7.$$

Set $a_5 = \kappa - 3\kappa_0 - 2$, $a_6 = \kappa - 5\kappa_0 - 3$, $H_1 = 7c_X K_0 \exp(3a_5)H$, and require

$$H \le 1, \quad 4HK_0c_X \exp(3a_5) \le 1, \quad z_S = 28H_1 \exp(\kappa_0)K_0 \exp(3a_6) \le \tfrac{1}{2}.$$

Claim. For any cube union $Y$, define

$$K(Y) = \sum_{\substack{\{X_i\} \text{ distinct} \\ \bigcup_i X_i = Y}} \prod_i (e^{V(X_i)} - 1), \quad \mathcal{H}(Y) = \int \mathrm{d}\mu_I(W)\, \mathrm{d}\widehat{\mu}(\zeta)\, K(Y).$$

Set $K(\varnothing) = 1$, hence $\mathcal{H}(\varnothing) = 1$. Let $u^T(Y_1, \dots, Y_n)$ be the alternating sum over connected spanning subgraphs of the incompatibility graph, with $u^T(Y_1) = 1$. For each nonempty connected $Y \in \mathcal{P}$, define

$$E^{\#}(Y) = \sum_{n\ge 1} \frac{1}{n!} \sum_{\substack{Y_1,\dots,Y_n \in \mathcal{P} \\ \bigcup_i Y_i = Y}} u^T(Y_1, \dots, Y_n) \prod_i \mathcal{H}(Y_i).$$

This series is absolutely convergent in the old norm and

$$\int \mathrm{d}\mu_I\, \mathrm{d}\widehat{\mu} \exp\{\sum_{X\in\mathcal{P}} V(X)\} = \exp\{\sum_{Y\in\mathcal{P}} E^{\#}(Y)\},$$

$$\|E^{\#}(Y)\|_w \le 14c_X^2 K_0^2 \exp(3\kappa) He^{-(\kappa - 5\kappa_0 - 4)d_M(Y)}.$$

The norm sum of terms with $n > N$ is at most $2^{-N}$ times this bound. Replace each $e^V - 1$ by $\sum_{1\le j\le N} V^j/j!$ and truncate the connected series at $n = N$. These polynomial approximants converge uniformly over all admitted families at each fixed torus side. Their finite-product error has factorial denominator $(N+1)!$ and constants depending on the finite number of polymers; no volume-uniform approximation error is asserted.

Proof work. Prove the formal connected-graph identity by component decomposition. Prove the tree bound by assigning each connected graph its least-labelled parent tree at preceding root-distance levels, and summing signs on the resulting graph intervals. Joining nets along incompatible pairs costs less than $3M$ per tree edge. Bound $K(Y)$ by $H_1 e^{-(\kappa - 3\kappa_0 - 3)d_M(Y)}$. Weighted Berezin integration and photon probability integration cost no norm; compatible supports factor. Sum a rooted tree from leaves to root using polymer counting, the factorial bound on child counts and Cayley's degree count. The remaining geometric series has ratio $z_S \le 1/2$. Absolute convergence justifies substitution of even Banach-algebra elements into the formal identity. Prove the polynomial approximation with its stated fixed-volume dependence.

Inputs. Weighted Berezin/Wick theorem and polymer geometry only. Exports: a local connected logarithm, its decay, geometric tail and polynomial approximants. For the chamber application take $V = F$, $H = 36c_X^2 K_0^2 e_k^{1/4-5\varepsilon}$ and $w_W = \sqrt{48}$ at zero.
Source. PDF pp. 149–152, §14.4, Proposition 14.5, (14.24)–(14.34) and its polynomial-approximant conclusion. TeX 13789–14107.

## S15. The symmetric correction and the exact nonlinear step

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S08, S11, S12, S13, S14

Objects and hypotheses. The activity input is the class $\mathcal{A}_k^{x,\mathbb{C}}$ defined in S09, including its explicit lattice, gauge, charge and reality actions. Real data belong to $\mathcal{A}_k^x$. Let local activities have independent ordinary and covariant pair legs, index measures $\eta^3/48, (\eta^3/48)^2$, and weights $h_k = (e_k^{-1/4}, e_k^{-1/4+\varepsilon})$. The norm is the weighted coefficient supremum times $\exp(\kappa d_M(X))$, over local-real-gauge analytic backgrounds and the external slot. The mass/activity windows are

$$w_k^m = e_k^{3/4-8\varepsilon}, \qquad w_k^E = e_k^{1/4-7\varepsilon}, \qquad |m| \le c w_k^m, \quad \|E\|_k \le c w_k^E.$$

At zero the input activity is zero. Assume the finite fluctuation construction has the support, component locality, affine-data dependence and bound

$$\|F(X)\|_{h_k/4;w_W} \le 36 c_X^2 K_0^2 e_k^{1/4-5\varepsilon} e^{-(\kappa-2\kappa_0-2)d_M(X)}.$$

Its photon and fermion primitives have the explicit lattice, real-gauge, transpose and reality covariance, with the identity Wick measure and the product photon box law. Use the connected logarithm $E_k^{\#}$ defined by the union series of integrated Mayer activities. Assume normalized scaling has factor $c_{\mathfrak{fl}} L^{-1/4+2\varepsilon}$, with field scaling $L^{-1}$ on ordinary legs and $L^{-1-\alpha}$ on pair legs, and the polymer reblocking bound. Require

$$c_S = 10^5 c_X^5 K_0^5 \exp(6\kappa), \quad c_S e_k^{1/4-5\varepsilon} \le 1, \quad c_S L^3 e_k^{\varepsilon} \le 1,$$
$$8 c_{\mathrm{rb}} L^3 C_d M^3 e_k^{2\varepsilon} \le 1, \qquad 12 \exp(-p_{0,k}^2/2) \le e_k^7.$$

This conditional assembly uses the following explicit inputs: S12's displayed normalized density equality and marginal normalization on the S09 domains, with $0 < \mu \le 1/L$, $e_{k+1} = \sqrt{L} e_k$; S13's displayed finite functions, locality and norm bounds; S14's connected-series identity and decay bound for $V = F$; and S11's scaled-domain and normalized-contraction conclusions. For the determinant input use S08's displayed $8 C_d M^3 e_k^{1/4} \exp[-\kappa d_M(X)]$ difference bound. The numerical requirements to realize those inputs are the finite inequalities explicitly stated in those cards, including S13's nine smallness conditions; the additional inequalities used in the present scaling and step assembly are displayed above. No tuned trajectory or physical stopping construction is a hypothesis of this conditional statement.

Claim. The logarithm has the lattice symmetry with $LM$-translations, independent local real-gauge covariance, charge conjugation and, for real data, reality. Each monomial is balanced. Define

$$E_k^*(m, E)(Y; A, C) = \sum_{\overline{X}^L = LY} S_L E_k^{\#}(m, E)(X; A_L, C_L).$$

It is a next-level activity and $\|E_k^*(m, E)\|_{k+1} \le e_k^{1/4-6\varepsilon}$ on the closed data ball of radius 4. It is norm-continuous on each affine data line's closed allowed set and norm-analytic in its interior. On the unit ball,

$$\|E_k^*(m_1, E_1) - E_k^*(m_2, E_2)\|_{k+1} \le e_k^{1/4-6\varepsilon} \left( \frac{|m_1 - m_2|}{w_k^m} + \frac{\|E_1 - E_2\|_k}{w_k^E} \right).$$

The activity quotient is absent at zero. Let $E_k^d$ be the even local determinant primitive and put $\delta E_k^{\det}(X;A,C) = E_k^d(X,A+C) - E_k^d(X,C)$. The exact output maps are

$$\begin{aligned} E^+ &= \mathcal{L}(RE)^{\mathrm{nat}} + \mathcal{L}\delta E_k^{\det} + E_k^*(m,E), \\ m^+ &= L(m+m(E)), \\ \varepsilon_v^+ &= L^3(\varepsilon_v + \varepsilon_v(E) + \delta\varepsilon_k^0), \\ P_{k+1}(Y,C') &= \sum_{\overline{X}^L = LY} [P_k(X,C'_L) + E_k^{\det}(X,C'_L)]. \end{aligned}$$

For real data in the unit ball they give the exact transformed normalized density, its exact box marginal, and

$$\begin{aligned} \|E^+\|_{k+1} &\le c_{\mathfrak{fl}} L^{-1/4+2\varepsilon}\|E\|_k + e_{k+1}^{1/4-2\varepsilon} + e_k^{1/4-6\varepsilon}, \\ |m^+ - Lm| &\le L c_{\mathfrak{fl}} e_k^{1/2}\|E\|_k, \\ |\varepsilon_v^+ - L^3\varepsilon_v| &\le L^3(c_{\mathfrak{fl}}\|E\|_k + e_k^7). \end{aligned}$$

At zero $m^+ = Lm$ and the vacuum increment is precisely $L^3\delta\varepsilon_0^0 \in [0, L^3 q^7]$. All operations commute with restriction of the external slot to zero.

Proof work. Pass lattice, gauge and reality actions through finite operations. For charge conjugation the asymmetric unbarred/barred fluctuation substitution is not directly covariant. At indicator parameters $1_U$, compare its Wick covariance

$$G_U(B) = H_k(1_U,B)\Gamma_k(1_U,B)\bar{H}_k(1_U,B)^T, \qquad G_U(-B)^T = -\mathsf{C}^{-1}G_U(B)\mathsf{C}^{-1}.$$

Derive equality of all transformed pair contractions and apply Wick; subset inversion then gives covariance of each integrated Mayer activity and hence of the local logarithm. Constant-gauge invariance forces balance. Use the connected bound and reblocking for $E_k^*$. Fixed-volume polynomial approximants prove norm analyticity; Cauchy at radii $3w$/data difference gives the Lipschitz bound. Finally evaluate the five finite expressions: their contour differences give exactly the density fluctuation exponent. Reconstruction uses $T_k H_k = 1$. Split the determinant quotient, retain adjustment's evaluated equality, and assign every fine polymer once under reblocking.

Inputs. Finite localization; connected logarithm; normalized contraction; exact density/current; determinant gain; weighted moments. This gives exact nonlinear maps with quantitative norm and parameter-dependence bounds. Equality across tori is a separate task.

Source. PDF pp. 153–159, §§14.5–14.7, Propositions 14.6–14.7, Theorem 14.8, (14.35)–(14.52). TeX 14108–14829.

## S16. Odd covers and one uniformly projected primitive system

Model-specific. Overview Target.

Proof dependencies. M20, S03, S04, S05, S06

Objects and hypotheses. Let $\ell_2 = J_c\ell_1$, $J_c$ odd, with both $\ell_i/(LM)$ odd and at least 13, at common mesh $\eta = L^{-k}$. The column character on the covering index set is $\chi(w) = 1$ for photon fields, coordinates and parameters, and $\chi(w) = (-1)^{w_1+w_2+w_3}$ for spinors and pair fields, where $w \in \mathbb{Z}^3$. Pullback $p^*$ identifies small-torus fields with the corresponding character subspace of the large torus. Matching column measures are unchanged: 1 on unit indices, $\eta^3$ on fine photon indices, and $\eta^3/48$ on chamber indices. A top parameter family is translation covariant under $\ell_1\mathbb{Z}^3$ on pulled-back data. For such a family put $\widetilde{O}(s, A) = O_2(p^*s, p^*A)$. For a polymer family put $\widetilde{O}(X, A) = \sum_{pX'=X} O_2(X', p^*A)$. Assume its row/column support diameter is at most $L$, $\gamma_w M \geq 1$, and the rooted polymer counting bound with constants $\kappa_0, K_0$.

General finite-cover claim. $\|p^*f\|_\infty = \|f\|_\infty$ and pairings multiply by $J_c^3$. Choose one integer representative of each class in $(\mathbb{Z}/J_c\mathbb{Z})^3$, forming a set $\mathcal{R}$. The projected operator has the exact kernel

$$O_1(t, t') = \sum_{w\in\mathcal{R}} \chi(w)\widetilde{O}(t, t' + \ell_1 w).$$

Use the character of the operator's column family. The combined summand, not the antiperiodic sign alone, is independent of the representative. Indeed replacing $w$ by $w + J_c v$ multiplies both its sign and its large-torus kernel by $\chi(v)$, so their product is unchanged. Intertwining passes to transposes and to inverses when defined. Projection of a represented net gives $d_M(pX') \leq d_M(X')$. The uniform image estimates are

$$\sum_w e^{-\gamma_w d_2(t,t'+\ell_1 w)} \leq 64e^{-\gamma_w d_1(t,t')},$$

$$\sum_{\substack{pX'=X\\ X'\supset\square_t}} e^{-(\kappa_w+\kappa_0)d_M(X')} \leq K_0 e^{-\kappa_w d_M(X)}.$$

Model verification and output. The exact photon minimizer, coarse form, axial coordinate map, fluctuation form and its resolvent/root, fluctuation injection and constrained block minimizer all intertwine with pullback. The exact chamber/fine/coarse fermion operators and their inverses do likewise, on the pulled-back local-real-gauge domain. A top weakened system with kernel constant $C_w^{(1)}$, rates $\gamma_w, \kappa_w + \kappa_0$, analytic parameter polydiscs, full-strength identities, active-component restriction, local read-dependence, $H_k$-row vanishing and the required covariances projects to the same properties with rates $\gamma_w, \kappa_w$ and common constant $C_w = 64K_0C_w^{(1)}$. At zero preserve the normalized chamber measure and $(1 + y)^{-1}$ resolvent factor.

A concrete top choice is furnished by the path-multiplied photon kernels and weighted fermion walks, with

$$\alpha_0 = 1/54, \qquad \gamma_w = \min(\gamma_F, \gamma_P/2),$$

$$C_P = 116C_P^{\mathrm{ph}}M^{402\alpha_0}, \qquad C_w^{(1)} = \max(C_F, C_P, 4\exp(2\gamma_w)M^{27\alpha_0}).$$

Here the top fermion walks reserve the additional $\kappa_0$ decay; their $M$ and charge conditions and the photon path-count condition are imposed before choosing the charge ceiling. The constants are independent of side, level, admitted mass, charge and real harmonic background.

Proof work. Sum a large fundamental domain over small translates to obtain the signed kernel formula with its unchanged measures. Use uniqueness of Euler problems for exact photon minimizers, then the positive resolvent integral for the root. For exponential image sums count nearby translates and the at most $26j^2$ translates at sup radius $j \geq 2$. Spend the extra $\kappa_0$ on polymer counting. Check active components, row vanishing and covariance after projection, not only decay. The level-zero diagonal parameter products have the same unique local lifts.

Inputs. Exact and weakened photon/fermion primitives; polymer geometry; finite-cover character algebra. Exports: one common projected system and its uniform constants. Independently chosen weakened systems on different tori are not asserted equal.
Source. PDF pp. 101–102,159–163, §§10.9,15.1–15.3, Propositions 15.1–15.2, (10.45)–(10.46), (15.1)–(15.9). TeX 9010–9071, 14830–15255.

## S17. Common lifted algebras and exact supported rows

Model-specific. Overview Target.

Proof dependencies. M20, S05, S06, S07, S10, S11, S16

Objects and hypotheses. Take a top torus and its odd smaller quotient with sides $\ell_2 = J_c\ell_1$, and the related projected kernel systems. Require $\ell_1 > (31L+16)M$. Put $d_U = 2L+1$, $c_l = 2L$. For a finite connected covering union $\widetilde{X}$ of $M$-cubes, let $\operatorname{diam}(\widetilde{X})$ be the largest side of its smallest rectangular cube-box. Call it admissible when

$$\operatorname{diam}(\widetilde{X}) \le \ell_1 - 14d_U M, \qquad N(\widetilde{X}) = \operatorname{box}(\widetilde{X})^{\sim 6d_U}.$$

Images of such connected lifts are flat polymers. Ordinary, pair, unit and coordinate indices have their geometric supports and matching column measures on the two tori. The weakened kernels are restricted to the linked active components when parameters outside are zero; their rows read only that component's connection. The stronger $H_k$-row vanishing margin is retained.

Claim. Projection is injective on $N(\widetilde{X})$; any nonzero period separates its sites by at least $2d_U M + \eta$. Supported indices and incidences of total support diameter less than $2d_U M$ have unique common lifts. Pair separations and their six coordinate paths agree. Identifying independent covering generators defines a norm-preserving local Grassmann algebra isomorphism $\iota_{\widetilde{X}}$, preserving measures, degrees and bar indices. It includes chamber $W$-modes at zero. Every small polymer $d_M \le L$ is flat; small polymers containing a fixed cube correspond across the two tori with identical volume and net length.

Let $\widetilde{Y}$ be admissible and $0 \le j \le 5d_U$. Parameters have equal lifts on cubes contained in $\widetilde{Y}$, and vanish outside their respective images. If the connections agree on $\widetilde{Y}^{\sim(j+d_U)}$, then every parameter-primitive row contained in $\widetilde{Y}^{\sim j}$ agrees exactly, has columns in $Y_i^{\sim(j+c_l)}$, and reads only bonds in $Y_i^{\sim(j+d_U)}$. For a connected $\widetilde{X} \subset \widetilde{Y}$, the local polymer kernels $M_k(X)$ and $S_{k,y}(X)$ also agree at common row/column lifts if the connections agree on bonds meeting $\widetilde{X}$.

The finite rows of the photon coordinate map, covariant pair quotient, nested averages, block resolvent $B_{k,y}$, interior and boundary reconstruction, mass form and first current agree as well. Their kernel equals the row coefficient divided by the common column measure. Adjustment, field scaling, weighted Wick integration and the normalized photon-coordinate marginal commute with $\iota$.

Proof work. Use the explicit injective margin to rule out any distinct period representative in a local incidence. The common independent generators give the free-algebra isomorphism. For a supported row first restrict to its attached active component. Apply the signed image formula for the projected operator. All possible nonzero column lifts are in the same separated neighborhood, so only image $w = 0$ survives. A connected preimage of a flat polymer is one period translate of its lift; row and column support kill both translation classes.

For finite rows check actual ranges and read bonds, including the reconstruction's boundary membership indicators. At zero the full chamber stencil and its identity kernel $48\delta$ are present. Common weighted Wick moments are the same ordering signs times $\nu^{-\#\text{pairs}}$; other modes integrate to one. The restricted photon law has the same product marginal on common coordinates.

Inputs. Common projected primitive system; component locality; polymer lifts; finite row formulas; weighted moments and scaled/adjusted local algebra. Exports: exact local row equality and the common free algebra. This is an equality statement, not an exponentially small finite-size error.

Source. PDF pp. 163–168, §§15.4–15.6, Propositions 15.3–15.4, (15.10)–(15.20). TeX 15256–15770.

## S18. Equality of local nonlinear steps across odd covers

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S08, S09, S13, S14, S15, S16, S17
Objects and hypotheses. Use one top system and its odd projected system on sides $\ell_2 = J_c\ell_1$, with both incoming and outgoing sides admissible and

$$\ell_1/L > (31L + 16)M.$$

A covering connected polymer is admissible if its box diameter is at most $\ell_1 - 14(2L + 1)M$; its images are flat. Compare their ordinary, pair and fluctuation generators using the norm-preserving common lifted algebra, with chamber measure $\eta^3/48$, and $1/48$ at the first step. The activities use exactly S09's class $\mathcal{A}_k^x$ on each torus, with its explicit analytic domains and generator transformation equations. The same real mass satisfies $|m| \le e_k^{3/4-8\varepsilon}$. The two input activities have norms at most $e_k^{1/4-7\varepsilon}$, vanish at $k = 0$, and agree on every flat lift when their two connection slots agree on its local read neighborhood. The exact inputs on both tori are the S15 maps with their stated density identity and bounds, S13's displayed finite contour definitions and domain inclusions, and S17's supported-row equalities. Use common $L, M, \alpha, \varepsilon, \kappa, e_k, \mu$ and fixed kernel constants. Their finite numerical prerequisites are explicitly listed in S05 (weakening/domain output), S08 (determinant gain), S09 (total connection), S11 (normalized scaling), S13 (finite localization) and S15 (scaled correction); the sole additional side requirement here is the one displayed above. The local construction consists of scalar extraction, shortest-chain quadratic adjustment, the two finite contour/parameter localizations, weighted fluctuation integration, the absolutely convergent connected union series, and scaling/reblocking.
Claim. The extracted scalar mass and vacuum contributions are equal, as is the photon box cost $\delta\varepsilon_k^0$. The local determinant primitive $E_k^d(X, B)$ is equal on each flat lift whenever $B$ agrees on the bonds meeting that lift. The finite local fluctuation activities, their integrated Mayer activities and their local connected logarithms agree. Consequently the outgoing activity maps

$$E_i^+ = \mathcal{L}(RE_i)^{\mathrm{nat}} + \mathcal{L}\delta E_k^{\mathrm{det},(i)} + E_k^{*,(i)}(m, E_i)$$

agree on every flat outgoing lift, where $\delta E_k^{\mathrm{det}}(X; A, C) = E_k^d(X, A + C) - E_k^d(X, C)$. The scalar recursion

$$P_{k+1}^{(i)}(Y, C') = \sum_{\overline{X}^L = LY} [P_k^{(i)}(X, C'_L) + E_k^{\mathrm{det},(i)}(X, C'_L)]$$

preserves this local agreement. All conclusions also hold without the external slot and commute with its restriction to zero. There is no error term or volume limit in this statement: the price is its explicit side margin and related primitive systems.
Proof work. Identify the extracted constant/quadratic coefficients and normalized integrals on each common small lift. For determinant traces all indices are supported in the flat polymer; the common finite $B, Q, F_0$ rows and local $S$ kernels agree. Paired antiperiodic signs cancel. Use the integrable majorants $(b_k^2 + y^2)^{-1}$ at positive levels and the split $(1 + y)^{-1}$ integrals at zero.

Track the successive neighborhoods of the composed photon field: the reconstruction columns are in $U^{\sim 2d_U}$, coordinate-map columns in $U^{\sim(1+2d_U)}$, and root columns in $U^{\sim 3d_U}$. All lie in $N(U)$, so photon fields agree on the read neighborhood of every contour background. Then identify the fermion rows, first contour functions, their localization and reconstruction, the second functions and localization, in that order. Common coordinate marginals and weighted Wick moments identify the integrated Mayer activities. Termwise equality and absolute convergence identify their connected logarithms.

For an outgoing lift $\widetilde{U}$, put $\widetilde{Y} = L\widetilde{U}$. Both reblocking sums have the same covering index set: connected $\widetilde{X} \subset \widetilde{Y}$ whose $LM$-closure is $\widetilde{Y}$. Compare the adjusted activity, determinant difference

and local logarithm on each lift, then apply the common scaling. Locality extends the comparison from pullback connections to all pairs with equal local lifts.

Inputs. Exact local row/algebra comparison; determinant primitives; extraction/adjustment; finite localizations; connected convergence; exact nonlinear step. Exports: equality of local steps and counterterms for the later common-pair tuning argument. No equality on wrapping nonflat polymers is asserted. Uniform norm estimates alone do not prove this claim.

Source. PDF pp. 168–172, §§15.7–15.9, Theorem 15.5, (15.21)–(15.32). TeX 15771–16242.

## T01. Stopping scales and a common tuned pair

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S09, S11, S15, S18

Objects and hypotheses. Fix an odd integer $L \ge 13$, a power $M$ of $L$ with $M \ge L^2$, $0 < \bar m \le 1$, and $0 < e\sqrt{h} \le e_1$, where

$$h = L^{\lfloor \log_L(1/\bar m)\rfloor}, \quad \eta = L^{-N}, \quad K = N + \log_L h, \quad a_j = L^{j-N}, \quad e_j = e\sqrt{a_j}, \quad \mu_j = \bar m a_j.$$

Put $v_j = -\log e_j$, $r_j = \min\{L^i : i \ge 0,\ L^i \ge v_j^r\}$, $p_j = v_j^p$, $p_{0,j} = v_j^{p_0}$, and $R_{\mathrm{top}} = hMr_K$, $R^* = R_{\mathrm{top}}/L$. For $R = hL^T \le R_{\mathrm{top}}$ set $\ell_j = R/a_j$ and $n_j = \log_L(\ell_j/(Mr_j))$. Assume $r \ge 2$, $0 < \varepsilon \le 1/200$, and, for $v \ge -\log e_1$,

$$v \ge 8, \quad (1 + 3\log L/(2v))^r < L, \quad v^r > 31L + 16, \quad (c_{\mathrm{fl}} + 2)e^{-\varepsilon v} \le (1 - L^{-\varepsilon/2})/16.$$

Let $E_j$ lie in specified real activity Banach spaces. Suppose the actual step maps give $m_{j+1} = L(m_j + m(E_j))$, $E_{j+1} = \Phi_j(m_j, E_j)$, and $\epsilon^{\mathrm{v}}_{j+1} = L^3(\epsilon^{\mathrm{v}}_j + \epsilon^{\mathrm{v}}(E_j) + \delta\epsilon_j)$. Write $w_j^m = e_j^{3/4-8\varepsilon}$ and $w_j^E = e_j^{1/4-7\varepsilon}$. The required normalized bounds are a backward mass factor at most $1/4$, activity contraction at most $1/16$, source at most $2e_{j+1}^{\varepsilon}$, and

$$\frac{|m(E) - m(\widetilde E)|}{w_j^m} \le c_{\mathrm{fl}} e_j^{\varepsilon} \frac{\|E - \widetilde E\|}{w_j^E}, \qquad \frac{\|\Phi_j(z) - \Phi_j(\widetilde z)\|}{w_{j+1}^E} \le \tfrac{1}{16} d_E + e_{j+1}^{\varepsilon}(d_m + d_E).$$

Here $d_m, d_E$ are the corresponding normalized differences. Assume the same continuation bounds for the external-field activity, the scalar extraction bounds of the step, and exact local agreement of steps under the prescribed odd covers.

Claim. For $N$ above a finite threshold, the first level $n_j = 0$ is a stop $k_*$ whose physical quantities are independent of $N$. On $R^*$,

$$a_* \in \{h/L, h/L^2\}, \quad \ell_* = Mr_*, \quad r_* > 31L + 16, \quad L^{-3} < \mu_* \le L^{-1}.$$

There is one pair selected on $R_{\mathrm{top}}$, with terminal mass and vacuum zero there, which is controlled on every permitted smaller torus:

$$|m_j| \le \tfrac{1}{8} e_j^{3/4-8\varepsilon}, \quad \|E_j\| \le \|E_j\|^x \le 4e_j^{1/4-6\varepsilon}, \quad |\epsilon^{\mathrm{v}}_j| \le 5c_{\mathrm{fl}} e_j^{1/4-6\varepsilon}.$$

Its fine mass shift is real, independent of the outer volume, twist and probe within the fixed construction, and has modulus $\le q^{3/4-8\varepsilon}/8 < 1/8$, where $q = e\sqrt{\eta}$. The total mass $\bar m\eta + m_{0,N}$ need not be positive.

Proof work. Use depth $K - j$ to find the last zero of an integer sequence with increments zero or one. Solve masses backwards and activities forwards on the whole finite sequence; the map contracts by $5/16$. Recover the vacuum by its backward scalar sum. Transfer couplings under covers, while bounding each smaller torus's full activity by its own continuation estimate. Smaller-volume terminal couplings need not vanish.

Inputs. Quantitative exact-step, activity-contraction, extraction, and local cover-comparison conclusions. No continuum or cumulant input.

Source. PDF pp. 173–178, §16, Propositions 16.1–16.2, Theorems 16.3–16.4, equations (16.1)–(16.20).

**T02. A compact probe with positive fourth coefficient**

INTERFACE. Overview Target.

Objects and hypotheses. On $L^2(\mathbb{R}^3;\mathbb{C}^4)$ let $\gamma_\mu = \mathrm{diag}(\sigma_\mu, -\sigma_\mu)$, using the Pauli matrices, and set

$$S_M = (M + \gamma\cdot\partial)^{-1}, \quad V_a = i\gamma\cdot a, \quad q_{4,M}(a) = -\tfrac{1}{4}\operatorname{Tr}(S_M V_a)^4.$$

Here $M > 0$ and $a$ is a real compactly supported smooth vector field. The available kernel is $S_M(z) = (M - \gamma\cdot\partial)e^{-M|z|}/(4\pi|z|)$.

Claim. The fourth power is trace class, its trace is real, and

$$\left| q_{4,M}(a) - \frac{1}{240\pi M^5}\int_{\mathbb{R}^3} |\operatorname{curl} a|^4\, dx \right| \le C_a M^{-7} \qquad (M \ge 1).$$

For each fixed $\bar m > 0$ there is a real $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ such that, for $f = dd^*\beta$, $b(f) := -6\operatorname{Tr}(S_{\bar m} i\gamma\cdot d^*\beta)^4 > 0$.

Proof work. First justify the five-propagator diagonal kernel obtained by differentiating in $M$. Taylor expansion, local gauge covariance and parity reduce its leading term to the antisymmetric first derivative of $a$. Compute that coefficient from the constant magnetic-field heat kernel, identifying formal inverse coefficients on Schwartz spinors. The diagonal mass derivative has leading term $-|\operatorname{curl} a|^4/(48\pi M^6)$ and error $O_a(M^{-8})$. Integrate in $M$ and dilate a compact probe: $\beta^{(\ell)}(x) = \ell^2\beta(x/\ell)$ gives

$$b(dd^*\beta^{(\ell)}) = \frac{\ell^3}{10\pi\bar m^5}\int |\operatorname{curl} d^*\beta|^4 + O_\beta(\ell\bar m^{-7}).$$

Choose a probe for which the integral is positive.

Inputs. The displayed continuum Dirac kernel, Clifford relations and finite Schatten/trace facts. No lattice or interacting estimate.

Source. PDF pp. 178–183, §17, Proposition 17.1, equations (17.1)–(17.18).

**T03. The fourth free chamber coefficient converges**

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S01

Objects and hypotheses. Fix $0 < \rho < 1$, $0 < \bar{m} \le 1$ and real compact smooth $\beta$. Put $a = d^*\beta$, $f = dd^*\beta$, and let $D_\beta$ be the side of a cube containing $\operatorname{supp}\beta$. Use centered cube averaging $J_\eta$ and $\delta_{\eta,\mu}g(y) = [g(y+\eta e_\mu/2) - g(y-\eta e_\mu/2)]/\eta$. Periodize $a^0_{\eta,\nu} = -\sum_{\mu\neq\nu}\delta_{\eta,\mu}J_\eta\beta_{\mu\nu}$ after midpoint sampling on a torus of physical side $R$. For the absorbed phase $B_t = t\eta a_\eta$, define

$$N_0(t) = \frac{\det(D^{\mathrm{ch}}[B_t] + \bar{m}\eta)}{\det(D^{\mathrm{ch}}[0] + \bar{m}\eta)}, \qquad b_{\mathrm{ch}} = (\log N_0)^{(4)}(0).$$

The massless chamber stencil has Bloch symbol $D_m(p) = D_{\mathrm{loc}} + m - \sum_\nu(e^{ip_\nu}T^+_\nu + e^{-ip_\nu}T^-_\nu)$ on a 192-dimensional fiber. Assume its verified chamber inputs: a constant-label isometry $J : \mathbb{C}^4 \to \mathbb{C}^{192}$, $J^*T^\pm_\nu J = (\rho \mp \gamma_\nu)/2$, and

$$D_m(p)^{-1} = J(m + i\gamma\cdot p)^{-1}J^* + E_m(p), \qquad \|E_m(p)\| \le C_\rho$$

on the Brillouin cube $[-\pi,\pi]^3$, $0 < m \le 1$. The symbol is $2\pi$-periodic, smooth away from $p = 0$ at $m = 0$, and uses Hermitian matrices $\gamma_\nu = \operatorname{diag}(\sigma_\nu, -\sigma_\nu)$. The further inputs, with constants depending only on $\rho$, are

$$\|D_m(p)^{-1}\| \le C_\rho(m + |p|)^{-1}, \qquad \|\partial^\alpha D_m(p)^{-1}\| \le C_{\alpha,\rho}(m+|p|)^{-1-|\alpha|} \quad (|\alpha| \le 4),$$
$$\operatorname{Re} D_m(p) \ge mI, \qquad \|T^\pm_\nu\| \le 3(1+\rho), \qquad \|\partial^\alpha D_m(p)\| \le 6 \quad (1 \le |\alpha| \le 4).$$

In the light–heavy decomposition relative to $JJ^*$, the heavy inverse is bounded by $(g_{\mathrm{h}} + m)^{-1}$ for a fixed $g_{\mathrm{h}} > 0$. The real-part and inverse estimates concern real $p$; the strip estimates needed for pressure are stated in T10. The determinant is a determinant of a linear map; the normalized label pairing does not insert a factor $1/48$ into its trace.

Claim. For $R \ge D_\beta + 1$, $0 < \eta \le 1$, $R/\eta \in \mathbb{N}$,

$$|b_{\mathrm{ch}}(\eta, R) - b_R(f)| \le C_{\rho,\bar{m},\beta}\eta^{1/4}, \qquad |b_R(f) - b(f)| \le C_{\bar{m},\beta}e^{-\bar{m}(R-D_\beta)}.$$

Here $b_R = -6\operatorname{Tr}(S_R i\gamma\cdot a_R)^4$ on the antiperiodic continuum torus and $b = -6\operatorname{Tr}(S_{\bar{m}}i\gamma\cdot a)^4$ on $\mathbb{R}^3$. Both traces exist and are real; constants are unchanged by real translation of the probe.

Proof work. With $D_j = \partial_t^j D^{\mathrm{ch}}[B_t]|_0$, $Y_j = GD_j$, retain

$$b_{\mathrm{ch}} = \operatorname{Tr} Y_4 - 4\operatorname{Tr} Y_1Y_3 - 3\operatorname{Tr} Y_2^2 + 12\operatorname{Tr} Y_1^2Y_2 - 6\operatorname{Tr} Y_1^4.$$

Bound the contact terms by $C_{\rho,\beta}\eta[1 + \log(1 + 5/(\bar{m}\eta))]$. Compare the four-first-vertex trace in five steps: remove the bounded inverse part, replace the vertex by multiplication, control Fourier aliases, replace the interpolant, remove high momenta. Their Schatten-four errors are respectively $O(\eta^{1/4}), O(\eta^{1/4}), O(\eta^2), O(\eta), O(\eta^{1/4})$. Localize the torus/infinite-space trace difference to the probe cube and sum nonzero Yukawa images.

Inputs. The stated chamber symbol/inverse conclusions and continuum Dirac kernel. The positive sign from T02 is not used in convergence.

Source. PDF pp. 183–190, §§18.1–18.5, Proposition 18.1, equations (18.1)–(18.28).

**T04. The two-photon cancellation gives uniform free coefficients**

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S01, T03

Objects and hypotheses. Fix $0<\rho<1$, $0<\bar m\le 1$, $0<\eta\le 1$ and $m=\bar m\eta$. Let $D_m(p)=D_{\mathrm{loc}}+m-\sum_\nu(e^{ip_\nu}T_\nu^+ + e^{-ip_\nu}T_\nu^-)$ be T03's chamber symbol, and set $R_m=D_m^{-1}$, $V_\nu=\partial_\nu D_m$ and $W_\nu=\partial_\nu^2 D_m$, with mixed second derivatives zero. Assume bounded nonzero symbol derivatives and

$$\|\partial^\alpha R_m(p)\|\le C_{\alpha,\rho}(\operatorname{dist}(p,2\pi\mathbb{Z}^3)+m)^{-1-|\alpha|}\quad(|\alpha|\le 4).$$

Assume an involution sends $D_m(p)$ to $D_m(-p)$ and $V_\nu(p)$ to $-V_\nu(-p)$. Fix $n=R/\eta\ge 2$ and the midpoint transform $h_{\eta,\nu}(K)=\eta^3\sum_{y\in\eta\mathbb{Z}^3+\eta e_\nu/2}a^0_{\eta,\nu}(y)e^{-iK\cdot y}$ of T03's compact smooth probe. The coupling is the nearest-neighbor Peierls coupling: if $(U_\nu^\pm\psi)(x)=\psi(x\pm e_\nu)$, the positively oriented bond has phase $tb_\nu^+(x)=t\eta A_\nu(x)$ and its reverse has phase $tb_\nu^-(x)=-t\eta A_\nu(x-e_\nu)$, where $A_\nu(x)=a^0_{\eta,\nu}(\eta x+\eta e_\nu/2)$, periodized. Thus

$$D_j:=\partial_t^j D^{\mathrm{ch}}[t\eta a_\eta]\big|_0=-\sum_{\nu,\pm}(ib_\nu^\pm)^j T_\nu^\pm U_\nu^\pm.$$

For the convention $\widehat{b}_\nu(k)e^{-ik_\nu/2}=\eta R^{-3}h_{\eta,\nu}(K)$, $k=\eta K$, the first two vertices are

$$D_1[p+k,p]=\sum_\nu \widehat{b}_\nu(k)e^{-ik_\nu/2}V_\nu(p+k/2),$$

$$D_1[p,p+k]=\sum_\nu \overline{\widehat{b}_\nu(k)e^{-ik_\nu/2}}V_\nu(p+k/2),$$

$$D_2[p,p]=\sum_{\nu,k}|\widehat{b}_\nu(k)|^2W_\nu(p).$$

Here $V_\nu=-i(e^{ip_\nu}T_\nu^+-e^{-ip_\nu}T_\nu^-)$ and $W_\nu=e^{ip_\nu}T_\nu^++e^{-ip_\nu}T_\nu^-$. For $N_0(t)=\det(D^{\mathrm{ch}}[t\eta a_\eta]+m)/\det(D^{\mathrm{ch}}[0]+m)$ put $c_j=N_0^{(j)}(0)$. The transpose symmetry gives $N_0(-t)=N_0(t)$, so $c_2=(\log N_0)''(0)$. Define

$$\begin{aligned}L_{\mu\nu}(p,k)&=\delta_{\mu\nu}\operatorname{tr}(R_m(p)W_\nu(p))\\&\quad-\operatorname{tr}(R_m(p)V_\mu(p+k/2)R_m(p+k)V_\nu(p+k/2)),\\ \Pi^{\eta,R}_{\mu\nu}(K)&=\eta^{-1}n^{-3}\sum_{p\in\mathcal{P}_n}L_{\mu\nu}(p,\eta K),\\ \Pi^{\eta}_{\mu\nu}(K)&=\eta^{-1}\int_{[-\pi,\pi]^3}L_{\mu\nu}(p,\eta K)\frac{dp}{(2\pi)^3},\end{aligned}$$

where $\mathcal{P}_n=(2\pi/n)(\mathbb{Z}+\frac12)^3$ modulo $2\pi$.

Claim. For $K$ in the bosonic momentum representatives, the exact ordinary second determinant coefficient is

$$c_2=R^{-3}\sum_K\sum_{\mu,\nu}\overline{h_{\eta,\mu}(K)}h_{\eta,\nu}(K)\Pi^{\eta,R}_{\mu\nu}(K).$$

Uniformly in the mesh and volume,

$$|\Pi^{\eta,R}-\Pi^\eta|\le C_\rho\bar m^{-3}R^{-4},\quad |\Pi^\eta(K)|\le C_\rho\bar m^{-1}|K|^2,\quad |\nabla_K\Pi^\eta(K)|\le C_\rho\bar m^{-1}|K|.$$

Thus $|c_2|\le C_{\rho,\bar m,\beta}$ for $R\ge 1$. Together with T03, $|c_4|$ is uniformly bounded for $R\ge D_\beta+1$, and $c_4=b_{\mathrm{ch}}+3c_2^2$. The coefficients are real, odd ones vanish, and none depends on the physical charge used later.

Proof work. Derive the midpoint normalization in the coefficient identity. At zero momentum the bubble and contact term form $\partial_\mu \operatorname{tr}(R_m V_\nu)$, whose period integral vanishes. Parity removes the first momentum derivative. Bound the Hessian by an integrable inverse fourth power. Four integrations by parts give quadrature error, using $\eta^{-1} n^{-4} m^{-3} = \bar{m}^{-3} R^{-4}$. Pair with the mesh-uniform transform decay obtained by discrete integration by parts. The two original traces must be combined.
Inputs. The displayed free-symbol hypotheses, determinant transpose symmetry, smooth probe bounds, and T03 for the fourth coefficient.
Source. PDF pp. 190–194, §§18.6–18.7, Propositions 18.2–18.3, equations (18.29)–(18.43).

**T05. Exact probe observables and separated cell sources**

MODEL-SPECIFIC. Overview Target.

Proof dependencies. T01
Objects and hypotheses. Fix real $\beta \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ and $f = dd^*\beta$. Let $Q_\beta$ have side $D_\beta^{\mathrm{buf}} = D_\beta + 4$ and contain $\operatorname{supp}\beta + [-2,2]^3$. With centered differences and cube average define

$$a'_{\eta,\nu} = -\sum_{\mu\neq\nu} \delta_{\eta,\mu}\beta_{\mu\nu}, \qquad f^\eta_{\mu\nu} = \delta_{\eta,\mu}a'_{\eta,\nu} - \delta_{\eta,\nu}a'_{\eta,\mu}.$$

On the physical lattice let $b_{\eta,\mu\nu}(c) = J_\eta\beta_{\mu\nu}(c+\eta(e_\mu+e_\nu)/2)$, $a^o_\eta = d^*_\eta b_\eta$, and $\widehat{a}^o(x,x+e_\nu) = \sqrt{\eta}\, a^o_{\eta,\nu}(\eta x)$. On a unit-lattice torus of side $n = R/\eta$, $\widehat{a}$ means the periodization $\widehat{a}(b) = \sum_{z\in n\mathbb{Z}^3}\widehat{a}^o(b+z)$; the buffered support condition below makes distinct supporting copies disjoint. Cell sources are periodizations of the reflected cover sources by this same rule. Here $d_\eta$ is the forward differential divided by $\eta$, and its adjoint uses $\eta^3$ times counting measure. For a smooth two-form $g$, set $g_{[\eta]}(x;\mu\nu) = \eta^{3/2}J_\eta g_{\mu\nu}(\eta x+\eta(e_\mu+e_\nu)/2)$. The field $F^\sharp$ has value $\eta^{-3/2}F^u(x;\mu\nu)$ on the cube centered at that plaquette center.

Let $R^* = hL^{T^*}$, $n_* = R^*/\eta$, $p = n_*/L$ odd, $w = p/2$, $\ell_c = R^*/(2L)$ and $N_* = 8L^3$. Cell copies are the ordinary covariant reflected copies in the open cubes $\ell_c(c+(0,1)^3)$. Require $D_\beta^{\mathrm{buf}} + 3\eta \le \ell_c$. For terminal estimates require the translated buffered cube to lie in $[\ell_c/8, 7\ell_c/8]^3$.
Claim. The source is transverse and

$$d\widehat{a} = (f^\eta)_{[\eta]}, \qquad \langle d\widehat{a}, F^u\rangle = F^\sharp(f^\eta), \qquad \|f^\eta - f\|_{H^s} \le \eta^2\|\beta\|_{H^{s+4}}.$$

Its cell copies satisfy $\langle d\widehat{a}_c, d\widehat{a}_{c'}\rangle = v_\eta \mathbf{1}_{c=c'}$, where

$$|\sqrt{v_\eta} - \|f\|_2| \le 21\eta|\beta|_{C^3}(D_\beta^{\mathrm{buf}})^{3/2} + \eta^2\|\beta\|_{H^4}.$$

At $a_* \in \{h/L, h/L^2\}$ rescale $a[j](b) = L^{j/2}a(L^j b)$. For $a(t) = e^{-1}\sum_c t_c\widehat{a}_c$,

$$|a(t)|_{\mathrm{ext}} \le C_\beta|t|_\infty/e, \quad v_\eta \le \mathbf{D}, \quad d_{\mathrm{sep}} \ge Mr_*/(8L) - 1, \quad |P_c^+| \le (D_\beta^{\mathrm{buf}}/a_* + 2M + 4)^3 =: p_\beta.$$

The external norm is stop sup norm plus first-difference sup norm plus its short $\alpha$-Hölder norm. One may bound $C_\beta \le 2\sqrt{h}(1+h+3h^2)|\beta|_{C^3}$ and $\mathbf{D} \le 48|\beta|^2_{C^2}(D_\beta^{\mathrm{buf}}+3)^3$. Here $P_c$ consists of unit-block centers containing a supporting bond endpoint and $P_c^+$ is its $M+1$ neighborhood.
Proof work. Check midpoint scaling, support buffers, nonoverlapping period copies, and polynomial Gaussian-growth bounds. Passing from endpoints to unit-block centers loses one in separation. Even cell indices give translations by $y \in (R^*/L)\mathbb{Z}^3$, so the observable is $F^\sharp(\tau_{v+y}f^\eta)$.
Inputs. T01's stop scales; elementary cochain adjoints and averaging.
Source. PDF pp. 194–197, §19, and pp. 205–207, §21.1, equations (19.1)–(19.11), (21.1)–(21.9).

**T06. Zero flux equals a harmonic-averaged Gaussian law**

INTERFACE. Overview Target.

Proof dependencies. M01

Objects and hypotheses. On a rectangular cubical torus with sides $n_\mu \geq 4$, use counting pairings and $V = n_1 n_2 n_3$. Fix $q > 0$, $r_q = 2\pi/q$. Let $W(B)$ be a continuous weight on real bond cochains, invariant under real exact shifts and under $r_q$ times integer cochains. Assume $W(B) \geq 0$ for real $B$ and that its zero-flux integral, with the Maxwell Gaussian and compact-link measure, is finite and strictly positive. In the application $W(B) = \det(D_B + \mu_{\mathrm{lat}})$ with real total mass; determinant positivity and the nonzero $z = 0$ integral verify these hypotheses. For complex response assume $W$ is entire and bounded on each imaginary strip uniformly in the real part. Let

$$V_g = dC^0, \quad V_h = \mathrm{span}\{\mathbf{1}_\mu\}, \quad V_t = (V_g + V_h)^\perp, \quad H_{\theta,\mu} = \theta_\mu/(qn_\mu), \quad \Delta_0 = d^*d.$$

For closed integer two-cochains, zero flux means zero sum on each coordinate two-torus. The photon curvature is $F = dA + r_q z$.

Claim. For exponentially bounded Borel $\Phi$,

$$Z_n^{(0)}[\Phi] = \mathcal{N}_n \int_{[-\pi,\pi]^3} \frac{d\theta}{(2\pi)^3} \int_{V_t} \Phi(dA) e^{-\|dA\|^2/2} W(A + H_\theta)\, d\sigma(A),$$

where

$$\mathcal{N}_n = r_q^{V+2} (\det{}' \Delta_0)^{1/2}, \quad G_n = \frac{(2\pi)^{V-1}}{\det{}' \Delta_0}, \quad Z_n^{\mathrm{ph}} = \mathcal{N}_n G_n.$$

Here $Z_n^{(0)}$ sums zero-flux integer $z$ and integrates each link of $A$ over $[0, r_q)$. If $\gamma_n$ is the standard Gaussian on $F_n = dC^1$ and $A(u)$ its transverse potential, then

$$Z_n^{(0)}[1] = Z_n^{\mathrm{ph}} z_{r,n}, \qquad z_{r,n} = \int \frac{d\theta}{(2\pi)^3} \int W(A(u) + H_\theta)\, d\gamma_n(u) > 0.$$

Normalizing gives equality of the two curvature probability laws.

Proof work. Prove real Fourier Hodge decomposition and integer zero-flux exactness separately. Decompose closed integer one-cochains into an integer gradient plus three seam classes. Unfold the compact-link fundamental domain, retaining the coupling of exact and harmonic parts. Their Jacobians are $r_q^{V-1}(\det{}' \Delta_0/V)^{1/2}$ and $V^{1/2} q^{-3} d\theta$. On $V_t$, $\det(d^*d) = (\det{}' \Delta_0)^2$. These compute all normalization factors. Establish absolute convergence before Fubini, and integrable strip majorants before holomorphic parameter integration.

Inputs. Finite-dimensional gauge/phase invariance and strict zero-sector normalization. For the chamber determinant, prove its degree $\leq 32$ in each oriented link phase and its imaginary-strip bound; the denominator free mass is not the total numerator mass.

Source. PDF pp. 197–203, §§20.1–20.5, Proposition 20.1, equations (20.1)–(20.29).

**T07. Sectorwise source shift and the full Wick derivative**

INTERFACE. Overview Target.

Proof dependencies. T05, T06

Objects and hypotheses. Use a finite cubical torus, $r_q = 2\pi/q$, and an entire weight $W$ with the integer-period and gauge invariances and strip bounds in T06. Let $Z_n^{(w)}$ be the photon integral in integer flux $w$. Define $Z_n^{x,(w)}(a)$ by replacing $W(A)$ with $W(A+a)$ while leaving the photon curvature $F = dA + r_q z$ unchanged. Pair complex cochains bilinearly, without conjugation. Let $\mathbb{E}_n^{\mathrm{av}}$ be the normalized zero-flux law. Choose real sources $a_c$ with $\langle da_c, da_{c'}\rangle = v\mathbf{1}_{c=c'}$, and put

$$X_c = \langle da_c, F\rangle, \quad a(t) = e^{-1}\sum_c t_c a_c, \quad \mathring{\Phi}(t) = Z_n^x(a(t))/Z_n^x(0),$$

where $Z_n^x$ is the harmonic-averaged transverse integral of T06.

Claim. For every complex source and every flux sector,

$$Z_n^{(w)}[e^{\langle da,F\rangle}] = e^{\langle da,da\rangle/2} Z_n^{x,(w)}(a).$$

The sector sums are absolutely convergent. Define $H_m(x;v) = \partial_z^m e^{zx - vz^2/2}|_{z=0}$. For $m = 2, 4$, every finite source set $C$, and real $c_m$,

$$\mathbb{E}_n^{\mathrm{av}} \prod_{c\in C} [e^{-m} H_m(X_c; v) - c_m] = \left[\prod_{c\in C} (\partial_{t_c}^m - c_m)\mathring{\Phi}\right](0).$$

In particular the reference quantity is the derivative of order $m$ in every cell variable, followed by an $|C|$th root. It is not a one-cell or logarithmic derivative.

Proof work. For a real source translate the compact-link fundamental domain and permute exact integer representatives within each fixed flux. Complete the photon square and extend to complex sources by entire parameter integrals. Normalize by the positive zero-source partition. Differentiate the exact identity

$$\mathbb{E}_n^{\mathrm{av}} \exp \sum_c (z_c X_c - v z_c^2/2) = \mathring{\Phi}(ez).$$

The factor $e$ in every marked variable accounts for all inverse charge powers. Exponential majorants justify differentiation and expansion of the centered product.

Inputs. T06's exact zero-sector equality and positivity, and T05's source orthogonality. No interacting estimate.

Source. PDF pp. 203–205, §20.6, Proposition 20.2, equations (20.30)–(20.36).

**T08. Terminal inverses, marked legs and coefficient norms**

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S02, S06, T01

Objects and hypotheses. Fine spinor blocks have total measure one per unit block, with four spin components and normalized chamber labels; coarse sites have four components and counting measure. Let $Q_A$ be fine-to-coarse averaging and $Q'_A$ its analytic transpose partner, equal to $Q_A^*$ for real $A$. Assume $Q_A Q_A^* = I$ and $\mathrm{Re}(D_A + \mu_*) \geq \mu_* I$ for real $A$. Set

$$F_* = D_A + \mu_* + b_* Q'_A Q_A, \quad S_* = F_*^{-1}, \quad H_* = b_* S_* Q'_A, \quad D_* = b_* - b_* Q_A H_*.$$

Assume $k_* \geq 1$, $\ell_* > 2M + 2$, $b_- = 1 - L^{-1} \leq b_* \leq 1$, $\mu_- = L^{-3} < \mu_* \leq L^{-1}$ and $\gamma^F M \geq 16$. On a symmetric open connection domain $W$, assume $e_* |\operatorname{Im} A| < 1/16$, $3M e_* |\operatorname{Im} A| < 1/4$, and the verified local inverse and short covariant-Hölder bounds $C^F e^{-\gamma^F d(y,y')}$ for $S_*, H_*$. Here $0 < \alpha < 1$, $\delta_{\alpha,A} v(x,x') = [U_A(x,x')v(x') - v(x)]/d_X(x,x')^\alpha$, where $U_A$ is the phase along the prescribed pair path. The short bound applies to $0 < d_X(x,x') < 1$, including distinct points within a chamber cluster, where the phase is one. Average phases use paths of length $\leq 3/2$ and pair phases of length $\leq 3d_X$; allowed long pairs have $1 \leq d_X \leq M$. Distances between unit blocks are periodic sup distances. For a block-localized operator put

$$|O|(y,y') = \sup_{\|v\|_\infty \leq 1} \|\mathbf{1}_y O \mathbf{1}_{y'} v\|_\infty,$$

$$\|O\|_\gamma = \max\left\{ \sup_y \sum_{y'} |O|(y,y') e^{\gamma d(y,y')}, \sup_{y'} \sum_y |O|(y,y') e^{\gamma d(y,y')} \right\}.$$

Fine variables are localized in their unit blocks and pair variables at their first endpoint; the sup norm is the maximum of spinor component moduli. For a direction $B$ define $\epsilon_B = e_* |B|_\infty$, $P_B$ by supporting bond endpoints' unit blocks, and $P_B^+ = \{y : d(y, P_B) \leq M + 1\}$.

Claim. First derivatives of $D_*$ are bounded by

$$226 (C^F)^2 \epsilon_B \sum_{z \in P_B} e^{-\gamma^F d(y,z)} e^{-\gamma^F d(z,y')}.$$

Put $\mathcal{S} = C^F (24/\gamma^F)^3$, $C_K = 226\mathcal{S}^2$, $C'_K = 1328\mathcal{S}^2$, $C_H = 203M(C^F)^2$. Writing $E(y,y') = e^{-\gamma^F d(y,y')}$, for $d(P_B, P_{B'}) \geq M + 2$ the mixed bound is

$$|\partial_B \partial_{B'} D_*|(y,y') \leq C_H^2 \epsilon_B \epsilon_{B'} \sum_{\substack{z \in P_B \\ z' \in P_{B'}}} [E(y,z)E(z,z')E(z',y') + E(y,z')E(z',z)E(z,y')].$$

Let $\mathrm{Leg}_A(u; y')$ be any coefficient of a field $H_*^\# \Psi$ or its covariant quotient, barred or unbarred. Set $\mathrm{pos}(u)$ equal to its first endpoint's block, adding the second endpoint for long pairs, so it has at most two elements. The leg bounds are

$$|\mathrm{Leg}_A(u; y')| \leq 3C^F \sum_{w \in \mathrm{pos}(u)} E(w, y'),$$

$$|\partial_B \mathrm{Leg}_A(u; y')| \leq C_H \epsilon_B \sum_{\substack{w \in \mathrm{pos}(u) \\ z \in P_B^+}} E(w,z)E(z,y'),$$

$$|\partial_B \partial_{B'} \mathrm{Leg}_A(u; y')| \leq C_H^2 \epsilon_B \epsilon_{B'} \sum_{\substack{w \in \mathrm{pos}(u) \\ z \in P_B^+, z' \in P_{B'}^+}} [E(w,z)E(z,z')E(z',y') + E(w,z')E(z',z)E(z,y')].$$

The first derivative weighted row/column norm is at most $C_K\epsilon_B$ at rate $\gamma^F/2$, and its total block sum at most $C_K\epsilon_B|P_B|$.

At every real field, without a small-field assumption,

$$D_*^{-1} = b_*^{-1} + Q_A(D_A+\mu_*)^{-1}Q_A^*, \qquad \|D_*^{-1}\|_2 \le b_-^{-1} + \mu_-^{-1}.$$

Its weighted component rows and columns are bounded by $\alpha_G = 2b_-^{-1} + 4\exp(1)\mu_-^{-1}(24/\theta)^3$, at rate $\vartheta_0 = \min(\theta/2, \gamma^F/8, 1/M)$, where $\theta = \mu_-/(12\exp(1))$. If $48e_*^{2\varepsilon} \le \mu_-$ and $2g_1C_K'e_*^{2\varepsilon} \le 1/2$, the corresponding marked complex neighborhood has inverse norm $\le g_1 = b_-^{-1} + 4/\mu_-$. Fix the calibration

$$c(\gamma) = \sum_{z\in\mathbb{Z}^3} e^{-\gamma|z|_\infty}, \qquad c_{\mathrm{bd}}(\gamma) = 3e^{3\gamma}c(\gamma),$$

$$C_0 = \max\{C_P^{\mathrm{ph}}c_{\mathrm{bd}}(\gamma_P),\, C^F,\, 4b_+ + 16C^F c(\gamma^F)\}.$$

The positive constants $C_P^{\mathrm{ph}}, \gamma_P$ bound the photon reconstruction kernel by $C_P^{\mathrm{ph}}e^{-\gamma_P d}$. For $a_2 = C_0^{-1}e_*^{-1/4+\varepsilon}$,

$$|F(H_*^{\#}\Psi)|_{a_2} \le \|F\|_{h_*}, \quad |\langle\bar\psi,\psi\rangle|_{a_2} \le 4e_*^{-1/2+2\varepsilon}\ell_*^3, \quad |\langle\bar\Psi, D_*\Psi\rangle|_{e_*^{-1/4}} \le C_0 e_*^{-1/2}\ell_*^3.$$

The coefficient norm $|\cdot|_a$ is the absolute coefficient sum weighted by $a$ per generator. The fine kernel norm $\|\cdot\|_{h_*}$ uses its fine integration weights and $h_* = (e_*^{-1/4}, e_*^{-1/4+\varepsilon})$ for spinor and covariant-Hölder generators, respectively.

Proof work. Differentiate local phases and inverses, retaining marked supports. Prove all-pair row bounds, including long and intra-cluster pairs. Use accretivity and finite-rank inversion for the real covariance, an exponential distance weight for decay, and a Neumann series for the complex neighborhood. The fine mass form has factor four; the label measure cancels the factor forty-eight.

Inputs. The displayed walk/average and photon kernel estimates with the three-part $C_0$ calibration. All marked paths must lie in the stated domains.

Source. PDF pp. 207–213, §§21.2–21.4, Proposition 21.1, equations (21.10)–(21.30); the calibration is (24.6).

**T09. Sector domination and the exact terminal measure**

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M03, S04, S12, S15, T01, T06, T08

Objects and hypotheses. Let $\bar{\mu}$ be the normalized harmonic-averaged zero-flux law with real lattice mass and coordinate sides $\geq 4$. Let $F_N = dC^1$ have its counting inner product and standard Gaussian $\gamma_N$. The following coordinate-slice trace is an input, in each cyclic choice of time direction. For time direction 1, let $\Sigma$ be the $n_2$ by $n_3$ slice, $r_q = 2\pi/q$, $v_s = n_2 n_3$, $\Gamma_s = r_q(\mathbb{Z}^{\Sigma^{(1)}} \cap \ker d_s)$, and $X_s = \mathbb{R}^{\Sigma^{(1)}}/\Gamma_s$. There are finite crossing spaces $\mathcal{F}', \mathcal{F}$, and

$$H^s = \bigoplus_{k\in\mathbb{Z}} L^2(X_s; \mathcal{F}'), \qquad H^l = \bigoplus_{k\in\mathbb{Z}} L^2(X_s; \mathcal{F}).$$

The index $k$ is the slice magnetic flux; $P_0$ is its zero projection. A bounded $r_q$-periodic half-slab matrix $\widehat{N}(A) : \mathcal{F}' \to \mathcal{F}$ defines

$$(M_0\phi)_k(A) = e^{-\|d_s A + r_q k \mathbf{1}_{\sigma_0}\|^2/4} \widehat{N}(A)\phi_k(A),$$

where $\sigma_0$ is one fixed slice plaquette. Here $F^{\mathrm{mag}}$ is multiplication by $d_s A + r_q k \mathbf{1}_{\sigma_0}$, and $F_j^{\mathrm{el}}$ is the unfolded slice increment $A_{j+1} - A_j$. The commuting self-adjoint operators $E_b$ generate translations $e^{-i\langle x, E\rangle}\phi(A) = \phi(A - x)$. If $u(a)$ is the crossing-space unitary with integer mode charges,

$$(P_G\phi)(A) = r_q^{-v_s} \int_{[0,r_q)^\Sigma} u(a)\phi(A - d_s a)\, da$$

defines the orthogonal gauge-average operator. It commutes with functions of $E$. For the two slice cycles $S_2, S_3$ and $\xi \in [0,1]^2$ set $b_\xi = q(\xi_2 \mathbf{1}_{S_2} + \xi_3 \mathbf{1}_{S_3})$. For bounded magnetic factors $u_j$ and sub-Gaussian electric factors $v_j$, let $K_j^\xi$ be the periodized convolution with kernel $v_j(A' - A)e^{i\langle b_\xi, A'-A\rangle} \prod_b g_1(A'(b) - A(b))$, where $g_1(t) = (2\pi)^{-1/2}e^{-t^2/2}$. The precise trace input is

$$Z^{(0)}\left[\prod_j u_j(F_j^{\mathrm{mag}})v_j(F_j^{\mathrm{el}})\right] = (2\pi)^{n_1 n_2 n_3} \int_{[0,1]^2} \mathrm{tr}\left[P_0 \prod_{j=n_1-1,\dots,0} r_q^{v_s} K_j^\xi P_G M_0 u_j(F^{\mathrm{mag}}) M_0^*\right] d\xi.$$

For a real two-cochain $\beta$ take $u_j = e^{-\langle \beta_j^{\mathrm{mag}}, F^{\mathrm{mag}}\rangle - \|\beta_j^{\mathrm{mag}}\|^2/2}$ and $v_j = e^{-\langle \beta_j^{\mathrm{el}}, F^{\mathrm{el}}\rangle - \|\beta_j^{\mathrm{el}}\|^2/2}$. The needed source factorization is

$$K_j^\xi = z_j U_j e^{-|E-b_\xi|^2/2}, \quad U_j = e^{i\langle \beta_j^{\mathrm{el}}, E\rangle}, \quad |z_j| = 1,$$
$$K_\xi = r_q^{v_s} e^{-|E-b_\xi|^2/2} P_G, \qquad Y_j = K_\xi^{1/2} M_0 \min(u_j, \delta^{-1})^{1/2}.$$

$Y_j$ is Hilbert–Schmidt uniformly in $\xi$ for every $\delta > 0$; all factors preserve magnetic flux. Cyclicity changes the truncated trace into $\prod_j z_j \,\mathrm{tr} \prod_j U_j P_0 Y_j (P_0 Y_j)^*$. These trace and factorization identities are assumptions to verify from the slab representation, rather than an assumption of Gaussian domination.

For a controlled flow with stop $k$, assume the exact photon minimum/fluctuation identity at every level, with linear fluctuation map $\mathrm{Sq}_j$. Recursively define $A_k^\downarrow = A$ and $A_j^\downarrow = A_j^{\min}(A_{j+1}^\downarrow) + \mathrm{Sq}_j\zeta_j$. Use the axial slice with coclosed mean-zero terminal $A$. The initial normalized determinant density is $w_N(B)$. For the fine transverse potential $A(F)$, define $d\nu^{\theta,a}(F) = w_N(A(F) + H_\theta + a)d\gamma_N(F)$, where $H_{\theta,\mu} = \theta_\mu/(qn)$. Its twist average, normalized by its positive mass, is $\bar{\mu}$. Assume the actual integrated RG density recursion

$$y_{j+1}(A'; a[j+1]) = \int_{\mathrm{Box}_j} y_j(A_j^{\min}(A') + \mathrm{Sq}_j\zeta_j; a[j])\, d\mu_I(\zeta_j), \quad y_0(B; a) = w_N(B + a),$$

on domains preserved by these boxes.

Claim. First establish, with the zero-sector projection retained,

$$\bar{\mu}(e^{\langle g,F\rangle}) \le e^{\|\Pr_{F_N} g\|^2/2}.$$

Curvature has bijective coordinates $(A_*;\zeta_0,\dots,\zeta_{k-1})$ with

$$\|F\|^2 = \|dH_k^{\mathrm{ph}} A_*\|^2 + \sum_{j<k} \|\zeta_j\|^2, \qquad \gamma_N \longmapsto m_*(dA_*) \prod_{j<k} d\mu_I(\zeta_j).$$

Fluctuation rows are orthonormal; each terminal plaquette row has norm $\le 1$. Their counts are respectively $\le 2\ell_*^3 L^{3(k-j)}$ and $3\ell_*^3$. Let $\mathfrak{G}$ require $|\zeta_j| \le p_{0,j}$ and $|dA_*| \le p_*$. Denote the terminal integrated fermion density by $Y_*^x$ and let $\nu^{\theta,a}$ be the fine determinant density against $\gamma_N$. For every bounded Borel coarse observable,

$$\nu^{\theta,a}(1_{\mathfrak{G}}\Phi(A_*)) = \int m_*(dA)\Phi(A)\chi_*(A)Y_*^x(A + c_\theta; a[k]).$$

The density is bounded, Borel, and jointly continuous on $\chi_* = 1$. Here $\chi_*(A) = 1_{\{|dA|\le p_*\}}$, $c_{\theta,\mu} = \theta_\mu/(e_*\ell_*)$, $p_* = v_*^p$ and $p_{0,j} = (v_* + (k-j)\log L/2)^{p_0}$. If $p_0 v_*^{p_0-1} \ge 28$, $v_* \ge 8$, $L \ge 5$, then

$$\bar{\mu}(\mathfrak{G}^c) \le \epsilon_3 := 6\ell_*^3(e^{-p_{0,*}/2} + e^{-p_*^2/(2C_0^2)}), \qquad m_*(\chi_* = 0) \le \epsilon_4 := 6\ell_*^3 e^{-p_*^2/(2C_0^2)}.$$

Proof work. In the displayed projected trace apply Schatten Hölder to its $2n_1$ Hilbert–Schmidt factors and then Hölder in $\xi$. The result is the geometric mean of zero-sector functionals with one magnetic factor repeated on every slice and no electric factor. Send $\delta$ to zero by monotone convergence. A cyclic coordinate change makes each repeated factor purely electric; the same inequality bounds it by $Z^{(0)}[1]$. This proves sector domination with its projection retained. Prove that the full photon tower is an isometry, including its normalized Jacobian and coordinate counts. Iterate the exact density over compact boxes; keep first-step $192n^3$ weighted pairs and terminal $4\ell_*^3$ weight-one pairs. Use the displayed Borel identity for arbitrary coarse events and the row/count bounds for the complement union bound.

Inputs. T01, T06; the explicitly stated slab trace/factorization, exact single-step photon and fermion density conclusions. No desired partition lower bound is an input.

Source. PDF pp. 213–219, Proposition 21.2, Propositions 22.1,22.3, Theorem 22.2, equations (21.31)–(22.17).

**T10. Free pressure and mesh-uniform block-penalty twists**

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S01, S06

Objects and hypotheses. For the 192-dimensional chamber symbol let $d_m(p) = \det D_m(p) > 0$ and $p^0_{\mathrm{ch}}(m) = \int \log d_m(p)\, dp/(2\pi)^3$. Use T03's periodic symbol and its light–heavy decomposition $D_m = \left(\begin{smallmatrix} D_{\mathrm{ll}} & D_{\mathrm{lh}} \\ D_{\mathrm{hl}} & D_{\mathrm{hh}} \end{smallmatrix}\right)$, with $B_m = D_{\mathrm{ll}} - D_{\mathrm{lh}} D_{\mathrm{hh}}^{-1} D_{\mathrm{hl}}$. For positive constants $g_{\mathrm{h}}, \theta_0, \kappa_{(0)}$ depending only on $\rho$, assume the following explicit strip bounds:

$$\|D_{\mathrm{hh}}(z)^{-1}\| \le 2/g_{\mathrm{h}} \quad (|\operatorname{Im} z_\nu| \le \theta_0),$$

$$\|\partial^\alpha B_m(p + ip'')^{-1}\| \le C_{\alpha,\rho}(|p|^2 + m^2)^{-(1+|\alpha|)/2} \quad \left( \begin{array}{c} |\alpha| \le 3,\ p \in [-\pi,\pi]^3, \\ |p''_\nu| \le \kappa_{(0)} m \end{array} \right),$$

$$\kappa_{(0)} m \le \theta_0/2, \qquad \|\partial^\beta B_m\| \le 24 K_s (2/\theta_0)^{|\beta|} \quad (1 \le |\beta| \le 4)$$

on the latter strip, with $K_s$ independent of $m \in (0,1]$. The matrices and derivative traces are holomorphic on neighborhoods of the closed strips; their periodicity cancels the vertical contour sides. At zero mass the coercivity input is $\|D_0(p)u\| \ge c_{\mathrm{nz}}|p|_2\|u\|$.

For the block statement take odd $N \ge 3$, $\tau = N^{-1}$, a fine chamber torus of side $\ell$ with boundary multipliers $-e^{i\theta_\mu}$, and unit-block averaging $Q$ with $QQ^* = 1$. For $0 < \mu \le 1$ and $0 < b_- \le b \le b_+$ define

$$T_\theta = D + \mu + bQ^*Q, \qquad D^{\mathrm{bl}}_\theta = b - b^2 Q T_\theta^{-1} Q^*.$$

The fine measure is $\tau^3/48$, and the coarse measure is counting.

Claim. There are $C_6, \kappa_{\mathrm{strip}} > 0$, depending only on the chamber parameter, such that

$$|\log\det(D^{\mathrm{ch}}[qH_\theta] + m) - n^3 p^0_{\mathrm{ch}}(m)| \le \epsilon_5(mn), \quad \epsilon_5(x) = C_6 \log(1 - e^{-\kappa_{\mathrm{strip}} x})^{-1}.$$

This is uniform in $n \ge 1$, $0 < m \le 1$ and real twist, including the harmonic average. At $m = \bar m \eta$ its argument is $\bar m R$. Moreover all three block determinants are positive,

$$(D^{\mathrm{bl}}_\theta)^{-1} = b^{-1} + Q(D + \mu)^{-1} Q^*, \quad \det D^{\mathrm{bl}}_\theta = b^{4\ell^3} \det(D + \mu)/\det T_\theta,$$

$$|\log(\det T_\theta/\det T_0)| \le C_\Sigma \ell^{-1} e^{-\gamma_\Sigma \ell},$$

with $C_\Sigma < \infty$, $\gamma_\Sigma > 0$ depending only on $\rho, b_-, b_+$, not on $\tau, \mu, \ell$. At the terminal scale this identifies the coarse twist ratio with the fine free determinant ratio times $\det T_0/\det T_\theta$.

Proof work. For pressure, shift differentiated Fourier coefficients and sum the modes selected by the torus; bound the light and heavy parts separately. For the penalty, prove a uniform one-cell inverse. In periodic gauge its symbol derivatives satisfy $\|\partial^j_\nu D(p)\| \le 9\tau^{j-1}$, $\|\partial^j_\nu (Q^*Q)(p)\| \le \exp(1)$ and rank $\le 4(j+1)$ for $1 \le j \le 4$. Use weighted Schatten estimates for the inverse to bound fourth derivatives of the logarithmic determinant independently of the growing fine dimension. Fourier decay and a strip shift then give the exponential twist error.

Inputs. The chamber coercivity, light/heavy symbol and holomorphic inverse estimates; finite-rank inverse/determinant identities.

Source. PDF pp. 219–224, §22.4, Propositions 22.4–22.5, equations (22.18)–(22.35).

### T11. Two-sided partitions and a common matter pressure

MODEL-SPECIFIC. Overview Target.

Proof dependencies. T01, T08, T09, T10

Objects and hypotheses. Use a controlled real flow on $R$, with positive running mass $\mu_*$, possibly nonzero terminal coupling $m_*$, vacuum $\epsilon_k^{\mathrm{v}}$, terminal density $Y_*$ and coarse twist $c_{\theta,\mu} = \theta_\mu/(e_*\ell_*)$. Let $R_\theta^{\mathrm{tw}} = \det D_*(c_\theta)/\det D_*(0) > 0$. Let $C_0$ bound terminal field substitutions, the quadratic coefficient norm and photon row sums; let $c_\kappa$ be the anchored polymer sum. Require

$$8C_0^2\ell_*^6(b_-^{-1}+\mu_*^{-1}) \le e_*^{-\varepsilon/2}, \quad 4(2C_0+c_\kappa+5c_{\mathrm{fl}}+6)\ell_*^3 \le e_*^{-\varepsilon},$$
$$e_*^{1/4-9\varepsilon} \le 1/8, \quad 4C_0\ell_*^2 p_* < e_*^{-\varepsilon},$$
$$12\ell_*^3(e^{-p_{0,*}/2}+e^{-p_*^2/(2C_0^2)}) \le e_*, \qquad C_\Sigma e^{-\gamma_\Sigma \ell_*} \le e_*.$$

Use $\epsilon_3, \epsilon_4$ from T09, $\epsilon_1 = e_*^{1/4-9\varepsilon}/2$, and $\epsilon_2 = C_\Sigma \ell_*^{-1} e^{-\gamma_\Sigma \ell_*}$.

Claim. For real mean-zero $A$ with $|dA| \le p_*$,

$$Y_*(A+c_\theta) = e^{-\epsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}}(1+r_*(\theta,A)), \qquad |r_*| \le \epsilon_1.$$

The positive fine measure then satisfies

$$e^{-\epsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}}(1-\epsilon_1)(1-\epsilon_4) \le \nu^\theta(\mathfrak{G}) \le e^{-\epsilon_k^{\mathrm{v}}\ell_*^3} R_\theta^{\mathrm{tw}}(1+\epsilon_1),$$

and $(1-\epsilon_3)\bar\nu(1) \le \bar\nu(\mathfrak{G}) \le \bar\nu(1)$. For the common pair chosen on $R_{\mathrm{top}}$, the positive matter factor $z_{r,n}$, defined by the twist-averaged transverse determinant with total mass $\bar m\eta + m_{0,N}$, has pressure $p_F = p_{\mathrm{ch}}^0(\bar m\eta) + \epsilon_{0,N}^{\mathrm{v}}$ satisfying

$$|\log z_{r,n} - p_F n^3| \le \epsilon_5(\bar m R) + 3e_*^{1/4-9\varepsilon}.$$

Consequently, on the two reference tori with the same total mass,

$$|L^3 \log z_{r,n_*} - \log z_{r,Ln_*}| \le C_{\mathrm{mat}} := (L^3+1)(\epsilon_5(1/L)+1).$$

Proof work. Gauge the real field into the analytic domain. Control the quadratic difference by coefficientwise Cauchy estimates and the mass/activity terms by weighted Grassmann norms. Normalize the Gaussian Grassmann integral by one-sided covariance substitution. Use the exact Borel density and positive complement bounds before taking logarithms. Cancel the common bulk pressure exactly; retain the small terminal vacuum on the smaller torus.

Inputs. T01, T08–T10. The inequalities above are later verified by T16.

Source. PDF pp. 224–227, §§22.5–22.6, Propositions 22.6–22.7, Theorem 22.8, equations (22.36)–(22.50).

## T12. The one-cell normalized logarithm

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S08, T01, T05, T08, T09

Objects and hypotheses. Let $A$ be a real mean-zero terminal coarse connection on a torus of side $\ell_*$, with $|dA|_\infty \le p_*$. Choose its gauge representative $A^g = A - d\lambda$ with $|A^g|_\infty \le 2\ell_*^2 p_*$, and define $A^0 = H_*^{\mathrm{ph}} A^g$ using the photon reconstruction. Set $c_{\theta,\mu} = \theta_\mu/(e_*\ell_*)$, $\theta \in [-\pi,\pi]^3$, and $C(t) = e^{-1}\sum_c t_c \widehat{a}_c[k_*]$. Here $|A^0|_{\mathrm{each}}$ is the maximum of its sup norm, its mesh first-difference sup norm and their short $\alpha$-Hölder norm. Take $|A^0|_{\mathrm{each}} < e_*^{-\varepsilon}/2$ and the uniform probe norm $C_\beta$ and support count $p_\beta$ of T05. Put

$$C_b = 1 + \sqrt{h}C_\beta, \quad t_x = e_*^{3\varepsilon}/C_b, \quad t_x^+ = e_*^{3\varepsilon}/(\sqrt{h}C_\beta), \quad r = t_x/4.$$

The two-field activity is analytic on an open set containing all $(A^0 + c_\theta, C(t))$ with $|t|_\infty < t_x^+$. The operator bounds hold on all paths $c_\theta + C(t) + sA^0$, $0 \le s \le 1$, and their stated complex twist neighborhoods. Rescaled marks lie in each local determinant domain. Define the terminal weight-one Berezin integral

$$Z_{\mathrm{in}}(A,C) = \int D\Psi \, \exp\left[-\langle\bar{\Psi}, D_*(A+C)\Psi\rangle - m_*\langle\bar{\psi},\psi\rangle + \sum_X E_*(X;A,C,\psi^\#)\right],$$

where $\psi^\# = H_*^\#(A+C)\Psi$ includes the barred fields and covariant Hölder quotients and the fine mass pairing retains its fine measure. Set

$$f_{\mathrm{int}}(t) = \frac{Z_{\mathrm{in}}(A^0 + c_\theta, C(t))}{\det D_*(c_\theta + C(t))}.$$

Assume $|m_*| \le e_*^{3/4-8\varepsilon}/8$, $\|E_*\|^x \le 4e_*^{1/4-6\varepsilon}$ and the local scalar determinant bound $|E_j^d(X,A)| \le C_d M^3 e^{-(\kappa+1)d_M(X)}$. Required finite scalar conditions are

$$4C_0\ell_*^3 g_1 \le e_*^{-1/4+\varepsilon}, \quad 4\ell_*^3(2C_K + K_0 + 4) \le e_*^{-2\varepsilon},$$
$$e_*^{1/4-9\varepsilon} \le 1/8, \quad 48e_*^{2\varepsilon} \le \mu_-, \quad 2g_1 C_K' e_*^{2\varepsilon} \le 1/2, \quad r_* > 31L + 16.$$

Also require $B_0 \le 1/4$, $c_\tau\ell_* \ge 1$ and $52B_1 e^{-c_\tau\ell_*} \le e_*^{1/4-9\varepsilon}$. Here $c_\tau = \min(1/8, \mu_-/48, (292\mathcal{S})^{-1})$, $B_1 = 16 g_1 C_K' p_\beta e_*^{2\varepsilon}$ and $B_0 = C_{\mathrm{fr}} e_*^{4\varepsilon} + B_{\mathrm{wr}} + B_1$, with

$$C_{\mathrm{fr}} = 2592 C_d K_0 C_\beta^2 [M^2 (D_\beta^{\mathrm{buf}})^3 L^4 h^{-2} + 8M^5 h],$$
$$B_{\mathrm{wr}} = 16\exp(6) C_d M^3 K_0 p_\beta e^{-r_*}.$$

Claim. The terminal ratio $R_{\mathrm{tm}}(t) = Y_*^x(A + c_\theta; C(t))/Y_*^x(A + c_\theta; 0)$ is nonzero, analytic, jointly continuous in real data, and has a unique normalized logarithm $G(0) = 0$ on $\{t : |t|_\infty < t_x^+\}$. The ratio uses the original coarse connection $A$; gauge invariance and the exact reconstructed density identify its interaction integral with $Z_{\mathrm{in}}(A^0 + c_\theta, C(t))$. Moreover $|f_{\mathrm{int}} - 1| \le e_*^{1/4-9\varepsilon}/2$ on this full polydisc. The one-cell free determinant ratio $N_0$ with positive reference mass has an even logarithm $\ell_0$, real on the real axis, independent of cell and admitted placement, analytic on $|t| < t_x^+$, with $|\ell_0| \le B_0$ on $|t| \le t_x$ and

$$|G(te_c;\theta,A) - \ell_0(t)| \le 3e_*^{1/4-9\varepsilon} \qquad (|t| < t_x^+).$$

Proof work. Prove the exact determinant telescoping along all levels. Before summing scalar determinant activities, remove the locally constant probe by gauge invariance and use evenness for a quadratic gain. Estimate wrapping polymers separately. Bound the coarse one-cell twist logarithm in a strip and obtain the exponential twist suppression. Use the nonzero normalized ratio to fix the logarithm.

Inputs. T01, T05, T08–T09, local determinant analyticity/locality/evenness. T16 verifies these conditional smallness hypotheses.
Source. PDF pp. 227–233, §§23.1–23.2, Propositions 23.1–23.2, equations (23.2)–(23.29).

### T13. A connected logarithm with separated marked cells

Model-specific. Overview Target.

Proof dependencies. M20, T05, T08, T12

Objects and hypotheses. In a finite Grassmann algebra let the real covariance $G$ have operator norm $\le g^2$ and absolute weighted row and column sums $\le \alpha_G$ at rate $\vartheta_0 > 0$. Put $h_G = 1 + g$. For even $F$ without constant term, with antisymmetric kernels $F_p$, define

$$\rho_G(F; w) = \sum_{p\ge 2} \frac{h_G^p}{(p-1)!} \sup_{u_1} \sum_{u_2,\dots,u_p} |F_p| w,$$

$$N_G(F; w) = \sum_{p\ge 2} \frac{h_G^p}{(p-1)!} \sum_{u_1,\dots,u_p} |F_p| w.$$

Distances use unit sites of the barred/spin indices. Let $w_\delta = e^{\vartheta_0 \max_{i,j} d(u_i,u_j)}$ and $w_P = e^{\vartheta_0 \max_i d(u_i,P)}$. Suppose $V(t) = V_0(t) + V_+(t)$ is analytic, $V_+$ has no constant, and $\alpha_G \rho_G(V_+; w_\delta) \le 1/2$.

Claim. The connected-tree series $W$ is analytic and $\int d\mu_G e^V = e^W$. For distinct marks,

$$|\partial_c \partial_{c'}(W - V_0)| \le 2N_G(\partial_c \partial_{c'} V_+; 1) + 2\alpha_G N_G(\partial_c V_+; w_P)\rho_G(\partial_{c'} V_+; w_{P'}) e^{-\vartheta_0 d(P,P')}.$$

For the terminal activity, first absorb its mass in single-cube activities $E^a$. Define

$$V(t) = -\langle \bar\Psi, [D_*(A + C(t)) - D_*(A)]\Psi\rangle + \sum_X E^a(X; t)[H_*^{\#}(A + C(t))\Psi],$$

and let $V_0$ be its constant part. Require the fine absolute kernel bound at weight $a_w$ to be $e_*^{1/4-7\varepsilon} e^{-\kappa d_M(X)}$ whenever $a_w \le e_*^{-1/4+\varepsilon}$. Here $C(t)$ consists of separated linear connection marks, and the operator/leg estimates of T08 are explicit inputs. Under

$$5h_{\mathrm{L}} h_G \le e_*^{-1/4+\varepsilon}, \quad (4 + M^3/2)e_*^{\varepsilon} \le 1, \quad \alpha_G[4h_G^2 C_K e_*^{2\varepsilon} + 650 K_0 e_*^{1/4-7\varepsilon}] \le 1/2$$

and $d_{\mathrm{sep}} \ge 6M + 8$, put $h_{\mathrm{L}} = 6C^F(24/\gamma^F)^3$, $q_x = 2/t_x$ and $C_V^{\mathrm{ins}} = 10^3 C_H^2 (24/\gamma^F)^{12} K_0$. With $\bar r = 4h_G^2 C_K e_*^{2\varepsilon} + 650 K_0 e_*^{1/4-7\varepsilon}$, the precise bounds are

$$\rho_G(V_+; w_\delta) \le \bar r \qquad (|t|_\infty \le t_x),$$

and, for $|t|_\infty \le t_x/2$,

$$N_G(\partial_c V_+; w_{P_c}) \le q_x C_V^{\mathrm{ins}} h_G^2 p_\beta,$$

$$\rho_G(\partial_c V_+; w_{P_c}) \le q_x C_V^{\mathrm{ins}} h_G^2,$$

$$N_G(\partial_c \partial_{c'} V_+; 1) + |\partial_c \partial_{c'} V_0| \le q_x^2 C_V^{\mathrm{ins}} h_G^2 p_\beta e^{-\vartheta_0(d_{\mathrm{sep}} - 4M - 6)} \qquad (c \ne c').$$

Consequently each variable-set component $G_S(t) = \sum_{T\subset S} (-1)^{|S\setminus T|} G(t_T)$, $|S| \ge 2$, satisfies

$$\sup_{|t|_\infty \le t_x/2} |G_S| \le 2^{N_*} C_W^{\mathrm{mix}} p_\beta e^{-\vartheta_0(d_{\mathrm{sep}} - 4M - 6)},$$

where $C_W^{\mathrm{mix}} = 3C_V^{\mathrm{ins}} h_G^2 + 2\alpha_G (C_V^{\mathrm{ins}})^2 h_G^4 + 4\exp(6) C_d M^3 K_0$. In this application $G(t) = \sum_X P_*(X, C(t)) + W(t) - W(0)$. The scalar activities $P_*$ vanish at zero, are local in $X$, and obey $|P_*(X, C)| \le 2C_d M^3 e^{-\kappa d_M(X)}$ for $d_M(X) \ge 1$. Only such polymers can meet two separated marked cells.

Proof work. Verify the differentiated kernel bounds using marked field legs, including kernels, single differentiated legs and mixed legs. Prove forest interpolation, its Gram property and the residual determinant estimate. Root the kernel sums; keep exponential decay along the unique path joining the two marked vertices. Finally represent two coordinate differences by integrals of mixed derivatives. A bound for singleton marks alone is insufficient.

Inputs. T05, T08, T12; activity locality and anchored polymer sums. The first displayed tree assertion is model-general.
Source. PDF pp. 233–238, §23.3, Propositions 23.3–23.5, equations (23.30)–(23.44).

### T14. Positive small-field measure and a small complex complement

Model-specific. Overview Target.

Proof dependencies. T05, T06, T07, T08, T09, T12, T13

Objects and hypotheses. Use orthogonal cell source curvatures of norm squared $v_\eta \le \mathbf{D}$ and the exact terminal Borel density. Let $\bar\nu^{a(t)}$ denote the finite analytic external determinant measure, with source shift as in T07 and the Borel terminal identity as in T09; its zero-source mass is strictly positive. Let $\mathfrak{G}$ be the small-field event and $\mathfrak{G}^-$ decrease every coordinate threshold by one. Write

$$\varpi_0 = 18\ell_*^3 e^{-p_{0,*}/2} + 6\ell_*^3 e^{-(p_*-1)^2/(2C_0^2)},$$

$$t_2 = e\min\{N_*^{-1/2}, C_\beta^{-1}, (2\sqrt{N_*\mathbf{D}})^{-1}\},$$

$$\varpi_L = \sqrt{\varpi_0}e^{\mathbf{D}/2}, \quad M_L = 2 + e^{\mathbf{D}/2}, \quad \lambda_L = (\varpi_L^{1/3} M_L^{2/3})^{1/N_*} 5(2N_* r/t_2)^4.$$

A zero denominator in the minimum means infinity. Let $r = t_x/4$, $\bar\kappa = 6e_*^{1/4-9\varepsilon}$ and assume

$$2\varpi_{\mathrm{sep}}^{+} \le \bar\kappa^{N_*}, \quad N_*\bar\kappa^{1-s} \le 1/2, \quad \varpi_0 \le 1/4, \quad 8N_* t_2 \le t_x, \quad 0 < s < 1,$$

where $\varpi_{\mathrm{sep}}^{+}$ is the T13 bound after substituting $d_{\mathrm{sep}} \ge Mr_*/(8L) - 1$.

Claim. There is a positive measure $\nu_{\mathrm{sf}}$ of mass in $[1-\varpi_0, 1]$ on real twists and the terminal cutoff set, and

$$\frac{\bar\nu^{a(t)}(\mathfrak{G})}{\bar\nu(1)} = \int e^{G(t;\theta,A)}\, d\nu_{\mathrm{sf}}.$$

The normalized complement $L(t) = \bar\nu^{a(t)}(\mathfrak{G}^c)/\bar\nu(1)$ is entire. Its jet of degree at most four in every mark has coefficient norm at radius $r$ at most $\lambda_L^{N_*}$.

Proof work. Prove positivity from the original nonnegative real determinant and the density identity for every nonnegative bounded Borel observable. Use continuity and the nondegenerate Gaussian to include boundary fields; a formal Berezin expression alone is not positive. For real $t = ez$ the source shift changes every photon coordinate by at most one. Sector Gaussian domination and Cauchy–Schwarz then give $0 \le L(t) \le \varpi_L$. For complex marks of radius $t_2$, the full generating function has modulus at most $e^{\mathbf{D}/2}$ and its small-field part at most two, so $|L| \le M_L$.

Prove the general holomorphic estimate: a function bounded by $M$ on a radius-$R$ polydisc in $\mathbb{C}^n$ and by $\delta$ on its real points is bounded by $\delta^{1/3}M^{2/3}$ on radius $R/(2n)$. Successive one-variable harmonic measure proves it. Cauchy's estimate then gives the displayed fourth-jet loss.

Inputs. T05–T09 and T12–T13. Uniform complement estimates must be applied to the zero-flux law actually identified there.

Source. PDF pp. 238–240, §23.4, Proposition 23.6, equations (23.45)–(23.50).

### T15. Free-factor annihilation in the fourth-jet ring

Interface. Overview Target.

Proof dependencies. T07, T12, T13, T14
Objects and hypotheses. For $N$ cells let $\mathcal{J}=\mathbb{C}[t_c:c\in C]/(t_c^5:c\in C)$ with $\|P\|_r=\sum_{\alpha_c\le 4}|P_\alpha|r^{|\alpha|}$. Let $\phi_0$ be the fourth jet of an even one-variable function $N_0$ with $N_0(0)=1$ and $c_m=N_0^{(m)}(0)$. Assume $\|\phi_0\|_r\le 2$, $|c_2|\le r^{-2}/16$, $|c_4|\le r^{-4}/16$, and

$$\mathring{\Phi}=\prod_c \phi_0(t_c)\int \prod_{\emptyset\neq S\subset C}(1+k_S)\,d\nu_{\mathrm{sf}}+L^{[4]}.$$

Here $\nu_{\mathrm{sf}}$ is positive of mass $\le 1$; every monomial of $k_S$ uses exactly the variables of $S$; $\|k_{\{c\}}\|_r\le\kappa$, $\|k_S\|_r\le\kappa^N$ for $|S|\ge 2$, and $\|L^{[4]}\|_r\le\lambda^N$. Require $0<s<1$, $N\kappa^{1-s}\le 1/2$ and $\lambda\le\kappa^s$.
Claim. For $m=2,4$,

$$\left|\left[\prod_c(\partial_{t_c}^m-c_m)\mathring{\Phi}\right](0)\right|^{1/N}\le m!r^{-m}(2\exp(1)\kappa^s+\lambda)\le 8m!r^{-m}\kappa^s.$$

For $N=N_*=8L^3$, $\kappa=6e_*^{1/4-9\varepsilon}$ and $r=e_*^{3\varepsilon}/(4C_b)$, put $\mathfrak{e}=s(1/4-9\varepsilon)-12\varepsilon>0$. The reference mixed derivatives obey

$$\Theta_4^{(0)}\le 3\cdot 10^5C_b^4h^{1/8}e^{\mathfrak{e}},\qquad (1+|c_2|)\Theta_2^{(0)}\le 3\cdot 10^4C_b^4h^{1/8}e^{\mathfrak{e}}.$$

Here $\Theta_m^{(0)}$ is the full $N_*$th-root derivative from T07.
Proof work. The tensor functional $\ell_c(P)=(\partial_{t_c}^m-c_m)P|_0$ annihilates $\phi_0$. Only families of error sets covering all cells survive. Give each singleton cost one and each larger set cost $N$. Keep a factor $\kappa^{sN}$ and sum all remaining families using the explicit product bound. Positivity and mass $\le 1$ of the measure permit integration of the coefficient estimates. Derive the exact jet representation from T12–T14 before applying this abstract lemma.
Inputs. T07 and the one-cell, mixed-cell and complement estimates of T12–T14. This algebraic lemma assumes no RG construction.
Source. PDF pp. 240–241, §23.5, Theorem 23.7, equations (23.51)–(23.54).

### T16. A consistent positive charge interval

MODEL-SPECIFIC. Overview Target.

Proof dependencies. S01, S02, S03, S04, S05, S06, S07, S08, S09, S10, S11, S12, S13, S14, S15, S16, S17, S18, T01, T05, T08, T09, T10, T11, T12, T13, T14, T15

Objects and hypotheses. Fix $0<\rho<1$ and the standard chamber weights. Set

$$\bar\varepsilon=1/200,\quad c_X=54,\quad \kappa_0=54\log 72,\quad K_0=72^{54}/36,\quad c_{\mathrm{rb}}=54K_0,\quad c_{\mathrm{fl}}=61c_XK_0.$$

Choose $1-\bar\varepsilon<\alpha<1$, $\kappa\ge 7\kappa_0+7$, and an odd $L\ge 5$ satisfying

$$L^{1/8-3\bar\varepsilon/2}\ge 2400,\qquad c_{\mathrm{fl}}L^{-3/8+6\bar\varepsilon}\le 1/16,$$
$$L(\kappa-6\kappa_0-4)-1\ge\kappa,\qquad c_{\mathrm{rb}}L^3e^{-L(\kappa-\kappa_0-1)+\kappa}\le 1/2.$$

After $L$, the unweakened primitive estimates supply finite $a',C_V,C_P^{\mathrm{ph}}>0$, positive decay rates $\gamma^F,\gamma_P$ and a finite $\kappa_w$, all independent of $M$, charge, mesh, volume and reference mass. Put $\gamma_w=\min(\gamma^F,\gamma_P/2)$, $\alpha_0=1/54$, $\kappa'=\kappa_w+\kappa_0$, $N_c=(2L+1)^3$, and $\widehat\kappa=N_c\kappa+\log(6c_X)$. Choose $M$ a power of $L$ with

$$M\ge L^2,\quad M>4L,\quad \gamma_wM\ge 16,\quad M^{\alpha_0}\ge 1+e^{\widehat\kappa},\quad 804\alpha_0\log M\le\gamma_PM,$$
$$M\ge\max\{4L+1,(108\pi C_Va')^2,108\pi C_Va'e^{27\kappa'}\}.$$

The following calibration constants are finite after $M$ is fixed. The component kernel constant $C^F$ is the common majorant in T08, including its all-pair field estimate. With $c(\gamma)=\sum_{z\in\mathbb{Z}^3}e^{-\gamma|z|_\infty}$, take

$$C_w=64K_0\max\{C^F,116C_P^{\mathrm{ph}}M^{402\alpha_0},4e^{2\gamma_w}M^{27\alpha_0}\},$$
$$c_\gamma=2L^3e^{(2L+1)\gamma_w}c(\gamma_w),\quad c_C=3L^{9/2},\quad b_-=1-L^{-1},\quad b_+=b=1,$$
$$C_T=2\exp(3/2)[1+b_-^{-1}(54L\exp(6)+12\exp(1)+1)],$$
$$C_d=16c_XC_w\max\{\tfrac{\pi}{2}\exp(1)(1+2\exp(1)),2+59\exp(1)\},\quad c_S=10^5c_X^5K_0^5e^{6\kappa}.$$

Let $C_B>0$ be the local-gauge majorant supplied by the photon estimate, depending only on $C_P^{\mathrm{ph}},\gamma_P,L,M$. Use T08's full calibration of $C_0$ and $1\le c_\kappa\le K_0$; let $C_\Sigma<\infty$, $\gamma_\Sigma>0$ be T10's charge- and mass-independent twist constants. All these constants precede the charge.

Fix $r=2$, $p_0=9$, $p=19$, $s=1/2$, and $1-\alpha<\varepsilon\le 1/200$. Here is the finite running-charge inventory. For $v=-\log e_j$ abbreviate $T(A,d,\lambda)=Av^d\exp(-\lambda v)$:

$$T(1,p_0,\varepsilon/4)\le 1,\qquad T(1,0,5\varepsilon/2)\le\tfrac12,$$
$$T(c_{\mathrm{fl}},0,\varepsilon)\le 1,\qquad T(3M,0,\varepsilon)\le 1,$$
$$T(2C_w^2c_\gamma^2c_C,0,\varepsilon/4)\le 1,\qquad T(16c_Xc_\gamma C_wC_T^2M^3,0,\varepsilon/2)\le 1,$$
$$T(96M^3,0,1/4)\le 1,\qquad T(32C_wC_T,0,3\varepsilon/2)\le 1,$$
$$T(88M^3,0,1/4-6\varepsilon)\le 1,\qquad T(112C_w^2c_\gamma,0,2\varepsilon)\le 3,$$
$$T(c_S,0,1/4-5\varepsilon)\le 1,\qquad T(c_SL^3,0,\varepsilon)\le 1,$$
$$T(36c_X^2K_0^2,0,1/4-5\varepsilon)\le 1,\qquad T(1,0,\varepsilon)\le\tfrac14,$$
$$T(160L^{1+3\varepsilon/2}M,0,\varepsilon)\le 1,\qquad T(90c_XLM,0,\varepsilon)\le 1,$$
$$T(9ML^\varepsilon,0,2\varepsilon)\le 1,\qquad T(8C_B,p,3/4-3\varepsilon)\le 1,$$
$$T(8,0,3/4-7\varepsilon/2)\le 1,\qquad T(6L,0,\varepsilon)\le 1,$$
$$T(4C_Va',0,\varepsilon)\le 1,\qquad T(8c_{\mathrm{rb}}L^3C_dM^3,0,2\varepsilon)\le 1.$$

The remaining four sum conditions are

$$T(3M,0,1/4+\varepsilon)+T(3M,0,2\varepsilon)\le 1,\quad T(1,0,1/4)+T(1,0,\varepsilon)\le 1,$$
$$T(2M,1,\varepsilon)+T(32M,0,\varepsilon)\le 1,\quad T(4,0,1/4+2\varepsilon)+T(2,0,3\varepsilon)\le 1.$$

Also require $v^{2p_0}/2-7v\ge\log 12$, $v^2\ge 10L$, $v\ge 9$. For the terminal ceiling write $\ell(v)=LMv^r$ and impose

$$(c_{\mathrm{fl}}+2)e^{-\varepsilon v}\le (1-L^{-\varepsilon/2})/16,$$
$$8C_0^2\ell(v)^6(b_-^{-1}+L\ell(v))\le e^{\varepsilon v/2},\quad 4(2C_0+c_\kappa+5c_{\mathrm{fl}}+6)\ell(v)^3\le e^{\varepsilon v},$$
$$e^{-(1/4-9\varepsilon)v}\le 1/8,\quad 4C_0\ell(v)^2v^p<e^{\varepsilon v},$$
$$12\ell(v)^3[e^{-v^{p_0}/2}+e^{-v^{2p}/(2C_0^2)}]\le e^{-v},\quad C_\Sigma e^{-\gamma_\Sigma Mv^r}\le e^{-v},$$
$$L^2e^{-2v}\ell(v)\le 1,\quad 6N_*Mv^re^{-v}(ve^{-v}+1/2)\le\pi^2,$$
$$v\ge\max(8,31L+17,2r),\qquad p_0v^{p_0-1}\ge 28.$$

In these exponentials $e^x=\exp(x)$. The response inventory is the explicitly displayed conditional inequalities of T11–T15, together with

$$Mr_*/(8L)-1\ge 6M+8,\qquad \lambda_L\le\bar\kappa^s,$$
$$D_\beta^{\mathrm{buf}}/2+1/2\le 3\ell_c/8,\qquad D_\beta^{\mathrm{buf}}+3\le\ell_c,\qquad e^{-\bar mR^*}\le e\sqrt{h}.$$

Here the response constants $C_\beta,\mathbf{D},D_\beta^{\mathrm{buf}}$ are fixed last as in T05; $r_*,\ell_*,\ell_c,R^*$ are the rounded stop scales of T01 and T05. The last right-hand side uses the physical charge. No coefficient in these inventories depends on that later charge.

Claim. All geometric choices exist in the indicated order. There are $e_0,e_1>0$, with $e_1\le\min(e_0,1/3)$, for which respectively the running and terminal inventories hold on the entire half-lines $v\ge-\log e_0$ and $v\ge-\log e_1$. These ceilings are independent of mesh, volume, twist and reference mass. Moreover

$$\mathfrak{e}=s(1/4-9\varepsilon)-12\varepsilon\ge 17/400,\quad \sigma=\mathfrak{e}-2a_L/N_*>0,\quad a_L=(L^3-1)/(L-1)^3,\quad N_*=8L^3.$$

After fixing $0<\bar m\le 1$ and $h=L^{\lfloor\log_L(1/\bar m)\rfloor}$, set

$$e_G=\begin{cases}e_1, & h=1,\\ \min(e_1/\sqrt{h},1/(\sqrt{h}\log h)), & h>1.\end{cases}$$

For each compact probe there is $e_\sharp(\beta)>0$ such that the response inventory and centered placements hold for every $0<e\le\min(e_G,e_\sharp(\beta))$, uniformly in $N\ge N_{\mathrm{ref}}(e)$. The final positive-cumulant restriction is made in T18 after $C_U$ has been obtained.

Proof work. Choose $L$ before the unweakened constants and $M$ after them. Verify that calibration after $M$ changes no earlier bound. For each finite sum $\sum_i A_iv^{d_i}e^{-\lambda_iv}$ with positive $\lambda_i$, split its exponential in half and maximize $v^{d_i}e^{-\lambda_iv/2}$ to obtain one explicit finite half-line. The terminal tails and twist tails obey the displayed inventory. For the response use

$$v_*^r\le r_*,\quad Mv_*^r\le\ell_*<LMv_*^r,\quad a_*\in\{h/L,h/L^2\},\quad v_*=-\log(e\sqrt{h})+O(1),$$

and $p_\beta\le(D_\beta^{\mathrm{buf}}L^2/h+2M+4)^3$. The separated-cell errors decay exponentially in $v_*^2$; the complement derivative loss obeys

$$\lambda_L\le A_8\exp[-v_*^2/(12N_*)+(r/(2N_*)+4)v_*]$$

for a finite probe-dependent $A_8$. Its quadratic negative term beats every fixed linear radius loss. The centered-cube conditions follow as $\ell_c\to\infty$. Use downward-closed sets of ceilings and half their supremum; rounded stop scales need not be monotone.

Inputs. The primitive kernel/local-gauge estimates supplying the declared finite constants; T01 and T08–T15 supply conditional conclusions with the explicit scalar inventory above. No output of the arbitrary-volume transfer or final cumulant-positivity choice is an input.
Source. PDF pp. 241–247, Proposition 23.8 and Proposition 24.1, equations (23.55)–(24.11); the reference exponential condition is recorded with (25.17).

### T17. Normalized transfer from two reference volumes

MODEL-SPECIFIC. Overview Target.

Proof dependencies. M03, M04, M05, T03, T04, T05, T06, T07, T11, T15, T16

Objects and hypotheses. For coordinate cells of width $w \in \frac{1}{2}\mathbb{Z}_{>0}$, let $Z_s(O)$ be the full reflected product of a local scalar integrand on a torus with even cell counts $s_\mu$ and sides $ws_\mu \geq 4$. Assume the proved rectangular chessboard estimate and, in each direction, $Z_s(O) = \mathrm{Tr}(Y^*Y)^{s_\mu/2}$ with $Y$ independent of that count. Set $s_0$ even, $n_r = ws_0 \geq 4$, $N_r = s_0^3$, $j \geq 2$ integer, and

$$D_j = j^3 \log Z_{n_r}(1) - \log Z_{jn_r}(1), \quad \delta_j = \frac{D_j}{(j-1)^3 N_r}, \quad \|O\|_* = \left(\frac{Z_{(s_0,s_0,s_0)}(O)}{Z_{n_r}(1)}\right)^{1/N_r}.$$

Claim. For $s = s_0$ or even $s \geq js_0$, and distinct cells,

$$D_j \geq 0, \qquad \left|\mathbb{E}_{C(ws)} \prod_{c\in C} O_c^{[c]}\right| \leq \prod_{c\in C} e^{\delta_j}\|O_c\|_*.$$

For the all-flux gauge law the defect separates into photon, matter and flux parts:

$$\begin{aligned} D_j^{\mathrm{mat}} &= j^3 \log z_{r,n} - \log z_{r,jn}, \\ D_j^{\mathrm{ph}} &\leq (j^3-1)\log(2\pi/(q^2 n)) + K_j, \\ D_j^{\mathrm{fl}} &\leq -j^3 \log(1-\varsigma_n), \qquad \varsigma_n \leq 7e^{-2\pi^2/(q^2 n)} < 1/8 \quad (q^2 n \leq 1). \end{aligned}$$

Here

$$G_j = (1 + j/3.2)\,\frac{2j^2}{2.56}, \qquad K_j = (1 + j + j^2)(\log j + jG_j).$$

These are independent of $n, q$ and mass. The scalar-Laplacian logarithm is

$$\mathcal{L}(n) = \sum_{k\in(\mathbb{Z}/n\mathbb{Z})^3\setminus\{0\}} \log[\sum_\mu (2 - 2\cos(2\pi k_\mu/n))].$$

For the field application assume $0 < e \leq 1/3$, $0 < \eta \leq 1$, $0 < \bar{m} \leq 1$, $h = L^{\lfloor \log_L(1/\bar{m})\rfloor}$, $R^* = hL^{T^*}$, and outer physical cutoff $R = 8hL^T$. Take $s_0 = 2L$, $j = L$, $N_* = 8L^3$, $n_* = R^*/\eta$, $w = n_*/(2L) \in \frac{1}{2}\mathbb{Z}_{>0}$ and $q = e\sqrt{\eta}$. Require

$$6N_* e^2 R^* \log(1/e) \leq \pi^2, \qquad \exp(-\bar{m}R^*) \leq e\sqrt{h}.$$

For $O_m = e^{-m}H_m(X; v_\eta) - c_m$, put

$$M_4 = 600 N_*^2 v_\eta^2 e^{-4} + |c_4|, \qquad M_2 = 12 N_* v_\eta e^{-2} + |c_2|,$$
$$u_m = \exp(\delta_L)(\Theta_m^{(0)} + 2e^6 M_m).$$

Here $H_2(x; v) = x^2 - v$, $H_4(x; v) = x^4 - 6vx^2 + 3v^2$, and $X$ is the curvature paired with one cell source. Its translates are $F^\sharp(\tau_y f^\eta)$, with common compact support, $\|f^\eta - f\|_2 = O(\eta^2)$ and $v_\eta \to \|f\|_2^2$. $\Theta_m^{(0)}$ is the $N_*$th-root full mixed derivative in T07. More precisely, assume the supplied reference estimates $D_L^{\mathrm{mat}} \leq C_{\mathrm{mat}}$, $\Theta_4^{(0)} \leq C_{\mathrm{ref}} e^{\mathfrak{e}}$ and $(1 + |c_2|)\Theta_2^{(0)} \leq C_{\mathrm{ref}} e^{\mathfrak{e}}$, with constants independent of charge, mesh and placement, together with the free coefficient bounds of T03–T04. These are T11 and T15. For $\sigma = \mathfrak{e} - 2a_L/N_* > 0$,

$$u_4 \leq C_U e^\sigma, \qquad (1 + |c_2|)u_2 \leq C_U e^\sigma, \qquad |c_4 - 3c_2^2 - b(f)| \leq C_U(e^\sigma + \eta^{1/4}).$$

The local exponent definitions are $\mathfrak{e} = s(1/4 - 9\varepsilon) - 12\varepsilon$ and $a_L = (L^3 - 1)/(L - 1)^3$. Every finite $Y \subset (R^*/L)\mathbb{Z}^3$ incongruent modulo the cutoff periods satisfies, for $N \geq N_{\mathrm{ref}}(e)$, $T \geq T^*(e) + 1$,

$$\left|\mathbb{E}_{N,T} \prod_{y\in Y} [e^{-m} H_m(F^\sharp(\tau_y f^\eta); v_\eta) - c_m]\right| \leq u_m^{|Y|}.$$

Moreover

$$|\kappa_4(F^\sharp(f)) - e^4 b(f)| \le (8C_U + 3C_U^2)e^{4+\sigma} + C_U(\eta^2 + e^4\eta^{1/4}).$$

$C_U$ depends on fixed construction, mass and probe, not on charge or cutoffs in their stated interval.
Proof work. First prove the general normalized inequality by sidewise convexity and Schatten monotonicity. For its denominator use the exact photon normalization, including $[\mathcal{L}(jn) - j^3\mathcal{L}(n)]/2 \le -(j^3-1)\log n + K_j$, and constant-cochain Gaussian bounds on flux probabilities. Split zero/nonzero flux for the reference Wick product using Cauchy–Schwarz and high moments. The normalization cost is $\exp(\delta_L) \le Ce^{-2a_L/N_*}$, an inverse charge power. Finally replace $f^\eta$ by $f$ using the multilinear cumulant bound $|K_4(g_1, \dots, g_4)| \le 15 \prod \|g_i\|$.
Inputs. Earlier slab/chessboard traces and all-flux domination; T03–T07, T11, T15–T16.
Source. PDF pp. 247–255, §25, Propositions 25.1–25.4, equations (25.1)–(25.37).

## T18. Wick-product bounds give component cumulants

INTERFACE. Overview Target.

Proof dependencies. M09, M10, M14, M19, T02, T17
Objects and hypotheses. Fix $0 < e \le 1$ and a translation-invariant weak limit $\mu$ of cutoff field laws. Translations act continuously on $\mathbb{R}^3$ and $\mu = \int \mu_\xi \, d\pi(\xi)$ is its translation-ergodic decomposition. Let real tests $f_i$ have one common compact support, $f_i \to f$ in $L^2$, and $v_i \to \|f\|_2^2$, with mesh $\eta_i \to 0$. Assume the moment convergence and quantitative $L^2$ test continuity supplied by Gaussian domination. Almost every component of this same law obeys the Gaussian bound.

Let $c_{2,i}, c_{4,i}$ be bounded real sequences. On one fixed full-rank lattice $\lambda\mathbb{Z}^3$, $\lambda > 0$, suppose the cutoff laws satisfy, eventually for each finite set $Y$ of distinct translates,

$$\left| \mathbb{E}_i \prod_{y \in Y} [e^{-m} H_m(F_i(\tau_y f_i); v_i) - c_{m,i}] \right| \le u_{m,i}^{|Y|} \qquad (m = 2, 4),$$

where $H_2(x; v) = x^2 - v$, $H_4(x; v) = x^4 - 6vx^2 + 3v^2$ and

$$u_{4,i} \le Ce^\sigma, \quad (1 + |c_{2,i}|)u_{2,i} \le Ce^\sigma, \quad |c_{4,i} - 3c_{2,i}^2 - b(f)| \le C(e^\sigma + \eta_i^{1/4}).$$

Here $C < \infty$ and $\sigma > 0$ are independent of $i$ and of $e$ in its admitted interval. The generic input M19 states: if a measure-preserving lattice action and $G \in \bigcap_{p<\infty} L^p$ satisfy all distinct-product bounds $|\mathbb{E} \prod_{j=1}^J U(y_j)G| \le u^J$, then $|P_{\mathrm{lat}}G| \le u$ almost surely; for the continuous translation-invariant projection $P$, $P = PP_{\mathrm{lat}}$. No ergodicity of the lattice action is assumed.
Claim. On one set of full component measure,

$$|\kappa_4^{\mu_\xi}(f) - e^4 b(f)| \le (8C + 3C^2)e^{4+\sigma}.$$

If $b(f) > 0$ and $(8C + 3C^2)e^\sigma \le b(f)/2$, then $\kappa_4^{\mu_\xi}(f) \ge e^4 b(f)/2 > 0$. After T17 fixes its uniform $C_U$, choose

$$0 < e_\beta \le \min\{e_G, e_\sharp(\beta), 1, [b(f)/(2C_f)]^{1/\sigma}\}, \qquad C_f = 8C_U + 3C_U^2.$$

This final positive restriction uses no later construction input.
Proof work. Extract a convergent subsequence of the bounded reference coefficients while retaining the already chosen parent law $\mu$. Expand each finite Wick product into monomials, replace $f_i$ by $f$ using the quantitative $L^2$ test estimate, and pass to this same law. All moments required by M19 are available from Gaussian domination. Apply M19 to $G_m = e^{-m} H_m(F(f); \|f\|_2^2) - c_m$, $m = 2, 4$. Intersect its two full-measure component sets with the Gaussian component set, on which the field mean is zero. If $d_m = \mu_\xi(G_m)$, then

$$e^{-4}\kappa_4^{\mu_\xi}(f) = c_4 + d_4 - 3(c_2 + d_2)^2.$$

Use the two conditional bounds, the bound for $c_4 - 3c_2^2 - b(f)$, and $e^{2\sigma} \le e^\sigma$ to obtain the stated error. The proof of the invariant-product lemma belongs to M19 and is not repeated here.
Inputs. M19, T17, and the moment convergence/component Gaussian-domination conclusions for the same parent law. For the final OS conclusion, intersect with the OS component set of the volume-first law selected by the independent OS branch.
Source. PDF pp. 255–257, §26, Proposition 26.2, equations (26.4)–(26.9); Lemma 26.1 is the separate task M19.